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Online Convex Optimization based tuning of Mixture of Experts for Sports Outcome Prediction

DIPLOMA THESIS

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ΤΟΜΕΑΣ ΤΗΛΕΠΙΚΟΙΝΩΝΙΩΝ

Κυρτή Βελτιστοποίηση σε Πραγματικό Χρόνο Βασισμένη στην Παραμετροποίηση Μείγματος Ειδικών για την Πρόβλεψη Αποτελεσμάτων Αθλητικών Αγώνων

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Abstract

This thesis predicts sports match outcomes (home / draw / away) as a problem of **online learning with expert advice**, rather than as the training of a statistical model. Each betting operator is treated as an expert quoting odds on the three outcomes, from which probabilities are derived. The task is to find a weight vector such that the mixture of experts outperforms each of them individually, learning not retrospectively over the past but match by match.

Three stages of **Online Convex Optimization** (OCO) are implemented. First, three algorithms are built (Hedge, Online Gradient Descent, Follow-The-Regularized-Leader), which handle partially covering operators through the **Sleeping Experts** reduction. A second OCO stage then calibrates the mixture via online temperature scaling. Finally the bet size is computed with the Kelly criterion, whose multiplier is itself learned online with Online Newton Step. Every comparison is tested with a moving block bootstrap. The dataset spans 22 leagues, 10 seasons and 76,584 matches.

The mixture significantly outperforms every individual bookmaker except the sharpest, with which it draws level in the configuration ultimately adopted. Under a deployment test with a genuine temporal cut (fitted through the end of 2024, evaluated on 2025–26) it significantly outperforms every opening price and draws level with the bookmakers' closing prices. The one source of odds that significantly outperforms it is Betfair Exchange, which is not a bookmaker but a betting exchange: a platform on which bettors bet against one another, so that its odds are set by the bettors themselves and carry no profit margin added by a company.

The three broader findings are that learning pays only where experts are heterogeneous in quality; that better prediction does not translate directly into profit, since the quality gain falls short of the market's margin; and that part of the advantage over the uniform average is due not to learning but to the geometry of the Sleeping Experts reduction.

Keywords: online convex optimization, prediction with expert advice, sleeping experts, regret, probability calibration, betting markets

Περίληψη

Η παρούσα εργασία προβλέπει τα αποτελέσματα αθλητικών αγώνων (1/X/2) ως πρόβλημα **online μάθησης με συμβουλή ειδικών**, και όχι ως εκπαίδευση στατιστικού μοντέλου. Κάθε στοιχηματικός πάροχος θεωρείται ένας expert που δίνει αποδόσεις για τα τρία ενδεχόμενα, από τις οποίες προκύπτουν πιθανότητες. Ζητούμενο είναι η εύρεση ενός διανύσματος βαρών έτσι ώστε το μείγμα των experts να τους ξεπερνά ατομικά, με μάθηση όχι αναδρομικά πάνω στο παρελθόν, αλλά αγώνα προς αγώνα.

Υλοποιούνται τρία στάδια Κυρτής Βελτιστοποίησης σε Πραγματικό Χρόνο (**Online Convex Optimization**, OCO). Αρχικά κατασκευάζονται τρεις αλγόριθμοι (Hedge, Online Gradient Descent, Follow-The-Regularized-Leader), που με την αναγωγή σε **Sleeping Experts** διαχειρίζονται τους παρόχους μερικής κάλυψης. Έπειτα, με δεύτερο στάδιο OCO, γίνεται βαθμονόμηση μέσω online temperature scaling. Τέλος υπολογίζεται το μέγεθος του στοιχήματος με το κριτήριο Kelly, όπου ο πολλαπλασιαστής του μαθαίνεται online με Online Newton Step. Κάθε σύγκριση ελέγχεται με moving block bootstrap. Το σύνολο δεδομένων καλύπτει 22 πρωταθλήματα, 10 σεζόν και 76.584 αγώνες.

Το μείγμα ξεπερνά στατιστικά σημαντικά κάθε μεμονωμένο bookmaker πλην του σκληρότερου, με τον οποίο ισοπαλεί στη διαμόρφωση που επιλέγεται τελικά. Σε έλεγχο ανάπτυξης με πραγματική χρονική τομή (εκπαίδευση έως το τέλος του 2024, μέτρηση στο 2025–26) κερδίζει σημαντικά κάθε τιμή ανοίγματος και ισοπαλεί με τις τιμές κλεισίματος των bookmakers. Η μόνη πηγή αποδόσεων που το ξεπερνά σημαντικά είναι το Betfair Exchange, το οποίο δεν είναι στοιχηματική εταιρεία αλλά ανταλλακτήριο στοιχημάτων: οι παίκτες στοιχηματίζουν ο ένας απέναντι στον άλλον, οπότε τις αποδόσεις τις διαμορφώνουν οι ίδιοι και δεν περιέχουν περιθώριο κέρδους κάποιας εταιρείας.

Τα τρία γενικότερα ευρήματα είναι ότι η μάθηση αποδίδει μόνο όταν οι experts έχουν ανομοιομορφία στην ποιότητα, ότι η καλύτερη πρόβλεψη δεν μεταφράζεται άμεσα σε κέρδος καθώς η διαφορά ποιότητας υπολείπεται του περιθωρίου της αγοράς, και ότι ένα μέρος της υπεροχής έναντι της ομοιόμορφης μέσης τιμής δεν οφείλεται στη μάθηση αλλά στη γεωμετρία της αναγωγής Sleeping Experts.

Λέξεις-κλειδιά: online κυρτή βελτιστοποίηση, πρόβλεψη με συμβουλή ειδικών, sleeping experts, regret, βαθμονόμηση πιθανοτήτων, αγορές στοιχηματισμού

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Chapter 1

Introduction

1.1 The problem

Predicting the outcome of a football match is usually treated as a supervised learning problem: features of teams and matches are collected, a model is trained, and it is evaluated on data it has not seen. This approach runs into a practical barrier. Betting companies already solve the same problem, with incomparably more data, with teams of analysts, and with a direct financial incentive to solve it correctly. A model trained within the scope of a thesis will struggle to compete with them on their own ground.

This thesis changes the question. Rather than building yet another forecaster, it treats *each betting company as an expert* that publishes, ahead of every match, its own probabilities (p_H, p_D, p_A) over the three outcomes: p_H that the home team wins, p_D that the match is drawn and p_A that the away team wins, with $p_H + p_D + p_A = 1$. The task then becomes: how are these opinions to be combined, online and without knowledge of the future, so that the mixture is a better forecaster than any one of them taken separately?

The framework that answers this is *prediction with expert advice* and, more generally, *Online Convex Optimization* (OCO). Its principal advantage here is that it yields regret guarantees **without any statistical assumption** about the distribution of the data: the outcomes are not required to come from a fixed distribution, nor the experts to be independent. That suits a market which changes over time, where companies appear and disappear and where prices are set strategically.

1.2 Why this is not trivial

Applying the framework to real data meets three difficulties, and these shape the structure of the thesis.

Partial coverage. The classical algorithms assume that every expert issues a prediction in every round. In the data used here, per-bookmaker coverage ranges from 6.9% to 99.8% of matches, and several companies appear or disappear midway through the decade. This calls for the *Sleeping Experts* reduction, which gives each expert a guarantee measured only over the rounds in which it was present.

Odds are not probabilities. Published odds embed the bookmaker's profit margin: their reciprocals sum to more than one. Converting them into probabilities (de-vigging) is a modelling choice that can influence the conclusions, and it is therefore checked with two independent methods.

The best log-loss is not the objective. A mixture may predict better than every bookmaker and still lose money, if its advantage is smaller than the market's margin. The thesis tests this gap explicitly, through a betting simulation based on the Kelly criterion.

1.3 Contributions

The thesis makes the following contributions:

- A complete implementation of three OCO algorithms under the Sleeping Experts reduction, over a dataset of 22 leagues, 10 seasons and 76,584 matches, with 26 bookmakers as experts.
- Two further OCO stages layered on the mixture: calibration via online temperature scaling, and bet sizing via a learned Kelly multiplier, so that the entire chain from odds to bet is online and causal. It is shown that *what* is learned in the third stage is decisive: learning the stake directly fails systematically, whereas learning the multiplier on top of the closed-form rule does not.
- Systematic statistical testing of every comparison with a moving block bootstrap, rather than comparison of point estimates.
- A deployment test with a genuine temporal cut, answering the question of what a model fitted through 2024 would do today.
- A series of negative findings that bound the scope of the method and are, in our view, as useful as the positive ones.

1.4 Thesis outline

The rest of the thesis is structured as follows. Chapter 2 reviews the related literature. Chapter 3 develops the theoretical background. Chapter 4 describes the data, its preprocessing and the experimental design. Chapter 5 gives the results and discusses them. Chapter 6 summarises, states the limits of the method and proposes future work. Appendix A gives the full result tables.

Chapter 2

Related work

This thesis sits at the confluence of three research traditions that developed largely independently: the theory of online learning, sports outcome prediction, and the economic study of betting markets. This chapter sketches each and locates the present contribution.

2.1 Online learning and regret theory

In *prediction with expert advice* a learner must issue a forecast at every round, and before doing so it sees the forecasts of a fixed set of N *experts*; in this thesis each expert is a betting company and its forecast is the probabilities implied by its odds. The learner's performance is measured by its *regret*: how much larger its total loss over T rounds is than that of the best single expert, or the best fixed combination of experts, chosen in hindsight. A regret that grows more slowly than T means that, averaged per round, the learner does as well as that best choice. The formal definitions are in Chapter 3.

The idea of prediction with expert advice predates modern machine learning. In its computational form it appears with the *weighted majority* algorithm of Littlestone & Warmuth (1994) [31], which showed that one can guarantee performance almost as good as that of the best expert in a set, with no assumption whatsoever about how the data are generated. The generalisation to continuous weights and convex losses led to *Hedge* (Freund & Schapire, 1997) [12], which keeps one weight per expert and after every round multiplies it by a factor that shrinks with that expert's loss, and more broadly to the family of *multiplicative updates*.

In parallel came the *aggregating algorithm* of Vovk (1990) [43], which exploits a stronger property of certain losses, *exp-concavity*. Exp-concavity is a curvature condition, stronger than convexity, which the logarithmic loss satisfies (it is defined in §3.3.4). Under it the regret can be bounded by $\ln N$, where N is the number of experts, that is *independently* of the *horizon* T, the total number of rounds, instead of growing as $O(\sqrt{T \log N})$. That result is theoretically stronger, and this thesis provides an empirical case in which it nevertheless fails, for reasons connected to the non-stationarity of the expert set.

The unification of the above under the common heading of *online convex optimization* is due in large part to Zinkevich (2003) [45], who showed that simple projected gradient descent attains $O(\sqrt{T})$ regret for any sequence of convex functions. Modern synthetic treatments of the field are given by Cesa-Bianchi & Lugosi (2006) [7] and Hazan (2016) [23].

Two extensions bear directly on our application. The first is the *sleeping experts* or *specialists* reduction (Freund, Schapire, Singer & Warmuth, 1997; Blum & Mansour, 2007) [13,

3], which handles experts that do not participate in every round. The second is *Fixed-Share* (Herbster & Warmuth, 1998) [24], which changes the comparator from the best fixed vector to the best sequence of vectors with a bounded number of switches, and therefore addresses precisely the problem of non-stationarity.

Finally, adaptive learning rates of the AdaGrad type (Duchi, Hazan & Singer, 2011) [11] propose per-coordinate scaling of the step according to accumulated gradient. The idea is appealing in a setting where experts differ greatly in how often they participate, and it is examined explicitly in §5.6.

2.2 Sports outcome prediction

The literature on football outcome prediction follows, roughly, three directions.

The first is *parametric statistical models*, starting from the observation that goals are approximated reasonably well by a Poisson distribution. Such models estimate attacking and defensive strength per team and produce a score distribution, from which the home/draw/away probabilities follow. Their main advantage is interpretability; their main drawback is that they require sufficient history per team and struggle with transfers, managerial changes and promotion or relegation.

The second is *machine learning over features*, where variables of form, ranking, head-to-head statistics and, in more recent work, player tracking data are assembled. Such models often perform well against parametric alternatives, but their comparison against the market is rarer and usually less rigorous.

The third, and the most relevant here, uses *the odds themselves as the prediction*. This approach recognises that the market price incorporates all available information, including items that never appear in public statistics. Kaunitz et al. (2017) [26] is the clearest empirical precedent: rather than building a competing forecast, they treat the consensus of the published odds as the probability estimate and bet where an individual bookmaker departs from it, a strategy profitable over a decade of simulation and over five months of real-money betting. The present work belongs to this category, with two differences. It does not use a fixed consensus but learns online how to weight the companies, with a regret guarantee; and the weighting is driven by a loss that each company pays on its own quoted matches, so a company covering a fraction of the fixtures can still contribute.

2.3 Efficiency of betting markets

Betting markets are a privileged field for the study of market efficiency, for one specific reason: unlike financial markets, every bet has *a fixed expiry date and an objectively verifiable outcome*. The fundamental value does not have to be estimated; it is revealed.

Levitt (2004) [30] documents that betting markets are organised differently from financial ones: the bookmaker does not merely act as an intermediary balancing the two sides, but takes an active position, exploiting systematic biases of bettors. That observation matters for interpreting the differences between companies: they reflect not only differing predictive ability but also differing business strategy.

Market microstructure theory supplies the framework for understanding how information becomes embedded in price. The models of Kyle (1985) [29] and Glosten & Milgrom

(1985) [16] show that, in the presence of informed traders, price movement is the mechanism through which private information becomes public, and that the bid–ask spread is the market maker's protection against them. That framework bears directly on the difference between opening and closing prices (§4.1.3).

2.4 Favourite–longshot bias and odds conversion

The most systematically documented anomaly of betting markets is the *favourite–longshot bias*: low-probability outcomes are systematically priced more expensively than their true frequency warrants. The phenomenon has been confirmed across multiple sports and decades (Thaler & Ziemba, 1988) [42].

Shin (1992, 1993) [38, 39] offered an explanation that treats it not as a psychological bias of bettors but as a *rational response* by the bookmaker to the existence of informed traders. The model generates the shape of the bias endogenously and, most importantly for us, provides an inversion procedure that recovers fair probabilities together with an estimate of the proportion z of informed traders. Štrumbelj (2014) [41] systematically compares methods for converting odds into probabilities and finds that Shin's method outperforms proportional normalisation in the quality of the resulting probabilities.

2.5 Probability calibration

The distinction between the *reliability* and the *resolution* of a probabilistic forecast goes back to Murphy (1973) [34], who showed that every proper scoring rule decomposes into these components. The observation has an immediate practical consequence: a forecaster may discriminate excellently between likely and unlikely outcomes and still report wrong probabilities, an error correctable with no additional information whatsoever.

In the modern machine learning literature the issue resurfaced with Guo et al. (2017) [18], who showed that deep neural networks are systematically overconfident and that a one-parameter correction (*temperature scaling*) suffices to calibrate them. The present work uses the same family of correction, with the difference that the parameter is not fitted on a validation set but learned *online*, as a second OCO stage on top of the first.

2.6 Forecast combination

The idea that combining many forecasts beats selecting one has a long history in econometrics and meteorology. A persistent empirical finding, known as the *forecast combination puzzle*, is that the simple average often proves more robust than weighted combinations whose weights are estimated from data: the gain from better weighting is cancelled by the estimation noise in the weights themselves.

The combination of online aggregation with the sleeping-experts reduction is established practice outside this domain. Pfitzner et al. (2025) [35] apply exactly that pairing to operational temperature forecasting, waking experts that are poor on average but good in identifiable circumstances, and obtain second-order regret bounds for the result. The reduction this thesis relies on is therefore not novel in itself; what is new is the observation

that in a betting panel the sleeping pattern is not a nuisance to be tolerated but carries information, since which companies quote a given fixture is itself a signal, and that the same reduction induces a movement of the weight vector that must be controlled for before any gain is read as learning (§5.5.1).

That finding bears directly on our results. This thesis shows that learned combination beats the simple average when the experts differ substantively in quality, and falls short of it when they do not. The puzzle is therefore not a general law but a statement about homogeneous panels: the condition it depends on is the spread in expert quality, and that condition is measurable in advance.

2.7 Position of the present work

Taken together, the above delineates the gap this thesis addresses. Online learning theory offers algorithms with guarantees but is more rarely applied to betting markets with genuine partial coverage. The sports prediction literature often uses odds as a benchmark, but less often as *a set of experts to be combined*. And the economic literature on markets explains why companies differ, but does not exploit those differences algorithmically.

Online convex optimisation has been brought to betting markets, but from the opposite side of the counter. Zhu et al. (2024) [44] place the *bookmaker* in the role of the learner, updating posted odds as bets arrive, and prove $O(\sqrt{T})$ regret for the resulting pricing algorithms. Their central result is worth stating here because it is the mirror image of one of ours: they show that maximising the operator's profit and eliciting information from bettors are fundamentally incompatible objectives. Read from the bettor's side, that is the same separation this thesis measures between predictive and economic performance (§5.4), and it supplies a reason to expect it rather than merely an observation that it occurred.

Our contribution is the systematic synthesis of the three: online algorithms with guarantees, applied to the full set of available companies with correct handling of partial coverage, with statistical testing of every comparison and with an explicit distinction between predictive and economic performance.

Chapter 3

Theoretical background

This chapter presents the algorithms the thesis uses, the guarantees that come with them and the quantities by which their output is evaluated, and explains what each one contributes. Proofs are standard and are left to the cited sources.

3.1 The online learning framework

3.1.1 Online learning and regret

In the online learning setting (Cesa-Bianchi & Lugosi, 2006; Shalev-Shwartz, 2012; Hazan, 2016) [7, 37, 23] there is no separation into a training phase and an application phase. The algorithm operates in rounds: at round t it issues a prediction based only on what it has seen up to round $t-1$, the true outcome is then revealed, it incurs a loss, and it updates its state. Every prediction is therefore out of sample by construction, with no need to split the data.

The measure of success is not the absolute loss but the *regret*: how much worse the algorithm did than the best *fixed* choice in hindsight.

Definition 3.1 (Regret). Let $\mathcal{K}$ be a convex set, $f_1, \dots, f_T$ a sequence of convex loss functions, and $x_1, \dots, x_T \in \mathcal{K}$ the algorithm's choices, where x_t depends only on $f_1, \dots, f_{t-1}$. The regret is defined as

$$\text{Regret}_T \;=\; \sum_{t=1}^{T} f_t(x_t) \;-\; \min_{x \in \mathcal{K}} \sum_{t=1}^{T} f_t(x).$$

An algorithm is considered successful when its regret grows sublinearly in T, that is $\text{Regret}_T / T \to 0$: in the long run its average loss approaches that of the best fixed choice. What matters for our application is that this guarantee holds **for every** sequence, with no assumption of independence or of a stationary distribution.

This point deserves to be stated precisely, because it is the reason the OCO framework is chosen here over the statistical one. In statistical learning theory the object of interest is a bound on the *risk* $\mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(h(x), y)]$, the expected loss ℓ of a predictor h that maps an input x to a forecast of the outcome y, which presupposes that a fixed distribution $\mathcal{D}$ exists and that samples are drawn independently from it. Neither assumption is harmless here: odds are set by a market that changes over the decade, and consecutive matches share matchday, league and season. Regret requires neither. The price is that the guarantee

becomes *relative*: it does not say that we shall predict well, only that we shall not predict substantially worse than the best fixed combination of the experts. If all the experts are bad, the guarantee holds and the result remains bad.

The protocol. Formally, online convex optimization is the following game between the algorithm and a (potentially adversarial) environment:

for $t = 1, 2, \ldots, T$:
the algorithm chooses $x_t \in \mathcal{K}$;
the environment reveals a convex $f_t : \mathcal{K} \to \mathbb{R}$;
the algorithm incurs loss $f_t(x_t)$ and observes f_t (or part of it).

The phrase "or part of it" is essential and returns in §3.3.8: the amount of information returned to the algorithm after each round defines different problems with different optimal bounds. In the full information model the whole of f_t is revealed, and hence its gradient at every point.

Are the bounds tight? The regret bounds proved for the algorithms below are *upper* bounds: each says that the regret is at most so large. An upper bound on its own does not say whether an algorithm is good or merely whether its analysis is loose. That is settled by a *lower* bound, which says that no algorithm whatsoever can do better, and Theorem 3.1 is such a bound. For the experts problem the answer is known, and it rules out any improvement in order of magnitude.

Theorem 3.1 (Lower bound for prediction with expert advice). *For every prediction algorithm with N experts and every sufficiently large T there exists a sequence of losses $\ell_t \in [0,1]^N$ such that*

$$\text{Regret}_T \;\geq\; (1 - o(1))\,\sqrt{\tfrac{1}{2}\,T \ln N}.$$

Hedge's bound (Theorem 3.6) is $M\sqrt{T \ln N/2}$, which therefore matches the lower bound up to the factor M expressing the scale of the losses. The algorithm is thus *optimal in order of magnitude* and there is no room for improvement in the worst case. This matters for interpreting the empirical results of §5.1: when it is found there that the theoretically optimal step is two orders of magnitude smaller than the empirically best one, that does *not* mean the theory is wrong. It means the data are not the worst case, and that the distance between the worst case and the particular sequence we happen to have is large.

The static comparator. A second limitation is worth noting, one that proves essential at several points of this thesis: the comparator is *static*. The guarantee concerns comparison with the best fixed weight vector, chosen in hindsight but remaining a single vector for the entire period. When the object itself changes (when, for example, an expert stops participating) an algorithm that converges quickly to one answer is punished for precisely that. The generalisation that lifts the restriction is defined as follows.

Definition 3.2 (Dynamic regret and tracking regret). For a comparator sequence $u_1, \dots, u_T \in \mathcal{K}$ the *dynamic regret* is defined as

$$\mathrm{Regret}_T^d(u_{1:T}) \;=\; \sum_{t=1}^{T} f_t(x_t) \;-\; \sum_{t=1}^{T} f_t(u_t).$$

When the sequence is restricted to those changing value at most m times, that is $|\{t : u_{t+1} \neq u_t\}| \leq m$, the quantity is called the *tracking regret* of order m.

Static regret is the special case $m = 0$. Without a restriction on m no non-trivial bound is possible: with u_t free to be the hindsight-best expert at every round, the dynamic regret can grow linearly. Fixed-Share, in §3.3.6, is precisely the algorithm that trades a worse constant in the static regret for a non-trivial bound at $m > 0$.

3.1.2 Prediction with expert advice

In the special case of interest here, $\mathcal{K}$ is the probability simplex

$$\Delta_N \;=\; \Big\{ w \in \mathbb{R}^N : w_i \geq 0,\ \textstyle\sum_{i=1}^{N} w_i = 1 \Big\},$$

where N is the number of experts. At round t each expert i publishes a prediction $p_{t,i} \in \Delta_3$ over the three outcomes, the algorithm chooses weights $w_t \in \Delta_N$, and predicts the mixture

$$\hat{p}_t \;=\; \sum_{i=1}^{N} w_{t,i}\, p_{t,i}.$$

The outcome $y_t \in \{H, D, A\}$ is revealed and the algorithm incurs the logarithmic loss $\ell(\hat{p}_t, y_t) = -\log \hat{p}_t[y_t]$.

The logarithmic loss is a *strictly proper scoring rule*: its expectation is uniquely minimised when the forecaster reports its true probabilities. That makes it appropriate here, in contrast to accuracy, which is not a proper rule and ignores entirely how confident a prediction was.

Theorem 3.2 (Propriety of the logarithmic loss). *Let p be the true outcome distribution and q the reported one. Then*

$$\mathbb{E}_{y\sim p}\big[-\log q_y\big] \;=\; H(p) \;+\; \mathrm{KL}(p \,\|\, q) \;\geq\; H(p),$$

with equality if and only if $q = p$.

Here $H(p) = -\sum_y p_y \log p_y$ is the Shannon entropy of p, the loss that remains even with perfect knowledge of the distribution, and $\mathrm{KL}(p \,\|\, q) = \sum_y p_y \log(p_y/q_y)$ is the Kullback–Leibler divergence of q from p, the extra loss paid for reporting q instead (Cover & Thomas, 2006) [9].

The corollary is that a forecaster has no incentive whatsoever to report probabilities other than those it genuinely holds. This property fails for accuracy: a forecaster interested only in the proportion of correct picks may report 100% confidence on every match at no penalty, provided it preserves the same ordering.

Linear versus logarithmic pooling

The definition $\hat{p}_t = \sum_i w_{t,i} p_{t,i}$ is a choice, not the only possible one. It is called *linear pooling* and is the natural choice here, because it is what makes the problem convex in w and therefore solvable with the tools of online convex optimization. The alternative is *logarithmic pooling*, the normalised geometric mean

$$\hat{p}_k^{\log} \propto \prod_{i=1}^{N} p_{t,i}[k]^{\,w_{t,i}}.$$

The two behave qualitatively differently, and that difference predicts one of the empirical findings of this thesis.

Proposition 3.3 (Linear pooling is under-confident). *Let $\hat{p} = \sum_i w_i p_i$ with $w \in \Delta_N$. Then:*

(a) $\max_k \hat{p}_k \leq \max_i \max_k p_{i,k}$*, with equality only if all experts with $w_i > 0$ attain their maximum at the same k with the same value;*

(b) $H(\hat{p}) \geq \sum_i w_i H(p_i)$*, where H is the Shannon entropy, with equality if and only if all experts with $w_i > 0$ agree exactly.*

In other words, the mixture is always *less* confident than the average confidence of its members, and the confidence it loses is exactly their disagreement. This is not a defect of the implementation but a property of the operator: linear pooling systematically pushes probabilities toward the centre of the simplex. Logarithmic pooling does not have this property (it can produce a distribution sharper than any of its members, provided the members agree), and for that reason it is said to be *externally Bayesian*.

The consequence is predictable in advance and is confirmed empirically: stratifying the predictions into ten probability bands, an individual bookmaker is better calibrated than the mixture in *all ten*, even when the mixture has the lower overall log-loss. This is no contradiction. The log-loss decomposes (§3.4.1) into a resolution term and a reliability term; pooling improves the former and worsens the latter.

3.2 From odds to probabilities

3.2.1 De-vigging

Before any algorithm there is a modelling step whose result depends on an assumption, and which can therefore affect every number that follows. Published odds are not probabilities: their reciprocals sum to more than one. If o_i is the decimal odds of outcome i and $\pi_i = 1/o_i$, then

$$B = \sum_i \pi_i > 1,$$

and the excess $B - 1$ is the *overround*, the bookmaker's profit margin. Converting the π_i into probabilities requires an assumption about *how* that margin is distributed across the three outcomes.

Proportional normalisation. The simplest choice assumes the margin is distributed proportionally:

$$p_i \;=\; \frac{\pi_i}{B}.$$

It is transparent and even-handed across outcomes, but it embeds a strong implicit assumption: that the bookmaker charges the same percentage margin on a favourite and on a longshot. The empirical literature does not bear this out. The well-known *favourite–longshot bias* (the systematic overpricing of unlikely outcomes in published odds; Thaler & Ziemba, 1988 [42]) means precisely that the margin is heavier on longshots.

Shin's method. Shin (1992, 1993) [38, 39] derives the distribution of the margin from a microeconomic argument rather than assuming it. The model posits that a proportion z of bettors hold private information and bet with certainty on the true outcome, while the remaining $1 - z$ bet according to public probabilities. The bookmaker, knowing that such bettors exist but not who they are, prices so as to protect itself. That protection is more expensive where private information pays off most, namely on low-probability outcomes, and so the model generates the favourite–longshot bias *endogenously*.

Inverting the model, the fair probabilities are given by

$$p_i(z) \;=\; \frac{\sqrt{z^2 + 4(1-z)\,\pi_i^2/B} \;-\; z}{2(1-z)},$$

with z determined by the requirement $\sum_i p_i(z) = 1$. The equation has no analytic solution; in the implementation z is found per match by bisection over the interval $[0,\, 0.4]$. The interval is a generous bracket rather than a modelling choice: z is a proportion of bettors, and on these data it comes out at 3.1% on average, with a 99th percentile of 6.3%. The upper end is reached by 495 of the $835{,}550$ quotes (0.06%), all but one of them opening prices of Betfair Exchange with overrounds of 68% and above, against a median of 6% for the data as a whole; no z in the interval satisfies the condition there, and the cap acts as a bound.

Why both are retained. The de-vig choice is an assumption, not a datum, and could in principle determine a conclusion. For that reason the entire dataset is built twice, once with each method, and every central result is checked under both. It is of further interest that the estimated parameter z is itself a quantity with an interpretation, which we exploit later as an indicator of information arrival.

3.2.2 Microstructure of the betting market

The de-vigging of §3.2.1 treats the overround as a technical obstacle to be removed. It is worth saying *why* it exists, because the answer explains two of the central findings of this thesis: the opening–closing difference, and why one particular firm is not beaten.

The market maker's problem. The bookmaker publishes prices before learning the outcome and accepts bets from two kinds of counterparty: the *uninformed*, who bet for reasons unrelated to information, and the *informed*, who bet only when the price is wrong in their favour. The second category creates *adverse selection*: the mere fact that someone accepts your price is bad news about your price.

The models of Glosten & Milgrom (1985) [16] and Kyle (1985) [29] show that the rational response is not to refuse to trade but to price with a *margin*: the maker charges enough that the profits from the uninformed cover the losses to the informed. The overround is exactly that margin, and its size is an increasing function of the estimated proportion of informed traders. In Kyle's model the price moves linearly with the *order flow*, the net volume of orders arriving on each side (here, the amount bet on each outcome), with a coefficient λ measuring market *depth*; each transaction reveals part of the private information and the price converges toward fundamental value as expiry approaches.

Remark 3.4 (Why the closing price is the better forecast). The consequence is immediate and testable: if price movement is the mechanism by which private information becomes public, then the closing price incorporates whatever the market learned between opening and kick-off, and must be systematically more accurate. More strongly: its advantage must be *larger where the price moved more*, since that is precisely where more information arrived. This is a quantitative prediction and not a mere qualitative expectation, and it is tested in §5.2.3.

The connection to Shin. Shin's model, which §3.2.1 uses as a de-vigging method, is the same idea in a form one can invert: the parameter z *is* the estimated proportion of informed traders, and the favourite–longshot bias arises endogenously because protection costs more where private information pays off more, namely on unlikely outcomes. The quantity z is therefore two things at once: a nuisance parameter for converting odds into probabilities, and *a measurable quantity with an economic interpretation.*

Two things the framework does not license. First, odds alone do not distinguish *private* from *public* information: a large movement may be due to a leaked line-up or to an announced injury. The claim concerns the *arrival* of information, not its nature, and any talk of "insider detection" would be an overstatement. Second, the model explains why a margin exists but does not predict its *size* for a particular firm; that depends on commercial strategy, cost of capital and risk tolerance, that is on factors outside our data (§6.2).

3.3 The algorithms

Three algorithms are used to learn the weights of the experts: *Hedge*, *Online Gradient Descent* (OGD) and *Follow-The-Regularized-Leader* (FTRL). They are presented in that order, followed by the *Aggregating Algorithm*, which exploits a stronger property of the loss, and by the reduction that lets all of them handle experts that do not quote every match.

3.3.1 Hedge

The Hedge, or multiplicative weights, algorithm (Freund & Schapire, 1997) [12] maintains one weight per expert and decays it exponentially in that expert's loss:

$$w_{t+1,i} \;\propto\; w_{t,i}\, e^{-\eta_t\, \ell_{t,i}}, \qquad \ell_{t,i} = -\log p_{t,i}[y_t].$$

The quantity $\eta_t > 0$ is the *step size*, also called the learning rate: it sets how far one round's loss is allowed to move the weights, and the whole of §5.1 is about choosing it.

With a suitable η_t the algorithm attains regret $O(\sqrt{T \log N})$. The dependence on N is only logarithmic, which is what makes large expert sets tractable.

Lemma 3.5 (Hoeffding, 1963 [25]). *Let X be a random variable taking values in an interval of width M. Then for every $\lambda \in \mathbb{R}$*

$$\ln \mathbb{E}\big[e^{\lambda X}\big] \;\le\; \lambda\, \mathbb{E}[X] \;+\; \frac{\lambda^2 M^2}{8}.$$

Theorem 3.6 (Hedge regret bound). *With a fixed $\eta > 0$ and losses $\ell_{t,i} \in [0, M]$, Hedge satisfies*

$$\sum_{t=1}^{T} \langle w_t, \ell_t \rangle \;-\; \min_i \sum_{t=1}^{T} \ell_{t,i} \;\le\; \frac{\ln N}{\eta} \;+\; \frac{\eta\, M^2 T}{8}.$$

With $\eta = \sqrt{8 \ln N / (M^2 T)}$ the bound becomes $M\sqrt{T \ln N / 2}$.

Note that the optimal η depends on the horizon T, which in an online setting is not known in advance. There are two established remedies. The first is the *doubling trick*: the algorithm runs with a fixed η tuned for horizon 2^j, and each time the elapsed time exceeds 2^j it restarts with $j+1$. The total regret is then the sum of a geometric series, hence of the same order with a multiplicative overhead of $(\sqrt{2}-1)^{-1} \approx 3.41$. The second, used here, is the decaying schedule $\eta_t \propto 1/\sqrt{t}$, which attains the same order without restarts.

Proposition 3.7 (Decaying schedule). *With $\eta_t = c/\sqrt{t}$ the second term of the bound becomes $\frac{cM^2}{8} \sum_{t=1}^{T} t^{-1/2} \le \frac{cM^2}{4}\sqrt{T}$, while the first becomes $\frac{\ln N}{c}\sqrt{T}$. The total bound remains $O(M\sqrt{T \ln N})$ for every T simultaneously, without the horizon being known.*

This choice has consequences examined empirically in §5.1, where the constant c dictated by theory is found to be, for these data, seriously wrong, too large for Hedge, and too small by two orders of magnitude for OGD.

3.3.2 Online Gradient Descent

OGD (Zinkevich, 2003) [45] takes a step in the direction opposite to the gradient and projects back onto the feasible set:

$$w_{t+1} \;=\; \Pi_{\Delta_N}\big(w_t - \eta_t \nabla f_t(w_t)\big),$$

where Π_{Δ_N} is the Euclidean projection onto the simplex, computable in $O(N \log N)$ (Duchi et al., 2008) [10]. For convex functions with bounded gradient the regret is $O(\sqrt{T})$.

Two ingredients of this scheme are used throughout the thesis and deserve explicit justification: the linearisation of the loss and the computation of the projection.

Why linearisation is permitted. The implementation does not use the gradient of the true loss $-\log\langle w, p_t\rangle$ but the vector of per-expert losses ℓ_t, that is the gradient of the *linear* function $\tilde{f}_t(w) = \langle w, \ell_t \rangle$. This substitution is not an approximation to be checked after the fact; it is safe by construction for any convex loss.

Lemma 3.8 (Linearisation does not loosen the bound). *Let f_t be convex and $g_t \in \partial f_t(x_t)$ any subgradient at the played point. Then for every $u \in \mathcal{K}$*

$$\sum_{t=1}^{T} f_t(x_t) - \sum_{t=1}^{T} f_t(u) \;\le\; \sum_{t=1}^{T} \langle g_t,\, x_t - u \rangle,$$

that is, the regret with respect to the true losses is bounded by the regret with respect to their linearisations. Any bound proved for linear losses therefore holds automatically for the original convex ones.

The direction of the inequality is the one we need: the linear problem is *harder*, so solving it well implies that the original was solved well too. The price is that linearisation discards the curvature, and with it the opportunity for the far stronger $O(\log T)$ bound permitted by the exp-concavity of the logarithmic loss (§3.3.4). This is a deliberate trade-off and not an oversight: §5.1 shows that the algorithm which does exploit the curvature performs worse on these data, for reasons that concern not the quality of the bound but the static comparator it refers to.

The projection onto the simplex. The projection is the only non-trivial operation in the step and it is executed tens of thousands of times, so its closed form matters.

Theorem 3.9 (Euclidean projection onto the simplex). *Let $v \in \mathbb{R}^n$ and $\Pi_{\Delta_n}(v) = \arg\min_{w \in \Delta_n} \|w - v\|_2^2$. Then*

$$\Pi_{\Delta_n}(v)_i \;=\; \max\{v_i - \theta,\, 0\},$$

where θ is the unique number satisfying $\sum_i \max\{v_i - \theta, 0\} = 1$. If u denotes the vector v with its entries sorted in decreasing order and $\rho = \max\{j : u_j - \frac{1}{j}(\sum_{r \le j} u_r - 1) > 0\}$, then $\theta = \frac{1}{\rho}\big(\sum_{r \le \rho} u_r - 1\big)$.

Remark 3.10 (The projection is not passive). The projection is not merely a feasibility mechanism. Because it zeroes every coordinate below the threshold θ, it is *sparsifying*: it imposes exactly zero weight on experts, not merely small weight. Under the Sleeping Experts reduction of §3.3.5 it is reapplied every time the awake set changes, and the empirical investigation of §5.3.2 shows that weight movement in the late stages of the period is due mainly to it and not to the gradient step: a finding that overturns the intuitive expectation that a decaying η_t "freezes" the algorithm.

Theorem 3.11 (OGD regret bound). *Let $\mathcal{K}$ be convex with diameter $D = \max_{u,v \in \mathcal{K}} \|u - v\|$ and bounded gradients, $\|g_t\| \le G$. With a fixed η,*

$$\mathrm{Regret}_T \;\le\; \frac{D^2}{2\eta} \;+\; \frac{\eta G^2 T}{2},$$

and with $\eta = D/(G\sqrt{T})$ it follows that $\mathrm{Regret}_T \le DG\sqrt{T}$.

In our case $\mathcal{K} = \Delta_N$, so $D = \sqrt{2}$, and G is bounded by the M of §5.1. This structure (that η multiplies the gradient of *one* round) explains why OGD responds to step scaling in the opposite way to Hedge.

Remark 3.12 (The structure of η and the direction of tuning). The reason Hedge and OGD respond in *opposite* directions to scaling of the step is structural and visible in the formulas. In Hedge, η sits in an *exponent* over a single round's loss: a smaller η means slower exponential decay, hence *longer memory*. In OGD, η *multiplies* a single round's gradient: a larger η simply means a larger step in the same direction. The two parameters, despite sharing a name, do not control the same quantity.

3.3.3 Follow-The-Regularized-Leader

FTRL (Shalev-Shwartz, 2012; Hazan, 2016) [37, 23] selects at each round the point minimising the entire accumulated loss, plus a regularisation term R preventing excessively abrupt movement:

$$w_{t+1} \;=\; \arg\min_{w\in\Delta_N} \Big\{ \eta_t \sum_{s=1}^{t} \langle \ell_s, w\rangle + R(w) \Big\}.$$

With $R(w) = \frac{1}{2}\|w - w_0\|^2$, where w_0 is the uniform starting vector, the solution is again given by projection onto the simplex. The difference from OGD is that FTRL is *"lazy"*: it rebuilds the weights from the whole history each time, instead of stepping from the previous position.

Why the regularisation term is needed. The obvious strategy without R (pick whatever was optimal so far, Follow-The-Leader) fails spectacularly. With two experts and losses alternating as $(\frac{1}{2}, 0), (0, 1), (1, 0), (0, 1), \ldots$, FTL picks at every round exactly the expert about to lose the next round and incurs linear regret, whereas the best fixed expert loses about $T/2$. The term R is what stabilises successive choices; the lemma below shows that stability is precisely what is required.

Lemma 3.13 (Be-The-Leader). *Let $w_t^\star = \arg\min_w\{\sum_{s=1}^{t}\langle \ell_s, w\rangle + R(w)/\eta\}$ be the FTRL choice. If the algorithm were allowed to play $w_t^\star$* after *seeing ℓ_t, then for every u*

$$\sum_{t=1}^{T} \langle \ell_t, w_t^\star\rangle \;\le\; \sum_{t=1}^{T} \langle \ell_t, u\rangle + \frac{R(u) - R(w_0^\star)}{\eta}.$$

Theorem 3.14 (FTRL regret bound). *With R strongly convex with respect to $\|\cdot\|_2$ with constant 1, losses satisfying $\|\ell_t\| \le G$, and R bounded on $\mathcal{K}$ by $D_R = \max_u R(u) - \min_u R(u)$, FTRL satisfies*

$$\mathrm{Regret}_T \;\le\; \frac{D_R}{\eta} \;+\; \eta\, G^2 T,$$

and with $\eta = \sqrt{D_R/(G^2T)}$ it follows that $\mathrm{Regret}_T \le 2G\sqrt{D_R\, T}$.

Remark 3.15 (OGD and FTRL as "greedy" and "lazy" projection). The two algorithms are not unrelated: with quadratic R, FTRL can be written as $w_{t+1} = \Pi_{\mathcal{K}}(w_0 - \eta \sum_{s\le t} \ell_s)$, that is a projection of the accumulated sum, whereas OGD is $w_{t+1} = \Pi_{\mathcal{K}}(w_t - \eta\ell_t)$, a projection after every step. The difference is *where* the projection intervenes: FTRL keeps the unprojected sum and projects at the end ("lazy"), OGD projects and continues from the projected point ("greedy"). Without constraints the two coincide; with constraints they

diverge, and precisely this divergence explains why they react in opposite ways to step scaling: in FTRL, η multiplies the entire accumulated history, so increasing it drives the algorithm towards unstable Follow-The-Leader, whereas in OGD it multiplies the gradient of a single round.

3.3.4 Exp-concavity and the Aggregating Algorithm

The bounds of Theorems 3.6, 3.11 and 3.14 grow as $O(\sqrt{T})$ and hold for any convex loss. For certain losses a substantially stronger result is attainable.

Definition 3.3 (Exp-concavity). A function f is called α-exp-concave if $e^{-\alpha f}$ is concave.

The logarithmic loss of the mixture is 1-exp-concave: if $f(w) = -\log\langle w, p\rangle$, then $e^{-f(w)} = \langle w, p\rangle$, which is *linear* in w and hence concave. This is a stronger property than convexity and a weaker one than strong convexity; its importance is that it permits a logarithmic, or even constant, bound instead of $O(\sqrt{T})$.

The Aggregating Algorithm. Vovk's (1990) [43] algorithm is not a variant of Hedge but a different construction. It maintains weights updated *Bayesianly* with the likelihood as the weight,

$$w_{t+1,i} \;\propto\; w_{t,i}\, e^{-\ell_{t,i}} \;=\; w_{t,i}\, p_{t,i}[y_t],$$

that is exactly Bayes' rule with a uniform prior, and predicts $\hat{p}_t = \sum_i w_{t,i}\, p_{t,i}$. For the logarithmic loss this scheme is identical to Hedge at $\eta = 1$, which is why the investigation of §5.1 examines it as the special case "constant $\eta = 1$".

Theorem 3.16 (Aggregating Algorithm bound). *For an α-exp-concave loss and N experts, the Aggregating Algorithm satisfies*

$$\mathrm{Regret}_T \;\le\; \frac{\ln N}{\alpha},$$

that is a bound independent of the horizon T.

For $N = 26$ experts this means regret at most $\ln 26 \approx 3.26$ *in total*, against hundreds for the $O(\sqrt{T})$ bounds at $T \approx 76{,}000$. Note that the result does not contradict the lower bound of Theorem 3.1: that one concerns *linear* losses, which are not exp-concave, and the difference between the two cases is exactly the curvature that the linearisation of Lemma 3.8 throws away.

This result is theoretically far stronger. It is therefore worth stating explicitly that on our data the corresponding algorithm performs *worse than the uniform average*. The explanation is not that the bound is wrong but that it concerns a *static* comparator: it guarantees that we will not do much worse than the best *fixed* expert. When the best expert stops participating in the middle of the period, the guarantee remains formally true but practically vacuous, and an algorithm that rapidly concentrates its mass on a few experts is penalised for precisely that.

3.3.5 The Sleeping Experts reduction

The algorithms above assume that every expert participates in every round. In our data this does not hold: many bookmakers cover only a small fraction of the matches. We use the classical *Sleeping Experts* or *Specialists* reduction (Freund, Schapire, Singer & Warmuth, 1997; Blum & Mansour, 2007) [13, 3]. Let $A(t) \subseteq \{1, \dots, N\}$ be the set of "awake" experts at round t. Then:

- each expert maintains its own persistent state (a weight for Hedge, a position for OGD, an accumulated loss for FTRL);
- the prediction is formed only from the awake ones, with their weights renormalised so as to constitute a distribution over $A(t)$ and nothing else;
- after the round only the state of the awake experts is updated; the sleeping ones remain *frozen* exactly as they were;
- a round in which nobody is awake is skipped entirely.

The reduction gives each expert i a regret guarantee measured *only over the rounds in which it was awake*, instead of requiring participation in all of them.

Theorem 3.17 (Per-expert regret under partial participation). *Let $T_i = \{t : i \in A(t)\}$ be the rounds in which expert i is awake and $n_i = |T_i|$. Hedge with the above reduction satisfies, for every i simultaneously,*

$$\sum_{t \in T_i} \Big(\langle \bar{w}_t, \ell_t \rangle - \ell_{t,i} \Big) \;\le\; \frac{\ln N}{\eta} \;+\; \frac{\eta M^2 n_i}{8},$$

where $\bar{w}_t$ are the weights renormalised over $A(t)$. With η tuned to n_i the bound becomes $M\sqrt{n_i \ln N/2}$.

The substantive consequence is that a bookmaker covering 7% of the matches does not "dilute" the guarantee of the rest: each expert is judged on its own domain. This is exactly what a panel with coverage ranging from 6.9% to 99.8% requires.

Remark 3.18 (A temptation the bound creates and the data reject). Theorem 3.17 has n_i where the original had T, so it suggests of its own accord that each expert's step should decay as $1/\sqrt{n_i}$ (on its own clock) rather than as $1/\sqrt{t}$ on the global one. The suggestion is theoretically sound and was tried; it was rejected empirically (§5.6). The reason is instructive: an expert with 7% coverage acquires a step roughly 14 times larger than one with full coverage at the same point in time, so the mixture chases the rare experts. The bound is not wrong; it is simply that the quantity it guarantees is not the one we are interested in minimising.

Two details prove critical in the implementation. First, the persistent weight vector must never be renormalised globally, only the copy used for the prediction. The reason lies in how Theorem 3.17 is established: the lower bound relies on the weight $w_{t,i}$ of a sleeping expert remaining *literally* unchanged, and a global renormalisation changes it. Second, the last row of the weight matrix is not "the final weight of each expert", since it is zero for anyone who did not participate in the last round; the interpretable quantity is the weight each one held the last time it was awake.

3.3.6 Fixed-Share

The Sleeping Experts reduction creates a side problem: an expert that sleeps for a long stretch keeps its weight frozen, and that weight is not eroded by the losses of the others. Rare experts can thus end up with disproportionate influence. The Fixed-Share algorithm (Herbster & Warmuth, 1998) [24] introduces a small leak α towards the uniform distribution after every update:

$$w_{t+1,i} \;=\; (1-\alpha)\,\tilde{w}_{t,i} \;+\; \frac{\alpha}{N}\sum_{j}\tilde{w}_{t,j}, \qquad \tilde{w}_{t,i} \;=\; w_{t,i}\,e^{-\eta\ell_{t,i}},$$

where the second form preserves the total mass exactly. Theoretically this changes the comparator from the best fixed vector to the best *sequence* of vectors with a bounded number of switches, that is to the tracking regret of Definition 3.2.

Theorem 3.19 (Fixed-Share tracking regret). *Let the comparison sequence select one expert at each round and switch at most m times. Fixed-Share with suitable η and $\alpha = m/(T-1)$ satisfies*

$$\mathrm{Regret}_T^{d} \;\le\; \frac{1}{\eta}\Big[\ln N + m\ln(N-1) + (T-1)\,\mathcal{H}(\alpha)\Big] + \frac{\eta M^2 T}{8},$$

where $\mathcal{H}(\alpha) = -\alpha\ln\alpha - (1-\alpha)\ln(1-\alpha)$ is the binary entropy. Optimising, the bound is $O\big(M\sqrt{T\,(m\ln N + m\ln\frac{T}{m})}\big)$.

The bound quantifies a trade-off: for $m = 0$ the term $(T-1)\mathcal{H}(\alpha)$ is pure overhead relative to plain Hedge, while for $m > 0$ it is the price that buys a guarantee Hedge does not offer at all. The empirical result of §5.6 is exactly what the trade-off predicts when the underlying sequence does not switch enough: every $\alpha > 0$ worsens the log-loss. More interestingly, the optimal value of α turns out to be 0 *once* the step has been tuned, because the two quantities control the same thing (how fast the algorithm forgets) and the step is the more efficient of the two instruments.

3.3.7 Markov chains: three different uses

The term "Markov chain OGD" is used in the literature for at least three different things, and this thesis tries all three. We separate them explicitly, because the negative results of one say nothing about the others.

(a) A chain on the weight vector. This generalises Fixed-Share along the lines of the mixing schemes of Bousquet & Warmuth (2002) [4]. After the gradient step and the projection, mass is moved by a row-stochastic matrix Q, that is a non-negative matrix whose every row sums to one (Q rather than M, which already denotes the bound on a single round's loss):

$$v_t = \Pi_\Delta(w_t), \qquad \hat{w} = v_t - \eta_t\ell_t, \qquad w_{t+1} = Q^\top\hat{w}.$$

The Fixed-Share of §3.3.6 is the special case $Q = (1-\alpha)I + \frac{\alpha}{N}\mathbf{1}\mathbf{1}^\top$, that is a uniform leak. A general Q is strictly more expressive, because it can say *where* the lost mass goes: e.g. to the opening/closing "twin" of the same house, or to peers in the same market phase. Row-stochasticity guarantees conservation of total mass, which is exactly what makes $\alpha = 0$ leave the weights exactly, not approximately, unchanged.

(b) A chain on a latent state (hidden Markov model). A hidden Markov model (Rabiner, 1989) [36] describes a system that moves between unobserved states according to a Markov chain, while only the effects of those states are observed. Here the hidden variable is not the weights but a state $k \in \{1, \dots, K\}$, and each state carries its own weight vector. The belief b_t over the states is updated by the *forward recursion*:

$$b_{t+1}(k) \;\propto\; b_t(k)\, L_t(k), \qquad L_t(k) = \langle w_t^{(k)}, p_t[y_t] \rangle,$$

followed by the transition $b \;\leftarrow\; (1-\tau)\, b + \tau/K$. The prediction is the mixture $\sum_k b_t(k)\, \langle w_t^{(k)}, p_t \rangle$ and each state is updated scaled by its *responsibility* $b_t(k)$.

Remark 3.20 (Why symmetry is a trap). If all states start close to one another and receive the *same* loss vector, they follow identical trajectories; then the $L_t(k)$ are equal for every k, the belief stays uniform, and the scheme degenerates into an OGD with step divided by K. The symmetric solution is a fixed point of the dynamics. An explicit differentiation mechanism is needed, sharpening of the belief so that it reacts to small per-round differences, forgetting so that it tracks the present instead of the sum of history, and a per-state clock so that none of them under-steps. §5.5.2 shows that without these the model fails in exactly this way.

(c) Markov Chain Gradient Descent: dependent sampling. This is what Sun, Sun & Yin (2018) [40] study, and it has nothing to do with weight transitions. The classical analysis of *stochastic gradient descent* (SGD), which updates on one randomly drawn sample at a time, assumes independent samples; when the samples are consecutive points of a Markov trajectory, after t rounds the algorithm has seen *less* than t independent units of information. If the dependence horizon is τ rounds, the effective number of samples is $\sim t/\tau$ and the schedule becomes

$$\eta_t \;=\; \frac{D}{M\sqrt{t/\tau + 1}} \;\approx\; \sqrt{\tau}\; \eta_t^{\text{iid}},$$

where η_t^{iid} is the schedule that would apply if the samples were independent and identically distributed (iid); that is, a dependent sequence *justifies* a step inflated by $\sqrt{\tau}$.

Remark 3.21 (An independent theoretical explanation of OGD's step multiplier). The empirical tuning of §5.1 finds that OGD performs best with a step 100 times larger than the textbook schedule (a multiplier of $\times 100$). The argument above predicts part of that inflation from theory alone, and it needs two numbers. The first is the dependence horizon τ. For it we take the block length of the moving block bootstrap (§3.4.2), which estimates the same quantity for an entirely different purpose: $b \approx n^{1/3}$, and with $n = 76{,}584$ rounds this gives $b \approx 42$, so the inflation is $\sqrt{42} \approx 6.5$. The second is the loss bound M in the schedule. Theory sets $M = \ln(1/\varepsilon) \approx 13.82$, the largest loss a probability clipped at $\varepsilon = 10^{-6}$ can produce (§4.2.4), whereas the largest loss actually observed in the data is 3.76; since the step is inversely proportional to M, using the observed bound inflates it by a further $13.82/3.76 \approx 3.7$. Together the two give $6.5 \times 3.7 \approx 24$. This does not reach the empirical $\times 100$, but it is a partial explanation obtained *without any grid search*, and for that reason worth reporting, even when the algorithm itself improves nothing.

3.3.8 Bandit online convex optimisation

Everything above assumes *full information*: after round t the algorithm learns the entire vector ℓ_t, that is how well *every* expert did, not only itself. In our application this is feasible because we score all 26 bookmakers after the fact. It is not, however, to be taken for granted. A forecaster that records only its own performance (and this is the usual situation in deployment) observes a single number per round:

$$f_t(w_t) \;=\; -\log\langle w_t, p_t[y_t]\rangle.$$

This model is called *bandit*, and it defines a genuinely harder problem. The difficulty is specific: to move in the direction that decreases a function one needs its gradient, and a single scalar value does not determine it. The solution of Flaxman, Kalai & McMahan (2005) [14] is to *construct* a stochastic estimate of the gradient from function values alone.

Lemma 3.22 (Gradient estimation from a single value). *Let $f : \mathbb{R}^d \to \mathbb{R}$ and $\hat f_\delta(x) = \mathbb{E}_{v\sim\mathbb{B}^d}[f(x+\delta v)]$ be its smoothing with v uniform on the unit* ball. *If u is distributed uniformly on the unit* sphere $\mathbb{S}^{d-1}$*, then*

$$\mathbb{E}_u\Big[\frac{d}{\delta}\, f(x+\delta u)\, u\Big] \;=\; \nabla \hat f_\delta(x).$$

The estimate is therefore *unbiased* for the gradient of the smoothed function, not of the original one, and the smoothing introduces an $O(\delta)$ error. At the same time its variance scales as $O(1/\delta^2)$: small δ gives small bias and enormous noise, large δ the converse. Optimising the trade-off gives the central result.

Theorem 3.23 (Flaxman, Kalai & McMahan, 2005). *For convex losses bounded in absolute value, Bandit Gradient Descent with $\delta \sim T^{-1/4}$ and suitable η achieves*

$$\mathbb{E}\big[\mathrm{Regret}_T\big] \;=\; O\big(T^{3/4}\big).$$

The comparison with the $O(\sqrt{T}) = O(T^{1/2})$ of full information is the quantitative price of ignorance. The bound improves to $O(\sqrt{T})$ if *two* queries to the same function are permitted per round (Agarwal, Dekel & Xiao, 2010) [1], with the estimate

$$\hat g_t \;=\; \frac{d}{2\delta}\big(f_t(x_t+\delta u) - f_t(x_t-\delta u)\big)\, u,$$

which is a finite difference along a random direction. The reason for the improvement is immediately visible: the subtraction cancels the constant term $f_t(x_t)$, which in the one-point estimate is multiplied by d/δ and dominates the variance. As $\delta \to 0$ the two-point estimate converges to the exact directional derivative, that is, the two-query model is not really bandit for smooth functions: an observation with a direct consequence for the measured results.

The geometry of the simplex. Applying this to our $\mathcal{K} = \Delta_N$ requires three adaptations that are not decorative:

- the simplex is $(N-1)$-dimensional (the constraint $\sum_i w_i = 1$ removes one of the N degrees of freedom), so the perturbation must lie in the *tangent space* $\{u : \sum_i u_i = 0\}$ and the factor in Lemma 3.22 is $d = N-1$, not N. A perturbation along a random direction of $\mathbb{R}^N$ would take the point off the plane $\sum_i w_i = 1$;

- the point about which we perturb must stay away from the boundary, otherwise the perturbation is infeasible. We use the *shrunk* simplex $\Delta^{\xi} = \{w \in \Delta_N : w_i \geq \xi/N\}$, where $\xi \in (0, 1)$ is a small shrinkage parameter that fixes the minimum weight, and the choice $\delta = \xi/N$ guarantees feasibility for free: since $|u_i| \leq 1$, every coordinate of $w \pm \delta u$ remains non-negative, while the sum stays exactly 1;
- the shrinkage introduces its own bias, since the optimum may lie on the boundary, and in our data it typically does, since the projection of Theorem 3.9 zeroes coordinates.

Under the Sleeping Experts reduction all of the above is restricted to $A(t)$, so d and δ vary from round to round with the number of awake experts. A round with a single awake expert has $d = 0$: there is no direction to perturb along and nothing is learned.

What "the loss" of a bandit algorithm measures. A bookkeeping detail that changes the meaning of the numbers. The algorithm publishes the perturbed point $w_t + \delta u_t$, not the centre w_t, so the loss it actually pays includes the cost of exploration; this is also the quantity Theorem 3.23 refers to. The centre w_t is, however, the prediction we would deploy, and it is the only quantity directly comparable with OGD across the whole battery of metrics. Both are reported (§5.5.1): the first is the honest bandit bookkeeping, the second answers the question "what did we actually learn".

3.4 Scoring and statistical testing

3.4.1 Proper scoring rules and calibration

Beyond log-loss we use the Brier score, $\frac{1}{T}\sum_t \|\hat{p}_t - e_{y_t}\|^2$, where e_{y_t} is the vector with 1 in the position of the outcome y_t that occurred and 0 elsewhere, so that the score is the mean squared distance of the forecast from what happened. It is also strictly proper. The general theory (Gneiting & Raftery, 2007) [17] shows that every proper scoring rule corresponds to a convex function G on the simplex via Savage's representation,

$$S(q, y) \;=\; G(q) \;+\; \langle \nabla G(q),\, e_y - q \rangle,$$

where the expected score $\mathbb{E}_{y\sim p}[S(q, y)]$ is minimised at $q = p$ precisely because G is convex. Log-loss arises for G the negative entropy and Brier for $G(q) = \|q\|^2$. The two are therefore not independent choices but two points of the same family: a fact that explains why, as observed empirically, they agree on 18 verdicts out of 18 (the comparisons of each algorithm against the plain average on every panel, §5.2.2 and Tables A.6–A.7), whereas accuracy, which does not belong to the family, reproduces none of them.

Murphy's decomposition. If predictions are grouped into classes sharing a common value, every proper rule decomposes into three terms:

$$\underbrace{\overline{S}}_{\text{score}} \;=\; \underbrace{\mathbb{E}\big[d(\hat{p}, \bar{y}_{\hat{p}})\big]}_{\text{unreliability}} \;-\; \underbrace{\mathbb{E}\big[d(\bar{y}_{\hat{p}}, \bar{y})\big]}_{\text{resolution}} \;+\; \underbrace{d(\bar{y}, \cdot)}_{\text{uncertainty}},$$

where $\bar{y}_{\hat{p}}$ is the actual frequency in the class with prediction $\hat{p}$, $\bar{y}$ the overall base rate and d the divergence induced by G. The third term is a property of the data and nobody

can improve it. The second requires genuine information. The first, however (unreliability) is correctable *without any additional information*, by monotonically rearranging the probabilities already in hand. This is exactly where the calibration stage intervenes, and Proposition 3.3 has already predicted that the aggregation operator systematically inflates precisely that term.

Temperature scaling. The correction we use raises the probabilities to an exponent $s = 1/T_{\text{temp}}$ and renormalises,

$$\tilde{p}_k \;=\; \frac{\hat{p}_k^{\,s}}{\sum_j \hat{p}_j^{\,s}}.$$

Values $s < 1$ "soften" the distribution, values $s > 1$ sharpen it. This is a one-parameter family which preserves the ranking of the outcomes and therefore cannot change any argmax decision: a property that explains why calibration improves log-loss substantially while leaving accuracy unchanged.

Proposition 3.24 (Convexity in the exponent). *Let $a_k = \log \hat{p}_k$ and let y be the realised outcome. The loss*

$$\phi(s) \;=\; -\log \tilde{p}_y \;=\; -s\, a_y \;+\; \log \sum_j e^{s\, a_j}$$

is convex in s, with $\phi'(s) = \mathbb{E}_{k\sim\tilde{p}^{(s)}}[a_k] - a_y$ and $\phi''(s) = \mathrm{Var}_{k\sim\tilde{p}^{(s)}}(a_k) \geq 0$.

Convexity is what makes s learnable by projected OGD, that is a *second OCO stage* on top of the output of the first, which remains fully causal: the temperature at round t depends only on outcomes up to round $t-1$. The form of the derivative has a further direct interpretation: the step is positive when the realised outcome had a log-probability below the average, i.e. the algorithm sharpens the distribution when it is surprised and softens it when it is not.

3.4.2 Statistical significance: the moving block bootstrap

Comparing two forecasters requires a test on the difference of their per-round losses. A simple t-test would be inappropriate, since consecutive losses are not independent: matches of the same round and the same season share difficulty. The consequence of ignoring this is specific and one-sided: for a positively correlated sequence,

$$\mathrm{Var}\Big(\tfrac{1}{n}\sum_t d_t\Big) \;=\; \frac{\sigma^2}{n}\Big(1 + 2\sum_{k\geq 1}\big(1 - \tfrac{k}{n}\big)\rho_k\Big) \;>\; \frac{\sigma^2}{n},$$

where d_t is the difference between the two forecasters' losses at round t, σ^2 its variance and ρ_k the correlation between differences k rounds apart. Since the ρ_k are positive, the independence formula *underestimates* the variance, confidence intervals come out far too narrow and results look significant when they are not.

We use the *moving block bootstrap* (Künsch, 1989) [28], which resamples with replacement *blocks* of consecutive observations of length b, thereby preserving the local correlation: each block carries its internal dependence structure with it, and only correlations at lags greater than b are destroyed. The 95% confidence interval is obtained from the quantiles of the bootstrap distribution.

Remark 3.25 (Why $b \approx n^{1/3}$). The choice of length is a bias–variance trade-off. Small b destroys dependence that is really there, leaving an $O(1/b)$ bias in the estimate of the large-sample variance; large b leaves few effectively independent blocks, namely n/b, and raises the variance of the estimate itself by $O(b/n)$. Minimising the mean squared error $O(b^{-2}) + O(b/n)$ gives $b \propto n^{1/3}$ (Hall, Horowitz & Jing, 1995) [19]. For $n \approx 76{,}000$ this gives $b \approx 42$, about one and a half match rounds: an order of magnitude consistent with the natural dependence horizon of the data. The same quantity reappears independently as a dependence horizon in the step-size study, selected there for an entirely different reason.

Sign convention. We define $d_t = \ell_t(A) - \ell_t(B)$, so a **negative** mean difference means the first is *better*. For log-loss, lower is better; for the logarithmic growth rate of capital the opposite holds.

3.5 Bet sizing

3.5.1 The Kelly criterion

Given an estimated probability p and decimal odds o, the fraction of capital maximising the expected logarithmic growth rate is the one of Kelly (1956) [27],

$$f^* \;=\; \frac{p\,(o-1) - (1-p)}{o-1} \;=\; p - \frac{1-p}{o-1}.$$

The criterion is not an arbitrary preference for the logarithm. Breiman (1961) [5] showed that the strategy maximising the expected logarithmic rate is *asymptotically optimal* in two strong senses: it maximises the growth rate of capital almost surely, and it minimises the expected time for the capital to reach any prescribed large level. Logarithmic utility therefore follows from the repeated structure of the problem, not from an assumption about the bettor's preferences.

Why fractional Kelly. Staking the whole fraction f^* is called *full Kelly*; staking a fixed share of it, λf^* with $0 < \lambda < 1$, is called *fractional Kelly*. Full Kelly is notorious for its volatility and for its sensitivity to errors in estimating p. The trade-off is quantified by the second-order expansion of g around zero. Writing $b = o - 1$ and $e = pb - (1-p)$ for the expected return per unit,

$$g(f) \;\approx\; e\,f \;-\; \tfrac{1}{2}\sigma^2 f^2, \qquad \sigma^2 = \mathbb{E}[b^2] \text{ approximately,}$$

a concave parabola with maximum at $f^* = e/\sigma^2$ and maximal value $g(f^*) = e^2/(2\sigma^2)$.

Proposition 3.26 (The cost of fractional Kelly). *In the quadratic approximation, betting a fraction λf^*, with the* Kelly multiplier *$\lambda \in (0, 1]$, retains a proportion $\lambda(2-\lambda)$ of the maximal growth rate, while the variance of the capital is reduced by a factor λ^2. (The symbol λ is used here and for the rest of the thesis in this sense only; it is unrelated to Kyle's depth coefficient of §3.2.2, which is written with the same letter by convention and appears nowhere else.)*

For $\lambda = \frac{1}{4}$, the value used here, one retains $\frac{1}{4} \cdot \frac{7}{4} = \frac{7}{16} \approx 44\%$ of the growth rate with $\frac{1}{16}$ of the variance: a very favourable trade. The argument is stronger than it looks, because p here is an *estimate*: if the true probability is lower than the estimated one, full Kelly overbets, and the damage from overshooting is asymmetrically larger than the gain from the corresponding undershoot, since g falls to $-\infty$ as $f \to 1$.

Betting when there are three possible outcomes. With three mutually exclusive outcomes the exact solution requires simultaneous optimisation of the vector of stakes. The implementation simplifies: at each round the outcome with the largest positive expected value is selected and only that one is backed, using the two-outcome formula. The simplification is conservative (a portfolio of simultaneous bets on mutually exclusive outcomes can only increase the rate, since it reduces the variance), and it matches how one bets in practice. We bet only when $p\,o > 1$, which requires our estimate to exceed the market's implied probability *after* the margin: the threshold is not "we are better than the bookmaker" but "we are better by more than its margin", and the difference between the two is exactly why improved log-loss does not imply profitability.

3.5.2 Stake size as a third OCO stage

The coefficient $\frac{1}{4}$ of Proposition 3.26 is a rule of thumb. Its value does not come from the data, and there is no reason for it to be the same against a bookmaker that prices tightly and one that does not. It is therefore a natural candidate to be learned online, as the weights and the temperature already were.

What exactly is learned, and why it matters. The obvious formulation (learn the fraction f of capital directly) is wrong, and the reason is the *comparator*. A regret bound for such a learner is a promise against the best *constant* f in hindsight. But the closed-form rule does not even belong to that class: it produces a different f every round, computed from $\hat{p}_t$ and the odds, both of which are known *before* the bet is placed. Even zero regret against the best constant stake does not reach a rule that varies from round to round. Moreover such a learner throws away $\hat{p}_t$, exactly the quantity the two preceding stages exist to produce, and relearns it from one bit per bet.

The correct formulation keeps the closed form and learns only the *multiplier*:

$$f_t \;=\; \lambda_t \cdot f_t^*, \qquad f_t^* = f^*(\hat{p}_t, o_t), \qquad \lambda_t \in [0, \lambda_{\max}].$$

The round's information stays inside f_t^* and only the shrinkage is learned. The comparator "best constant λ" now becomes a meaningful object: it is precisely the quantity that the coefficient $\frac{1}{4}$ pins to a constant.

Proposition 3.27 (The loss in the multiplier). *With $w \in \{0, 1\}$ the win indicator, the loss of a single round*

$$\ell_t(\lambda) \;=\; -\log\left(1 - \lambda f_t^* + \lambda f_t^* o_t w\right)$$

is convex in λ, with derivative

$$\ell_t'(\lambda) \;=\; \begin{cases} -\,f_t^*(o_t - 1)/g_t, & w = 1, \\ \;\;f_t^*/g_t, & w = 0, \end{cases} \qquad g_t = 1 - \lambda f_t^* + \lambda f_t^* o_t w.$$

It is moreover 1-exp-concave on any interval where g_t stays bounded away from zero, that is where $\lambda_{\max} \max_t f_t^* < 1$.

The constraint $\lambda_{\max} \max_t f_t^* < 1$ is not a technical detail but a *solvency* condition: above it a single loss wipes out the capital and the loss ceases to be bounded. The admissible interval is therefore determined by the data and not by choice.

Which algorithm. Exp-concavity is the critical point. Projected OGD gives $O(\sqrt{T})$ here, whereas the Online Newton Step of Hazan, Agarwal & Kale (2007) [22] exploits the curvature and gives $O(\log T)$. In one dimension the scheme is unusually short,

$$A_t \;=\; A_{t-1} + \ell_t'(\lambda_t)^2, \qquad \lambda_{t+1} \;=\; \Pi_{[0,\lambda_{\max}]}\Big(\lambda_t - \tfrac{1}{\gamma}\,\frac{\ell_t'(\lambda_t)}{A_t}\Big),$$

where the projection in the norm defined by A_t coincides with simple clipping, since in one dimension that norm is a positive rescaling of the Euclidean one.

Where it starts. The initial value $\lambda_0 = 1$, that is full Kelly, is not a neutral choice but the theoretically correct one: if $\hat{p}$ were exact, full Kelly would be optimal, and any shrinkage below one is a response to *estimation error*. Starting from there, the algorithm is asked to discover from the data how wrong its own estimate is: a quantity no a-priori constant knows.

Why Universal Portfolios does not carry over. The pure Online Portfolio Selection formulation (Cover, 1991) [8] compares against the best constant rebalanced portfolio over a *fixed* set of assets. Here the "assets" are the H/D/A of a different match every round, so a fixed weight vector corresponds to nothing interpretable. The reduction to a scalar multiplier is the way to retain a meaningful comparator in a problem where the set of available bets changes completely from round to round.

3.5.3 A Bayesian version of Kelly and Markov chain Monte Carlo sampling

Both the closed form and the learned multiplier treat $\hat{p}$ as *known*. It is not: it is an estimate, and how uncertain it is varies from match to match: when the awake experts agree the estimate is reliable, and when they disagree it is not. The natural formulation replaces the point maximum with the *posterior expected* one:

$$f^{\mathrm{B}} \;=\; \arg\max_{f}\; \mathbb{E}_{\theta\sim\pi(\cdot\,|\,\mathcal{D})}\big[\, g(f,\theta)\,\big],$$

$$g(f,\theta) \;=\; \theta_k \log\big(1 + f(o_k - 1)\big) \;+\; (1-\theta_k)\log(1-f).$$

Here $\theta = (\theta_H, \theta_D, \theta_A)$ is the unknown true outcome distribution of the match, $\pi(\cdot \mid \mathcal{D})$ is the posterior distribution of θ given the data $\mathcal{D}$ (the experts' forecasts for that match), k is the outcome being backed, and $g(f,\theta)$ is the expected logarithmic growth of capital when a fraction f is staked on it at odds o_k, the quantity the Kelly criterion maximises. Maximising its posterior expectation, rather than its value at a point estimate, is the Bayesian Kelly criterion of Browne & Whitt (1996) [6].

Proposition 3.28 (The Bayesian version is always more conservative). *For every non-degenerate posterior π with mean $\bar{\theta}$ we have $f^{\mathrm{B}} \le f^*(\bar{\theta})$, where f^* is the closed form at the point $\bar{\theta}$. The inequality is strict when π has positive variance.*

The corollary is conceptually attractive: the *shrinkage emerges* from the uncertainty instead of being imposed by an arbitrary constant $\frac{1}{4}$, and its magnitude varies from round to round according to how much the experts agree.

The model. The prior is taken to be $\theta \sim \text{Dirichlet}(1, 1, 1)$, i.e. uniform on Δ_3. Each awake expert i contributes its prediction p_i as evidence, modelled as $p_i \sim \text{Dirichlet}(\kappa\,\theta)$ and weighted by the learned weight w_i. The parameter κ controls how tightly the experts are assumed to cluster around the truth: large κ means a concentrated posterior and returns the method to the closed form, small κ means a wide posterior and strong shrinkage.

Why Markov chain Monte Carlo (MCMC) is needed. The resulting posterior is not conjugate (θ appears inside the Dirichlet parameters of the evidence), so there is no closed form either for it or for the integral $\mathbb{E}_\pi[g(f, \theta)]$. We use random-walk *Metropolis–Hastings* sampling (Hastings, 1970) [21]: from the current state θ a proposal θ' is made and accepted with probability

$$a(\theta \to \theta') \;=\; \min\left\{1,\; \frac{\pi(\theta' \mid \mathcal{D})\, q(\theta \mid \theta')}{\pi(\theta \mid \mathcal{D})\, q(\theta' \mid \theta)}\right\}.$$

With a symmetric proposal ($q(\theta' \mid \theta) = q(\theta \mid \theta')$) the ratio simplifies to the ratio of posterior densities, so the *normalising constant is not needed*, which is exactly the method's reason for being here. This rule satisfies *detailed balance* $\pi(\theta)P(\theta \to \theta') = \pi(\theta')P(\theta' \to \theta)$, so π is the stationary distribution of the chain; with ergodicity, time averages converge to expectations under π.

Remark 3.29 (Parallel chains instead of one long one). The implementation runs N *parallel* chains for a few iterations instead of one long one. The two options are equivalent with respect to the stationary distribution, but not with respect to cost: parallel chains turn a scalar Python loop into vectorised operations of the NumPy library (Harris et al., 2020) [20], which is what makes the per-bet cost manageable over tens of thousands of bets. The price is that no chain has time to forget its initialisation completely, so the approximation rests on the burn-in being sufficient: an assumption not formally checked here and reported as a limitation.

Theory therefore clearly favours this formulation over the fixed $\frac{1}{4}$. §5.6 reports that on these data the measurement does not, which is itself part of the thesis's argument: the theoretical superiority of a formulation guarantees nothing when its assumptions (here, that the Dirichlet model really describes how bookmakers' disagreements are generated) do not hold.

Chapter 4

Problem formulation and experimental design

This chapter states what the data are and how every number in Chapter 5 was produced. The guiding principle throughout is that every quantity reported must have been computed without access to information that would not have been available at the moment of prediction, and must be compared against quantities measured on exactly the same data.

4.1 The data

4.1.1 Source and scope

Table 4.1 summarises the result. The data come from `football-data.co.uk` [15], section "Main Leagues", which publishes for each match the result and the odds of a number of betting companies. The "Extra Leagues" section has a different format and usually does not contain odds, so it is out of scope.

Leagues	22
Seasons	10 (2016/17 – 2025/26)
Matches with $\geq$1 usable price	76,584
Bookmakers as experts	26 (13 opening / 13 closing)
Coverage range per bookmaker	6.9% – 99.8%

Table 4.1: The dataset after preprocessing, that is after the cleaning steps of §4.1.2: aggregate columns removed, rebranded companies merged into one identity, and matches without a single usable price dropped.

The 22 leagues are distributed over England ($\times$5), Scotland ($\times$4), Germany / Italy / Spain / France ($\times$2 each) and the Netherlands / Belgium / Portugal / Turkey / Greece ($\times$1). The combination of top and lower divisions is deliberate: lower divisions attract smaller betting volume and therefore less attention from the companies' analysts, so the heterogeneity among the experts is greater there.

4.1.2 Cleaning

Aggregate columns. Beyond the individual companies, football-data.co.uk also publishes columns with the maximum and the average of the market (`Max`, `Avg`), as well as the equivalents computed by Betbrain, an odds-comparison service, for the older seasons [15]. These are excluded: they are not independent experts but functions of the rest, and their presence would violate the framework's assumption that the experts give separate opinions.

Company rebranding. The columns `VC` (VC Bet) and `BV` (BetVictor) correspond to the same company before and after a rebrand, and their presence does not overlap in time. They are merged into a single continuous identity, so that the expert is not "reborn" halfway through the decade losing its learned history. The merge is inferred from the non-overlapping presence, not from an official source, and is noted as a limitation.

Companies with partial temporal presence. Several columns do not cover the whole decade: some start late and others stop early. None is excluded for that reason: the Sleeping Experts reduction (§3.3.5) exists precisely to handle them, and each expert's guarantee is measured only over the rounds in which it was active. Temporal presence does, however, have a consequence for *coverage thresholds*: a threshold computed over the decade says very little about a company's presence in a narrower window, which is why every restricted-window analysis recomputes its own threshold inside the window it examines (§5.3.1).

4.1.3 Opening and closing odds

For each company the source publishes two prices: the *opening* one, when the market opens, and the *closing* one, shortly before kick-off. The naming convention is that the suffix `C` denotes the closing price of the identically named company: `B365` and `B365C`, `PS` and `PSC`, and so on. The panel of 26 therefore splits cleanly into 13 opening and 13 closing.

The distinction is not a technical detail but a substantive variable. The closing price has absorbed whatever information reached the market in the interim (team-sheet announcements, injuries, moves by large bettors) and is for that reason considered in the literature the most efficient available signal. The distinction also imposes a causality constraint: when we bet into an opening price, no closing price exists yet, not even that of the same company.

4.2 Experimental design

4.2.1 The expert tensor

The basic structure is a tensor $P \in \mathbb{R}^{T \times N \times 3}$ with T rounds (matches in chronological order), N experts and three outcomes, together with a binary matrix $A \in \{0,1\}^{T \times N}$ indicating which expert quoted which match. The chronological ordering is essential: the entire framework presupposes that round t precedes $t+1$.

4.2.2 The three OCO stages

The chain from raw odds to the bet consists of three successive stages, all online and causal.

1. **Weighting the experts**. The three algorithms of Chapter 3 with the Sleeping Experts reduction, plus the uniform average as a baseline. They produce a mixture $\hat{p}_t$ per round.
2. **Calibration**. Online temperature scaling on top of $\hat{p}_t$ (§3.4.1), with the temperature learned by projected OGD.
3. **Stake size**. The Kelly criterion on the calibrated probabilities, with the *multiplier* learned online instead of being pinned to a constant (§3.5.2). The fixed coefficient $\frac{1}{4}$ is retained as a reference point for comparison.

All three stages are the same recipe applied to a different object: convex loss, causal update, projection onto the feasible set. The only difference between them is which quantity is treated as unknown (the weight vector, the temperature, the multiplier) and which loss class each one belongs to, which also determines the algorithm appropriate for each.

The uniform average is not decorative. It is the critical baseline: if an algorithm does not beat it, then learned weighting offers nothing over simply taking the market average, and the complexity is not justified.

4.2.3 Hyperparameter selection protocol

The framework does not require a train/test split, since every prediction is already out of sample. It does, however, require care about whatever is *selected* by looking at the data. Two such quantities exist: the step multiplier of each algorithm and the learning rate of the calibrator. Both are selected by the same protocol:

- a chronological prefix of 75% of each series' rounds as *train*;
- a grid sweep over candidate values, evaluated exclusively on the prefix;
- a single final causal pass over the whole history with the selected value frozen.

Wherever a *configuration choice* is made on the basis of reported numbers (for example the choice between panels) a chronological 25% tail that the choice never saw is additionally reported, so that it is visible whether the choice was fitting noise.

4.2.4 Tuning the step

The theoretically optimal step of all three algorithms contains a bound M on the maximum loss of a single round. Because probabilities are clipped at $\varepsilon = 10^{-6}$ so that the log-loss stays finite, this gives

$$M \;=\; \ln(1/\varepsilon) \;=\; \ln 10^{6} \;=\; 6 \ln 10 \;\approx\; 13.82.$$

This bound presupposes a bookmaker pricing the outcome that actually occurred at one in a million. The largest loss actually observed in the data is 3.76, that is the worst-case step is an order of magnitude more conservative than the data warrant. The multipliers are therefore tuned empirically, with the protocol of §4.2.3 and over the same grid for all three algorithms, so that the comparison between them stays fair.

4.2.5 Comparing series with different coverage

Partial coverage creates a trap that determines a large part of the design. A log-loss computed over one set of matches is not comparable with a log-loss computed over another: the matches differ in difficulty, and a series covering fewer of them may look better simply because it happened to cover the easier ones.

Wherever series with different coverage are compared, therefore:

- **fitting** uses each series' full available support;
- **scoring** is restricted to the intersection of the rounds in which *all* compared series are defined;
- **the paired tests** run on that same set.

This principle has a cost. The more series enter a comparison, the smaller the intersection, and in extreme cases requiring everyone to participate drives the statistical power to zero. Where that happens, the low-coverage series is reported separately, in a one-to-one comparison over its own rounds, rather than being imposed on the common table.

4.2.6 Causality controls in value betting

The betting simulation uses a *leave-one-bookmaker-out* design: the bookmaker we bet against is removed from the panel before fitting, so that the mixture does not "predict" using the very price it is asked to beat. This is not enough. When the target is an opening price, the *whole* closing side of the panel must be removed, because no closing price exists yet at that moment, not even that of the same company. A panel that leaves them in bets knowing where the market will close, and ends up producing final bankrolls of order 10^6–10^9, that is a pure artefact of construction.

The training panel therefore matches the phase of the target: a closing panel for a closing target, an opening panel for an opening target.

4.2.7 Deployment test

Beyond the above, the thesis includes a test that answers a different question from regret: *what would a model trained up to yesterday do today?* The cut is a real date rather than a percentage of the series, fitting is done on the rounds before it, and measurement on those after. The opponents are selected by their coverage *inside the test window*, so that the comparison concerns the market that exists then and not the one that existed ten years ago.

Two readings are reported, because they answer different questions: one in which the mixture keeps learning inside the test window, which is the realistic deployment scenario, and one in which the weights are frozen at the cut, which is the strict reading of "trained once".

Chapter 5

Results

5.1 Tuning the step

The step multipliers are tuned first, since they determine the algorithms with which all the remaining results are produced. The multipliers of all three algorithms needed changing, and (this is the non-obvious part) **in opposite directions**. Table 5.1 lists the multipliers that are in force everywhere else in this chapter.

	Mult.	**train**	**full**	**tail**	**train min.**	**tail min.**
OGD	$\times 100$	0.99739	0.99745	0.99765	$\times 110$	$\times 100$
Hedge	$\times 0.25$	0.99828	0.99842	0.99883	$\times 0.25$	$\times 0.05$
FTRL	$\times 0.05$	0.99828	0.99843	0.99889	$\times 0.05$	$\times 0.005$

Table 5.1: Step multipliers in force. **Mult.** is the factor applied to the textbook step schedule. **train**, **full** and **tail** are the log-loss obtained with it on the chronological 75% prefix, on the whole sample and on the held-out 25% tail. **train min.** and **tail min.** are the multipliers that would minimise the log-loss on the prefix and on the tail respectively. Selection is made exclusively on the chronological prefix, so the decisive column is **train min.**: for all three algorithms it coincides with the multiplier actually used, the one exception being OGD's $\times 110$, which gives a tail identical to $\times 100$ (0.99765). The **tail min.** column is reported for comparison and not as a criterion; the difference between the two is discussed below.

As Figure 5.1 shows, OGD wants a step a hundred times larger than the theoretical one, with a genuine interior optimum: a plateau over the range $\times 75$ to $\times 110$, with $\times 200$, $\times 500$ and $\times 1000$ clearly worse. Hedge and FTRL want a *smaller* step, by factors of 4 and 20 respectively.

A caveat that turned out to be unfounded. The *tail* minimum for Hedge and FTRL lies below the values used, at $\times 0.05$ and $\times 0.005$ respectively, and for a time this thesis carried the caveat that those two algorithms were reported with a sub-optimal step. The caveat was wrong, and the reason is worth stating: **selecting on the tail is selecting on the test set**, which is exactly what the protocol of §4.2.3 exists to prevent. Because the log-loss is a plain mean over rounds, the prefix figure is recoverable exactly from the full sample and

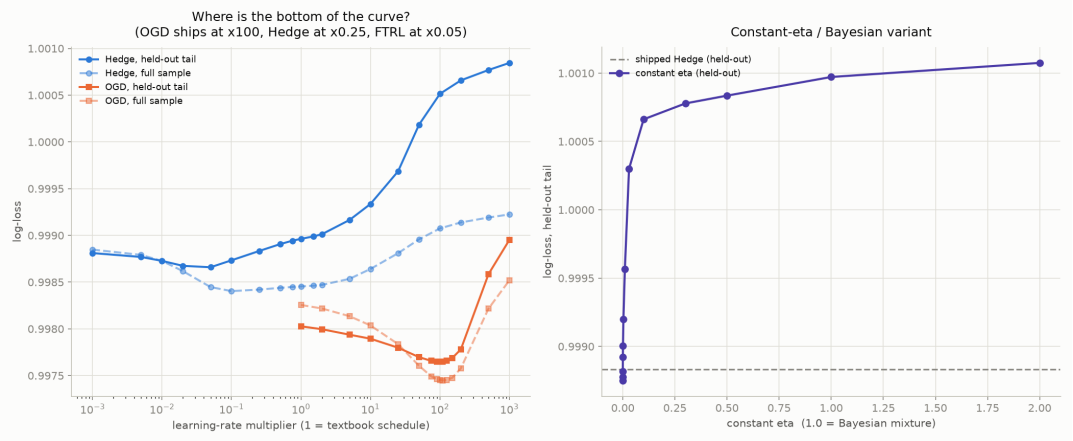


Figure 5.1: Sweep of the step multiplier for all three algorithms. Horizontal axis the step multiplier on a logarithmic scale, with 1 the textbook schedule; vertical axis the resulting log-loss, lower better. One curve per algorithm, and for each a second curve showing the held-out tail. Selection is made on the chronological prefix; the tail is reported as verification. OGD exhibits a genuine interior optimum. The tail curves for Hedge and FTRL keep falling towards the left, but the selection is not made on them; see the remark on the caveat that turned out to be unfounded.

the tail, as $(\text{full} \cdot T - \text{tail} \cdot n_{\text{tail}})/n_{\text{train}}$, and under the protocol the answer reverses: Hedge selects $\times 0.25$ and FTRL $\times 0.05$, that is *exactly* the multipliers in force.

The numbers also show why. For Hedge, $\times 0.05$ gives a train figure of 0.99838 against 0.99828 for $\times 0.25$; for FTRL, $\times 0.005$ gives 0.99862 against 0.99828 for $\times 0.05$. In both cases the ordering *reverses* between prefix and tail, which is the signature of fitting the tail rather than evidence of better tuning. OGD remains optimal at $\times 100$, with $\times 110$ giving an identical tail: a tie inside the plateau already described.

A constant step: the one schedule idea that had not been tested properly. The decaying $1/\sqrt{t}$ is derived for a *static* comparator, whereas the panel here is not static, so a step that never stops moving is the theoretically indicated alternative. It had been tried only for Hedge and only over $[0.1, 2]$, while Hedge's decaying schedule passes $\eta \approx 0.033$ at the first round and $\eta \approx 1.7 \cdot 10^{-4}$ by the middle of the period, i.e. the entire interesting region lay *below* the old grid $[0.1, 2]$. The sweep was extended over five decades and to all three algorithms, with selection on the prefix and a paired test on the tail. Table 5.2 gives the comparison.

	constant η	tail	decaying	p	
Hedge	$3 \cdot 10^{-4}$	0.99892	0.99883	< 0.001	decaying
OGD	0.1	0.99795	0.99765	0.021	decaying
FTRL	10^{-5}	0.99879	0.99889	0.001	**constant**

Table 5.2: Constant against decaying step. η is selected on the chronological prefix and the comparison is made on the tail with a moving block bootstrap.

The result is negative for two of them and **positive for one**. For Hedge and OGD the

constant step loses significantly, by $+0.000089$ and $+0.000299$ respectively. For FTRL, however, it wins significantly, by -0.000096 ($p = 0.001$), and the critical point is that the two schemes have an *identical* score on the prefix (0.99828 for both). With nothing to separate the two on the training data, the selection cannot have favoured the constant step in order to flatter the tail, and the advantage emerges only out of sample. The explanation is the same asymmetry that explains the opposite tuning directions: FTRL's η multiplies the entire accumulated loss, not one round's gradient, so it is the one of the three for which "non-decaying" means something qualitatively different.

The sweep needed extending here too, for the same reason as before: FTRL's minimum initially sat on the grid's lower endpoint. Extended down to 10^{-9} it became genuinely interior, and the extension simultaneously confirmed the theoretical limit: as $\eta \to 0$ FTRL degenerates to the uniform average, and indeed at $\eta = 10^{-9}$ it gives 0.99882, the uniform average's value on the tail.

A methodological error worth reporting. The original sweep had the grid $1, 2, 5, \ldots, 1000$ and concluded that Hedge and FTRL were "already optimal at $\times 1$". But 1 was the *lower endpoint of the grid*, not an optimum, and nobody had looked to the left of it. Extending downwards revealed genuine interior minima, with gains of $+0.00013$ and $+0.00018$ respectively on the tail. The conclusion generalises: when a sweep's answer sits at the endpoint of the grid, the grid must be extended before the answer is believed.

The opposite direction has a structural explanation. OGD's η multiplies the gradient of *one* round, so increasing it means moving further. Hedge's η sits in an exponent over one round's loss, so decreasing it *lengthens the memory*. FTRL's η multiplies the entire accumulated loss, so increasing it collapses it towards follow-the-leader.

5.2 Ranking and the choice of panel

5.2.1 Ranking on the full panel

		phase	log-loss (raw / cal.)	Brier (raw / cal.)
PSC (Pinnacle)	bookmaker	closing	0.99714 / 0.99685	0.59592 / 0.59577
OGD	algorithm	—	0.99745 / 0.99693	0.59612 / 0.59586
Hedge	algorithm	—	0.99842 / 0.99773	0.59674 / 0.59637
FTRL	algorithm	—	0.99843 / 0.99774	0.59674 / 0.59638
Uniform	baseline	—	0.99886 / 0.99813	0.59703 / 0.59665
B365C	bookmaker	closing	0.99953 / 0.99910	0.59732 / 0.59711
BWC	bookmaker	closing	0.99998 / 0.99919	0.59769 / 0.59728
VC_BV	bookmaker	opening	1.00075 / 1.00026	0.59833 / 0.59806
WH	bookmaker	opening	1.00119 / 1.00048	0.59860 / 0.59818
PS (Pinnacle)	bookmaker	opening	1.00083 / 1.00055	0.59838 / 0.59824
IW	bookmaker	opening	1.00186 / 1.00061	0.59899 / 0.59826
BW	bookmaker	opening	1.00140 / 1.00065	0.59873 / 0.59833
B365	bookmaker	opening	1.00113 / 1.00073	0.59852 / 0.59832

Table 5.3: Ranking by calibrated log-loss, bookmakers with coverage $\geq$60%.

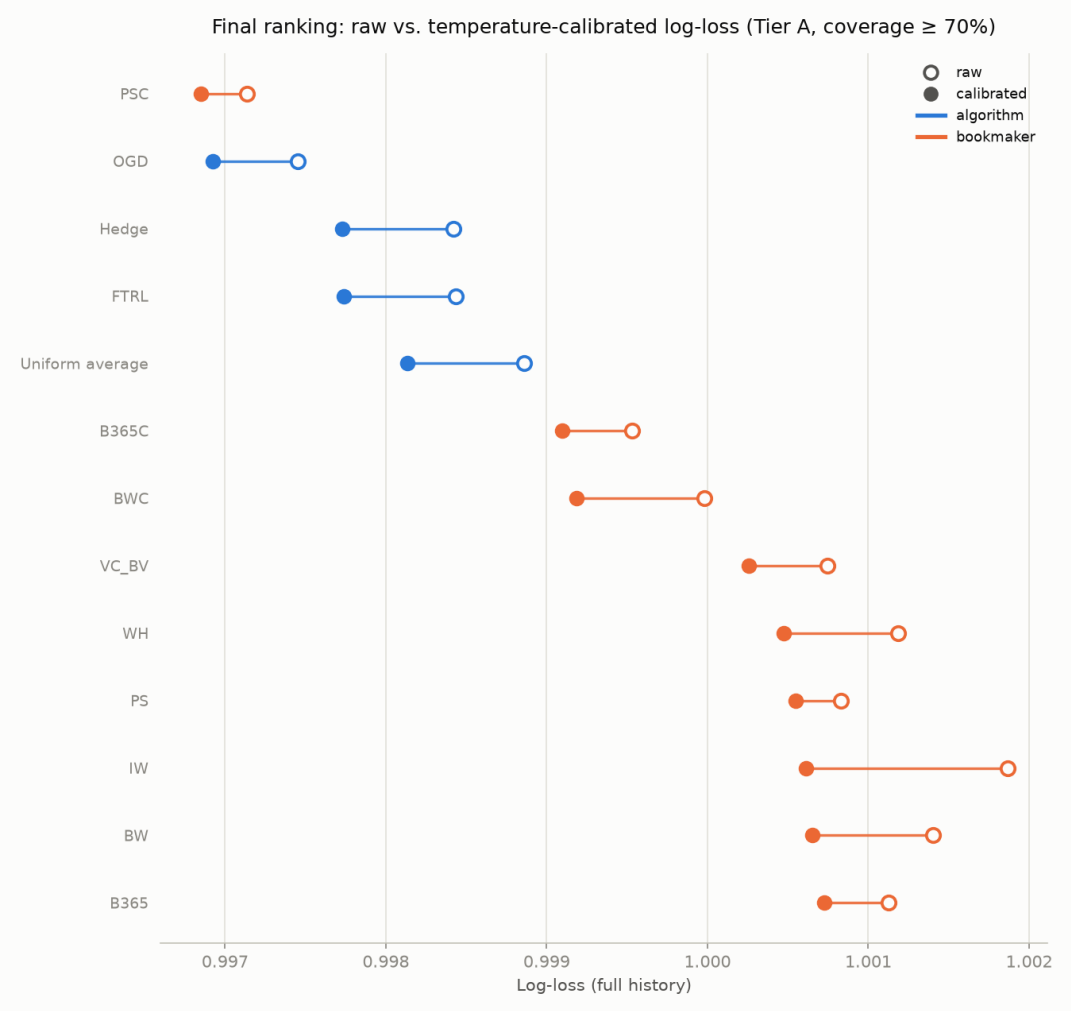


Figure 5.2: Horizontal axis log-loss, lower better; one row per series, ordered best at the top. Each row is a short segment: the hollow circle is the raw log-loss and the filled one the calibrated value, so the segment's length is what calibration bought. Orange marks a bookmaker and blue an algorithm, here and in the figures that follow. The choice of a dot/slope plot rather than a bar chart is deliberate, since log-loss values cluster around 1.0 with no meaningful zero and a bar chart with a truncated axis would visually exaggerate minute differences.

Table 5.3 gives the ranking and Figure 5.2 plots it. The paired tests give:

- All three algorithms significantly beat the uniform average: OGD -0.00141, Hedge -0.00044, FTRL -0.00043, all $p < 0.001$. OGD's margin is about threefold.
- OGD significantly beats Hedge (-0.00097), FTRL (-0.00098) and B365C (-0.00063).
- Calibration significantly improves all four methods, by $+0.00053$ to $+0.00073$.
- PSC remains significantly ahead of OGD in this configuration: $+0.00043$ calibrated, $p < 0.001$, over the 71,818 common matches.

A necessary caveat. Three of the twelve closing columns clear the 60% coverage floor (PSC at 93.8%, B365C at 69.6%, BWC at 65.0%); the rest start around 2019/20 and do not

reach it over the decade. The ranking therefore compares the mixture against three closing prices and six opening ones, which is enough for the comparison that matters but not the whole closing market. The fuller treatment is in §5.3.1 and §5.3.2, where the window is short enough for most closing columns to qualify.

A second caveat applies to the table as a whole and is more serious. Each series is scored over the rounds it actually quotes, and those differ: 76,584 for the algorithms, 71,818 for PSC, 53,267 for B365C, 49,755 for BWC. The columns are therefore not strictly comparable with one another, which is why every claim below rests on a *paired* test over the rounds the two compared series share, never on the difference of two table entries (§4.2.5).

Note that Interwetten occupies the *last* place in the raw log-loss ranking (1.00186) despite having the seventh best coverage. Its addition therefore makes the statement "we beat everyone except PSC" somewhat cheaper, and for that reason the substantive comparison remains the one against PSC and B365C.

Note also that log-loss and Brier give almost identical orderings, the only exception being the last two places. Accuracy gives a different ordering, but none of its differences is statistically significant, as expected, since it is not a proper scoring rule.

5.2.2 Which panel

	opening (13)		**all** (26)		**closing** (13)	
	basic	Shin	basic	Shin	basic	Shin
OGD	1.00076	1.00024	0.99745	0.99709	0.99722	0.99689
Hedge	1.00088	1.00033	0.99842	0.99781	0.99762	0.99714
FTRL	1.00088	1.00033	0.99843	0.99782	0.99761	0.99714
Uniform	1.00080	1.00022	0.99886	0.99824	0.99748	0.99703

Table 5.4: Log-loss by panel and de-vig method.

Table 5.4 reports the grid and, as Figure 5.3 makes visible, the ordering closing $<$ all $<$ opening holds in every cell, and in all 24 comparisons (4 algorithms $\times$ 3 comparisons $\times$ 2 normalisations) the difference is statistically significant. That closing prices beat opening ones is expected. What is unexpected is that the closing-only panel also beats the *full* one: the opening bookmakers do not act as useful diversification but as pure dead weight in the mixture.

Comparing each algorithm with the plain average *inside* the same panel explains the mechanism and is, in our view, the most generally useful finding of the thesis:

- On the **full** panel all three algorithms significantly beat the average.
- On the **restricted** panels Hedge and FTRL *significantly lose* to the average, in 8 tests out of 8. OGD wins on closing and ties on opening.

The full panel contains 13 good closing and 13 systematically worse opening experts, that is a clear quality difference available to be learned. In a homogeneous panel that difference is absent, and for two of the three algorithms the adaptation costs more than it returns.

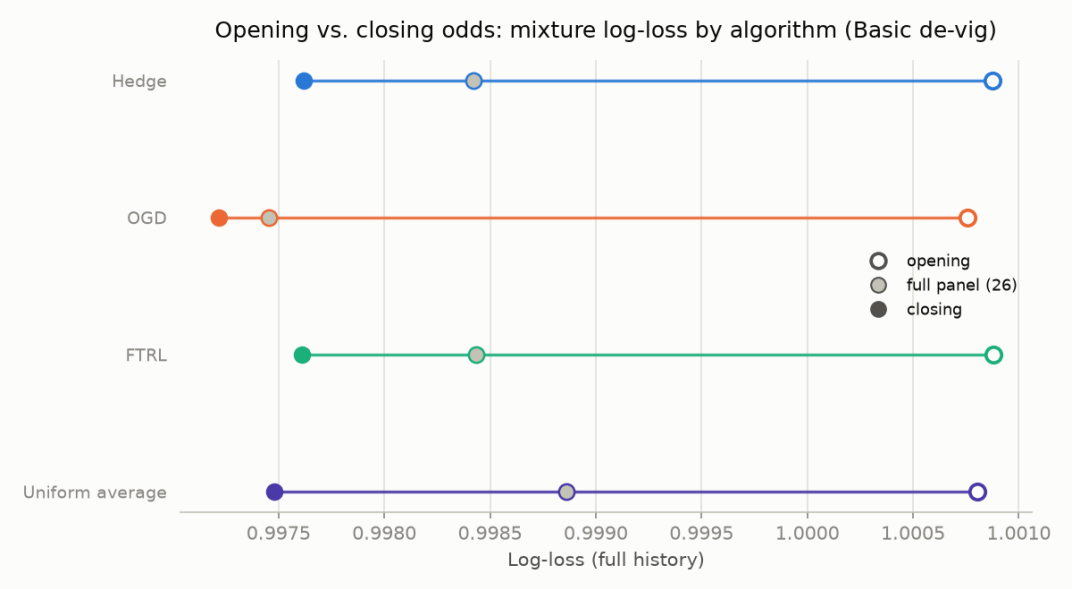


Figure 5.3: Comparison of the three panels for each algorithm. Horizontal axis log-loss over the full history, lower better; one row per algorithm, and the three marks on a row are the same algorithm fitted on the opening-only panel, on the full 26-expert panel and on the closing-only panel. The spread within a row is therefore the effect of the panel, not of the method. The ordering closing $<$ all $<$ opening holds in every case.

5.2.3 Where exactly closing prices are superior

§5.2.2 shows that the closing panel wins by ~ 0.0035 in log-loss. That is an average, and Remark 3.4 makes a sharper prediction: if the superiority is due to information arrival, it must be concentrated where the price *actually moved*. To test this, each match is characterised by the total variation distance between the initial and the final probability distribution of each house, averaged over the 13 available opening/closing pairs, and the matches are split into quintiles. Table 5.5 reports the result and Figure 5.4 plots it beside the same exercise run on Shin's z.

Movement quintile	Q1	Q2	Q3	Q4	Q5
Mean movement	0.0082	0.0144	0.0215	0.0324	0.0598
log-loss opening	0.99161	1.00581	1.00068	1.00666	1.00790
log-loss closing	0.99118	1.00509	0.99868	1.00214	0.99632
closing − opening	−0.00044	−0.00072	−0.00200	−0.00452	**−0.01158**
p	0.052	0.050	< 0.001	< 0.001	< 0.001

Table 5.5: Closing advantage by quintile of line movement, 10,649–10,650 matches per quintile.

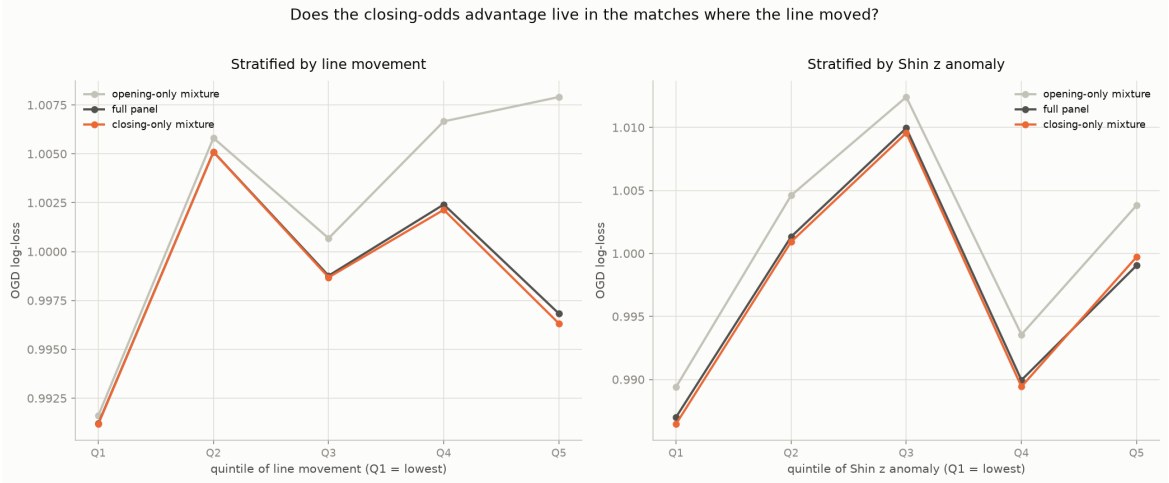


Figure 5.4: Left, the stratification by line movement; right, by anomaly of Shin's z. Horizontal axis the quintile, Q1 quietest and Q5 noisiest; vertical axis the log-loss of OGD, lower better. The three curves are the mixture fitted on the opening-only panel (grey), on the full panel (black) and on the closing-only panel (blue). The vertical gap between the grey and blue curves is the closing advantage, and it is that gap which widens from left to right.

The prediction is confirmed, with a 26-fold gradient. The closing advantage runs from -0.00044 in the quiet fifth of the matches to -0.01158 in the noisy one. In Q1 it is in fact **not even statistically significant** ($p = 0.052$): in the fifth of matches where the line barely moved, there is no measurable closing advantage. The "superiority of closing prices" is therefore not a general property of final prices but a phenomenon of the top quintile.

A necessary caveat. Movement is measured from the *same* closing prices that the closing panel uses, so the null result in Q1 is partly mechanical: if the two prices nearly coincide, the two mixtures cannot differ much. The non-trivial part is the *direction* at the other extreme. A large movement could perfectly well go the wrong way; that the later price is systematically the better one by 0.0116 is exactly the market-efficiency claim, localised where it actually operates. A useful control: the *difficulty* of the matches is not monotone across the quintiles (the full panel's log-loss runs 0.99120, 1.00510, 0.99877, 1.00241, 0.99683), whereas the advantage is, so the gradient is not an artefact of the form "harder matches, larger differences".

Shin's z is a much weaker indicator. Stratifying instead by the anomaly of z (the quantity that §3.2.2 identifies with the estimated proportion of informed traders) the gradient is only -0.00295 to -0.00498, that is a factor of 1.7 against 26, with all quintiles significant. So z does detect something, but the actual price movement is an incomparably cleaner indicator that information has arrived, reasonably so, since z is estimated from a single moment of the market whereas movement observes the difference of two.

5.2.4 The configuration that is selected

Three improvements have been measured separately (the corrected step, the closing-only panel, Shin) and are combined here. Every configuration is scored on the same 71,818 matches, the intersection of all of them with PSC, following the principle of §4.2.5. Table 5.6 stacks them.

Configuration	full	train	25% tail
PSC alone	0.99685	0.99746	0.99502
closing-only + Shin	0.99692	0.99754	0.99507
closing-only, proportional	0.99693	0.99756	0.99506
closing-only + Shin, no single-season	0.99692	0.99754	0.99509
full panel, proportional	0.99728	0.99801	0.99508
full panel, Shin	0.99731	0.99804	0.99512

Table 5.6: Calibrated log-loss of OGD by configuration, the same 71,818 matches throughout.

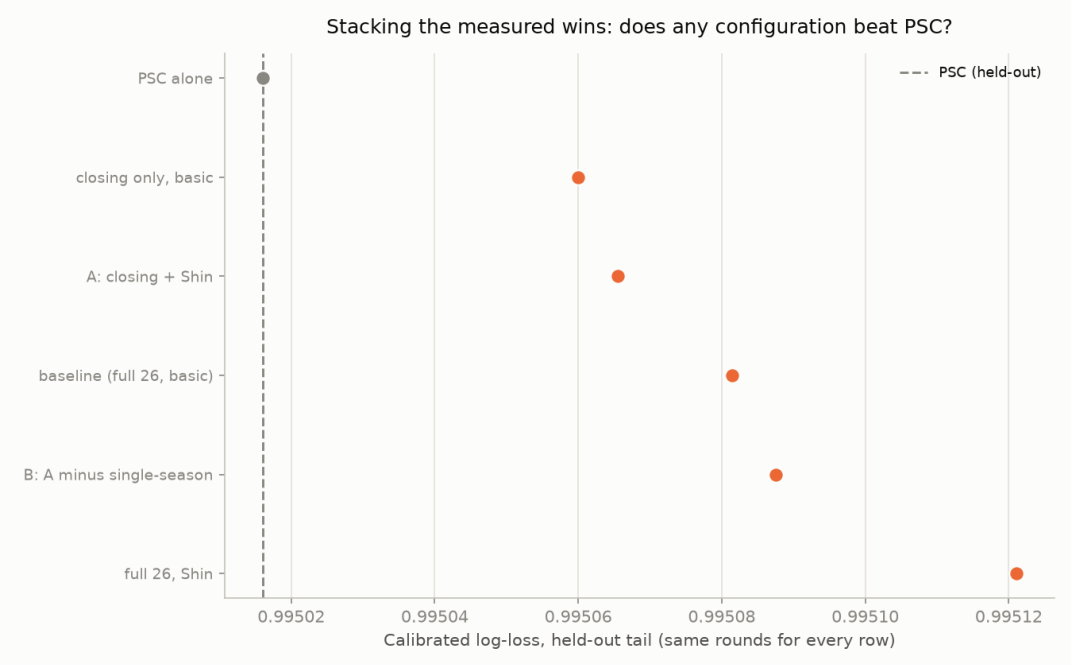


Figure 5.5: The configurations on the held-out tail, with PSC as the reference line. Horizontal axis the calibrated log-loss on the held-out 25%, lower better; one dot per configuration. The dashed vertical line is PSC scored on the same rounds, so a dot to its left beats Pinnacle and a dot to its right does not. All of them cluster within 0.00005.

- **The panel does all the work**. Closing-only + Shin beats the full panel by -0.00036 ($p < 0.001$), and closing-only with proportional normalisation by -0.00034 ($p < 0.001$).
- **Shin on its own offers nothing measurable** over the full panel ($+0.00004$, $p =$

0.181), consistent with Shin and temperature scaling correcting the same favourite-longshot bias, so that the second absorbs the first.

- **Against PSC**: the full panel significantly loses ($+0.00043$, $p < 0.001$), while closing-only + Shin *ties* ($+0.00007$, $p = 0.234$) and closing-only with proportional normalisation likewise ($+0.00009$, $p = 0.143$).

The 25% tail does not produce a winner: all six rows gather within less than 0.00005 (Table 5.6, Figure 5.5). It thus confirms the tie, and it would not be honest to present it as a win. Note also that in every closing configuration PSC is *inside* the mixture: "we reach PSC" reads as "the mixture reaches its own best member".

5.2.5 Without Pinnacle

The obvious counter-argument is that the mixture merely reproduces Pinnacle, which is inside it anyway. Removing both of its columns and refitting from scratch on three panels gives Table 5.7:

Panel	OGD rank	log-loss	First	Paired
full (24)	#3 of 9	0.99782	FTRL 0.99771	5 wins, 0 losses
closing-only (12)	#1 of 9	0.99919	OGD (itself)	3 wins, 2 ties
opening-only (12)	#5 of 9	0.99999	VC_BV 0.99987	3 wins, **1 loss**, 1 tie

Table 5.7: Each panel is refitted from scratch and compared against the bookmakers of its own phase.

On the **full** panel the result holds comfortably (Figure 5.6): all four methods sit above every bookmaker, and OGD significantly beats all *five*, with margins from -0.00181 (VC_BV) to -0.00315 (IW) and $p < 0.001$ throughout. All of them, however, are opening prices.

On the two **homogeneous** panels the result changes. On closing, OGD beats three (WHC -0.00088, IWC -0.00077, BWC -0.00051) and ties with B365C ($p = 0.505$) and VC_BVC ($p = 0.284$); it does rank first, but with the plain average second at a distance of 0.00002, i.e. practically indistinguishable. On opening it again beats three, ties with B365 ($p = 0.203$), but **significantly loses to VC_BV** ($+0.00030$, $p = 0.004$): the only case in the whole thesis where a *bookmaker* other than Pinnacle significantly beats us, and there the plain average moves ahead of OGD. The statement is deliberately restricted to bookmakers: in the deployment test (§5.3.2) Betfair Exchange also significantly beats us, but that is not a bookmaker but an exchange.

The pattern is stable enough not to be attributed to chance, and it repeats the finding of §5.2.2: once Pinnacle leaves, the remaining members of the same phase resemble one another so closely that no quality difference is left to learn. The configuration that is selected is therefore the best partly *because* it contains the market's best odds-setter.

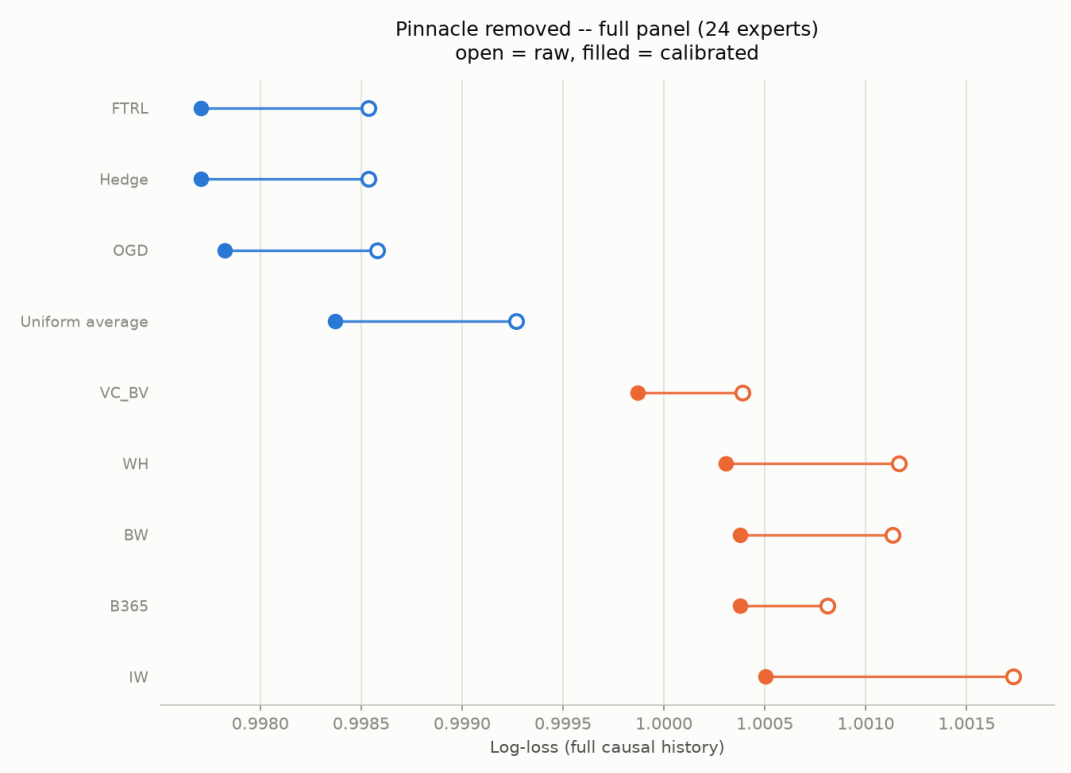


Figure 5.6: Full panel without any Pinnacle column. Horizontal axis log-loss, lower better; one row per series, drawn as a short segment whose hollow end is the raw value and whose filled end is the calibrated one, so the length of a segment is what calibration bought. Orange is a bookmaker, blue an algorithm. All four methods sit above all remaining bookmakers.

5.3 Robustness over time

5.3.1 Recent window

The coverage threshold is defined over the decade, which favours whoever was quoting long ago. Restricting to the last five seasons (38,749 matches) brings VC_BVC and WHC above it as well, so the mixture is measured against five closing prices instead of three and the comparison becomes correspondingly more demanding (Table 5.8).

#	Series	phase	log-loss
1	PSC	closing	0.99439
2	**OGD**	—	**0.99495**
3	VC_BVC	closing	0.99498
4	WHC	closing	0.99502
5	BWC	closing	0.99504
6	B365C	closing	0.99543
7	Uniform	—	0.99591
8	Hedge	—	0.99604
9	FTRL	—	0.99607
10–14	WH, PS, VC_BV, BW, B365	opening	0.99771 and below

Table 5.8: Last five seasons, full panel, calibrated log-loss.

OGD sits at #2 (Figure 5.7), ahead of every closing bookmaker other than Pinnacle, and the gap to PSC shrinks to $+0.00036$ ($p = 0.014$) from the decade's $+0.00043$. Two findings, however, *reverse* and are reported explicitly: Hedge and FTRL lose to the plain average on *all* panels here, not only the restricted ones ($+0.00012$ and $+0.00015$, $p = 0.006$ and $p = 0.002$), and log-loss and Brier cease to agree on the ordering.

5.3.2 Deployment test

The test with a real date cut: fitting on 65,009 rounds up to 31/12/2024, measurement on 11,574 rounds of 2025–26, of which 8,151 are common. Table 5.9 gives the ranking and Figure 5.8 plots it.

#	Series	phase	log-loss
1	BFEC (Betfair Exchange)	closing	1.00040
2	**OGD closing-only, online**	—	**1.00110**
3	OGD full panel, online	—	1.00129
4	BWC	closing	1.00146
5	OGD closing-only, frozen	—	1.00151
6	B365C	closing	1.00165
7	OGD full panel, frozen	—	1.00183
8–10	B365, BW, BFE	opening	1.00371 – 1.00758

Table 5.9: Trained up to 31/12/2024, measured on 2025–26.

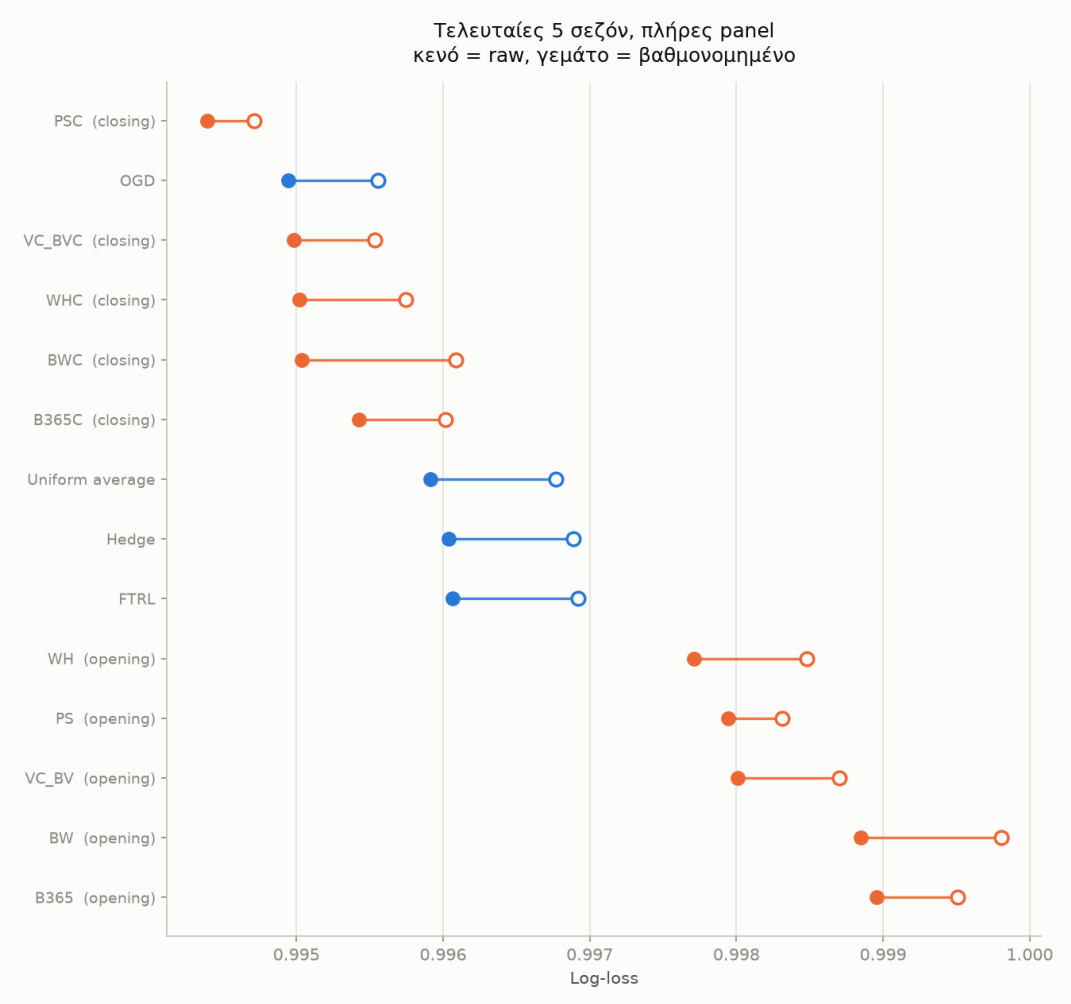


Figure 5.7: Last five seasons. Five closing bookmakers clear the coverage threshold inside this window against three over the decade, which makes the comparison more demanding. Horizontal axis log-loss, lower better; one row per series, hollow end raw and filled end calibrated, orange for a bookmaker and blue for an algorithm.

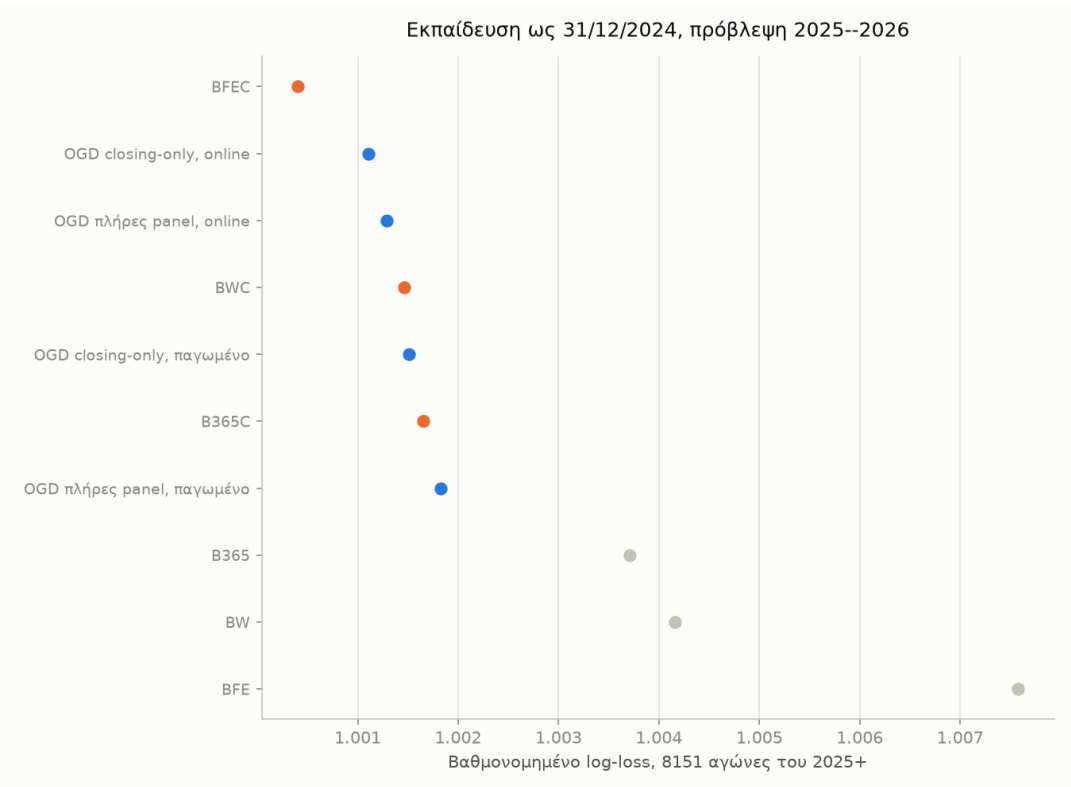


Figure 5.8: Deployment test. Horizontal axis the calibrated log-loss over the 2025–26 rounds, lower better; one dot per series, best at the top. Blue dots are the mixture in its four variants, orange dots the bookmakers still quoting in the test window. The opening prices sit clearly lower, while the closing prices and the variants of the mixture cluster together.

- **Every opening price is significantly beaten**, in every variant: -0.0019 to -0.0065, all $p \leq 0.019$.
- **The closing prices are not beaten**. The best outcome is a tie with all three, and only in the closing-only / online variant: B365C $p = 0.193$, BWC $p = 0.361$, BFEC $p = 0.052$. In the full-panel variants **BFEC significantly beats us** ($+0.00089$, $p = 0.013$ online; $+0.00143$, $p = 0.002$ frozen), so the tie with every closing price holds only for the closing-only variant.
- The online variant consistently beats the frozen one (1.00110 against 1.00151), so the precise statement is not "trained up to 2024" but "runs continuously".

Pinnacle in this window. Its coverage drops to 59% after 2025 and it stops quoting on 14/01/2026, so it does not pass the threshold. Forcing it into the common table drops the common matches from 8,151 to 2,860 and no result remains significant. It is therefore compared one to one, over its own rounds: we significantly beat PS (-0.0022 to -0.0023, p from 0.007 to 0.019, over 6,857 matches), and we tie with PSC (p from 0.532 to 0.796, over 6,892 matches).

The best forecaster in this window is not a bookmaker but **Betfair Exchange**: an exchange where players bet against one another, that is a market price without an operator's margin.

5.4 Value betting and the learned stake size

A Kelly simulation with calibrated probabilities and a panel matched to the target's phase. The stake size is not pinned to a fixed fraction but learned online, with the multiplier of §3.5.2; the fixed coefficient $\frac{1}{4}$ is retained as a reference point.

Correctness check and selection. A constant $\lambda = 0.25$ reproduces the implemented simulation *exactly*, at every round of deployment and over the same set of bets, so the learned multiplier is the only difference between the columns that follow. The rate, the initial value and the upper bound of λ are selected on a chronological 75% prefix of the *bets*, jointly for all targets, and the remaining tail stays outside the selection. Neither of the two preceding stages is re-tuned here.

The selection gives $\lambda_0 = 0.5$ and $\lambda_{\max} = 1.5$ for Online Newton Step, with a tail of $+0.000106$ in log growth rate per bet, and $\lambda_0 = 0.25$, $\lambda_{\max} = 1$ for OGD, with a tail of $+0.000101$. One point is worth noting that contradicts every other tuning in the thesis: **Online Newton Step selects a rate multiplier equal to** 1, that is the constant the theory supplies needs no correction at all, whereas OGD needs $\times 10$ and in one setting the grid runs off the edge without stabilising. Where the structure of the loss is exploited correctly, the a-priori bound turns out to be correctly calibrated as well. Table 5.10 gives the closing targets and Figure 5.9 the bankroll paths behind them.

Against	Bets	Kelly $\frac{1}{4}$	OCO λ	Full Kelly	mean λ
WHC	7,253	$84.2\times$	**6,494**$\times$	$29{,}342\times$	0.79
BWC	6,920	$3.09\times$	**7.08**$\times$	$13.77\times$	0.62
VC_BVC	4,561	$1.58\times$	$1.54\times$	$1.74\times$	0.57
B365C	7,643	$1.05\times$	$0.67\times$	$0.24\times$	0.39
PSC	19,497	$0.54\times$	$0.25\times$	$0.0035\times$	0.37

Table 5.10: Final bankroll starting from 1, calibrated probabilities, proportional normalisation. Every column plays the *same* bets, since the choice of outcome does not depend on the size. The last column is the mean multiplier learned by Online Newton Step.

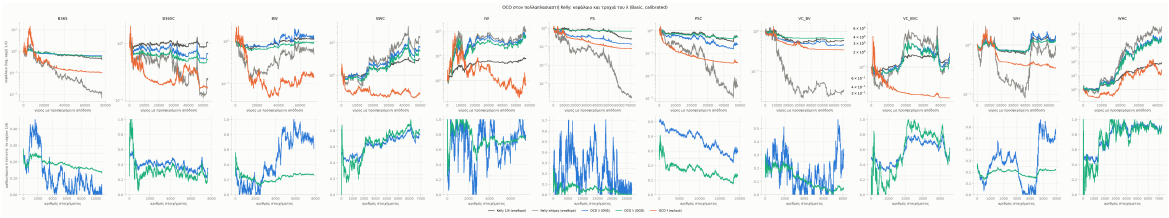


Figure 5.9: One column per target. Above, the path of the bankroll against the number of rounds on which a bet was offered, on a logarithmic vertical axis with a dotted line at the starting bankroll of 1; below, the trajectory of the learned λ over the same rounds, with a dotted line at the $\frac{1}{4}$ in force.

The result has two parts that do not say the same thing. On the closing targets, in the per-bet paired test the learned multiplier **ties** with the fixed $\frac{1}{4}$: over twenty cells (two normalisations $\times$ two stages $\times$ five targets) there are 17 ties, two wins and one loss. In final bankroll it is ahead on two of the five targets, and on WHC by seventy-seven times. The two reconcile numerically: the per-bet difference is $+0.00060$, minute, but it compounds over $7{,}253$ bets, while the uncertainty remains wide enough that the test does not certify it ($p = 0.099$). It is the same pattern as the "sharp" anchor-bookmaker result: systematic superiority in the point estimate, without statistical certification.

What the multiplier learns. Against the closing targets it sits almost everywhere above $\frac{1}{4}$, and its value tracks how profitable the target is: 0.37 against the loss-making PSC, 0.79 against the profitable WHC. It therefore raises the risk where there is an edge and pulls it back where there is none, behaviour no constant can express, and which emerges without its being given any information about who the opponent is.

The formulation was the problem, not the idea. The same question had been posed earlier with the learner learning the *fraction* of capital directly. That version lost systematically: 9 losses out of 20 cells, no wins. The cause is exactly the one §3.5.2 predicts; the comparator "best constant stake" does not contain the closed form, and the learner throws away the $\hat{p}_t$ that the two preceding stages have just produced. Changing only what is learned, with the same data and the same test, the result goes from systematic loss to a tie with a point-estimate advantage.

A side finding: $\frac{1}{4}$ is excessively conservative. *Full* Kelly ties in 16 of the 20 cells, wins in 2 and loses in only 2. Its risk is concentrated almost entirely on PSC, where it wipes out the bankroll (0.0035). On the profitable targets, that is, the fixed $\frac{1}{4}$ leaves a substantial part of the growth unused. The learned multiplier is the middle road: it keeps most of the upside and avoids the catastrophe. The paired tests are in Table 5.11.

Where there is an edge. An *edge* here means an advantage over a bookmaker's price large enough to be detected statistically, that is a positive expected growth of capital when betting against it. Which bookmakers are beatable at all in this sense is mapped over eleven targets with the fixed $\frac{1}{4}$, so that the rows stay comparable with one another.

Against	Phase	Bets	Bankroll	p	
WHC	closing	6,565	$38.52\times$	< 0.001	profit
IW	opening	7,443	$7.89\times$	0.008	profit
BWC	closing	6,607	$1.81\times$	0.247	—
WH	opening	5,931	$1.38\times$	0.545	—
BW	opening	7,841	$1.09\times$	0.820	—
VC_BVC	closing	4,653	$0.75\times$	0.455	—
VC_BV	opening	8,083	$0.60\times$	0.245	—
B365C	closing	7,970	$0.58\times$	0.328	—
B365	opening	10,829	$0.41\times$	0.101	—
PS	opening	26,675	$0.26\times$	0.055	—
PSC	closing	39,997	$0.0002\times$	< 0.001	loss

Table 5.11: The p column concerns the mean log growth rate per bet. The final bankrolls are not comparable across rows, since the number of bets differs by an order of magnitude.

Over eleven bookmakers there are *two* significant profits (both against "soft", recreational-facing houses) and one significant loss, against PSC, which is the best forecaster in the panel and therefore exactly the opponent an inferior model ought to lose to. The test thus works in both directions. Phase matching is essential: with the full panel instead of the closing-only one, the result gets worse on *every* closing target.

The extension to opening targets. The learned multiplier is applied to all six opening targets as well, with the corresponding opening-only panel. The causality requirement is stricter here: removing the target is not enough, the *whole* closing side must go, since no closing price exists at the moment the bet is placed (Table 5.12). The hyperparameters are selected separately per phase, on that phase's own prefix of bets, so that the opening result does not depend on closing bets it is supposed to be independent of.

Against	Bets	Kelly $\frac{1}{4}$	OCO λ	Full Kelly	mean λ
IW	7,443	$7.89\times$	**64.4**$\times$	$109.6\times$	0.67
WH	5,931	$1.38\times$	$1.39\times$	$0.78\times$	0.28
BW	7,841	$1.09\times$	$1.10\times$	$0.50\times$	0.45
VC_BV	8,083	$0.60\times$	$0.67\times$	$0.03\times$	0.17
B365	10,829	$0.41\times$	$0.61\times$	$0.007\times$	0.12
PS	26,675	$0.26\times$	**0.68**$\times$	$0.0003\times$	0.05

Table 5.12: Opening targets: final bankroll starting from 1, calibrated probabilities, proportional normalisation. The OCO λ column gives the better of the two variants (ONS / OGD) and the mean λ corresponds to it.

The finding is that the learned multiplier pays off *more* on opening prices than on closing ones: 8 wins and 2 losses over 48 paired tests, against 4 wins and 3 losses on the closing side. Significant wins arise against PS ($p < 0.001$), IW ($p = 0.027$) and VC_BV ($p = 0.009$), while the two losses are minute (-0.000028 and -0.000080).

More interesting than the result is the *mechanism*, which becomes visible here. The mean λ ranks the targets almost perfectly by how beatable they are, without the algorithm being given any information about who the opponent is: against IW, the only profitable opening target, it raises λ to 0.67, that is nearly full Kelly; against PS, the hardest, it lowers it to 0.05 and essentially stops betting, thereby more than doubling the surviving capital, from $0.26\times$ to $0.68\times$. The opening λ values (0.05–0.75) are systematically lower than the closing ones (0.17–0.79): the opening market is harder and the algorithm discovers this by itself.

The same appears from the opposite side. *Full* Kelly, which tied on the closing targets (2 wins, 2 losses), collapses on the opening ones: **11 significant losses against 2 wins**. The opening side genuinely does need shrinkage, and the question is not whether to impose it but how much, which is precisely the question a fixed coefficient of $\frac{1}{4}$ cannot answer and a learned one can.

The conclusion is that improved log-loss does not translate into profitability. The quality difference gained is smaller than the market's margin, and whatever edge exists concerns *specific* bookmakers, not a strategy that works generally. Stake size is orthogonal to this: it changes how fast the bankroll grows where an edge exists, and how fast it is lost where none does, without creating an edge from nothing.

5.5 How much information does learning need?

5.5.1 The bandit experiment

Everything above presupposes full information: after each match all 26 bookmakers are scored. The question of §3.3.8 is what remains when the algorithm sees only *one* number per round, the loss of its own prediction. The two gradient estimators, one-point and two-point, are implemented, with ξ and the step selected on the same chronological train prefix as every other hyperparameter, and with ten different random seeds, since the algorithm is stochastic.

Two correctness checks come first. With full information and a linearised loss the code reproduces the implemented OGD *exactly*, and the two-point estimator is identified

with the true tangential gradient with relative error 0.6% and cosine 0.99998 over $200{,}000$ samples. Table 5.13 decomposes the margin and Figure 5.10 shows the sweep it rests on.

Series	Cal. log-loss	Raw log-loss	Difference
OGD, full info., exact gradient	0.996928	0.997455	$+0.000002$
OGD, full info., linearised	0.996926	0.997454	—
Two-point BGD ($\xi = 0.0003$)	0.997368	0.998004	$+0.000443$
No learning ($\eta = 0$)	0.997632	0.998296	$+0.000707$
One-point BGD ($\xi = 0.02$)	0.997631	0.998297	$+0.000706$
Uniform average	0.998132	0.998862	$+0.001206$

Table 5.13: Bandit against full information, the same 76,584 rounds, the same calibration. The last column is the paired difference from the implemented OGD in calibrated log-loss; positive means worse than it. The first row is a tie ($p = 0.823$), the rest are significant at $p < 0.001$.

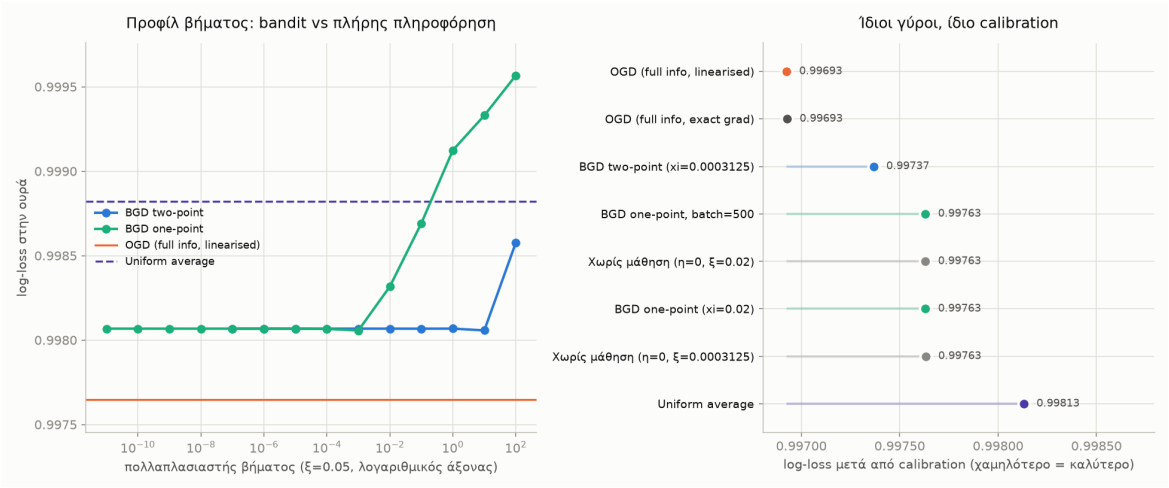


Figure 5.10: Left, the step profile on which the selection is made: horizontal axis the step multiplier on a logarithmic scale, vertical axis the log-loss on the held-out tail. Right, the final comparison: horizontal axis the calibrated log-loss, lower better, one bar per method. The flatness of the left panel is itself the result, since a profile with no interior minimum means there is nothing for the estimator to select.

The control that makes the numbers interpretable. Write BGD for *bandit gradient descent*, the estimator of §3.3.8 run on the loss of the chosen mixture alone. The obvious reading ("BGD beats the uniform average, so the bandit feedback taught it something") is wrong. Under the Sleeping Experts reduction the weight vector moves even with a *zero* step: every time the awake set changes, the persistent state is reprojected onto a simplex of different dimension, and that churn alone produces non-uniform weights. Measured directly: at $\eta = 0$ the weights are as far as 0.69 from uniform in max norm. The correct benchmark is therefore not the uniform average but *the same algorithm with the step set to zero*: same shrinkage, same perturbations, same projection dynamics, no learning. Whatever lies between that control and BGD is what the bandit signal actually bought.

	Projection without learning	+ bandit learning
Two-point BGD	$+0.000500$ (41.4%)	$+0.000264$ (21.9%)
One-point BGD	$+0.000501$ (41.5%)	$\mathbf{-0.000000}$ (0.0%)

Table 5.14: Decomposition of the full-information margin (uniform average 0.998132 against OGD 0.996926, i.e. 0.001206) into what the projection gives with no learning at all and what the bandit feedback adds on top.

The one-point estimator learns nothing here. Selection on train drives its step to $1.4 \cdot 10^{-14}$, and the whole region from 10^{-14} to $9 \cdot 10^{-11}$ ties at the same train score 0.998375: this is the plateau of not learning, and the selector simply takes its endpoint. Any positive amount of learning hurts. The cause is quantified by the alignment of a single round's estimate with the true gradient: for the two-point estimator the mean cosine is $+0.27$, whereas for the one-point estimator it is $+0.002$, that is zero. The signal is 0.2% of the size of the estimate; summing over T rounds reduces the noise by $\sqrt{T} \approx 277$, which leaves an effective ratio of about 0.55, still below one. **The horizon is not long enough** for a one-point estimator on a 25-dimensional simplex.

The one legitimate fix was tried and changes nothing. The variance argument dictates a specific intervention: accumulating the estimate for m rounds before stepping, which raises the effective ratio by $\sqrt{m}$ and keeps the point in the interior of the simplex, so that the projection does not asymmetrically clip the noise. m was swept over $\{1, 20, 100, 500\}$ across five orders of magnitude of the step. *No* combination produces an interior optimum: every cell is either on the plateau or worse, and batching does exactly what the theory predicts (it tolerates a larger step without damage) while never producing a gain. The signal is not there to be amplified.

A caveat about the two-point estimator. Its shrinkage parameter was driven by the selection to the lower end of the admissible interval. This is not an accident but what is to be expected: as $\delta \to 0$ the finite difference converges to the exact directional derivative, so the two-point estimator *ceases to be bandit.* The 21.9% of Table 5.14 must therefore be read as an upper bound on what the two-query approach achieves, not as performance in a genuine bandit environment. The one-point estimator is the honest experiment, and its answer is zero.

What remains from the experiment. Two things, and neither is what it was looking for. First, that one-point feedback is insufficient at this horizon, with a quantitative explanation of why and with the obvious fix tried and rejected. Second, and more useful for the rest of the thesis, that 41% **of the mixture's margin over the uniform average does not come from learning at all**, but from the reprojection forced by the changing composition of the panel. This is an independent finding about Sleeping Experts dynamics and it agrees with the measurement of §5.3.2, where weight movement in the late stages turned out to be dominated by the projection and not by the gradient step.

A side finding in favour of linearisation. The full-information control with the *exact* gradient of the non-linear loss ties with the implemented, linearised OGD ($+0.000002$, $p =$

0.934). The simplification of Lemma 3.8, that is, costs nothing measurable on these data: a confirmation of a design choice that until now rested only on the theoretical argument that the linear problem is harder.

5.5.2 Markovian switching of market regimes

The Markov chain on the *weight vector* has already been tried and rejected: every transition kernel selects $\alpha = 0$ on train, i.e. collapses to plain OGD. Here the chain runs over a different object, over *market states*. We maintain K latent regimes, one weight vector per regime under Sleeping Experts, and a belief updated by the forward recursion of a hidden Markov model; each regime steps in proportion to its responsibility. $K = 1$, a single regime, reproduces the plain Sleeping-Experts OGD of §3.3.5 exactly.

The motivation is measured and not hypothetical: §5.2.2 and the information arrival analysis show that the closing advantage varies 26-fold depending on how much the line moved. There is therefore a genuine latent state in the data. The regime is deliberately left *latent*, so that it can be checked afterwards whether the chain discovers it on its own.

The first version failed in a diagnosable way. The mean belief stayed at exactly 0.500 for both regimes. The regimes *never differentiated*: they receive the same loss vector, so from a symmetric starting point they follow identical trajectories and their likelihoods stay equal. The result was simply OGD with the step divided by K, and it lost significantly ($+0.000128$, $p < 0.001$). The symmetric solution is a fixed point of the dynamics, and an explicit differentiation mechanism is required.

After the fix: a tie. A belief sharpness β, a forgetting factor γ and a per-regime clock were added. Selection on train gives $K = 3$, $\beta = 30$, $\gamma = 0$, $\tau = 0$, with calibrated log-loss 0.996909 against OGD's 0.996926: a difference of -0.000016, $p = 0.211$, **a tie**. The $\beta = 30$ is a genuine interior optimum; the grid was extended to 1000 and the values 100, 300, 1000 are all worse on train.

Two findings that nullify any claim that "the chain works". First, the tail is *flat*: from $\beta = 3$ to $\beta = 1000$ it moves between 0.997647 and 0.997649, a range of two millionths over three orders of magnitude of the parameter, while the train differences driving the selection are ten times larger. The selection is fitting noise, exactly as in the joint sweep of step and calibration rate (§5.6). Second, and more decisive, $\tau = 0$ is selected (no transition), and the standard deviation of the belief is 0.0000: it freezes early and never moves again. There is therefore no Markovian switching. What remains is a fixed convex combination of three OGDs with different initialisations, that is *ensembling*, and the small point-estimate gain must be attributed to that.

And the chain discovers nothing. The correlation of the belief with the measured line movement is undefined, precisely because the belief is constant. The latent regime corresponds to no measurable property of the market. The answer to the question the experiment posed (does a model with no side information discover the structure that was measured directly?) is negative and clean. Figure 5.11 shows both halves of it.

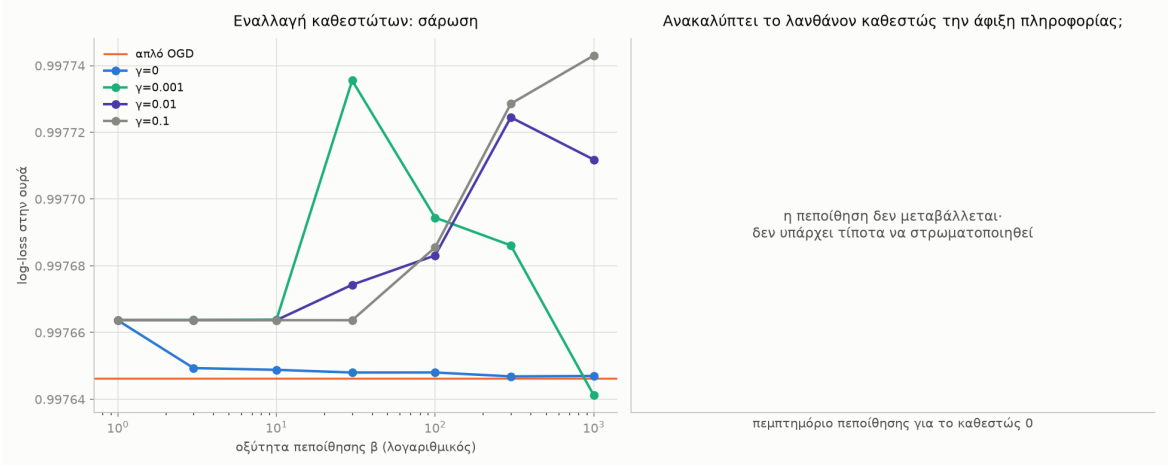


Figure 5.11: Left, the sweep of the belief sharpness. Right, the check of whether the latent regime corresponds to the measured information arrival.

This is the third independent instance of the strongest pattern in the thesis: once the step is tuned, no movement of mass over the experts offers anything measurable. Fixed-Share, transition kernels, and now regime switching. The difference here is that the failure was *diagnosed* rather than merely observed.

5.6 Negative results

A series of plausible hypotheses was tested and rejected. We report them explicitly, since each one bounds the method's scope.

Per-league specialisation does not help. A separate model per league loses to the single model in *every* one of the six leagues tested. Two plausible causes: each league has only 2,231–3,800 matches, far too few for the online algorithm to converge on its own, and the specialists end up trusting largely the *same* experts as the unified model, i.e. they do not discover anything local.

Fixed-Share does not help. Every $\alpha > 0$ worsens the log-loss (0.99842 at $\alpha = 0$ against 0.99885 and above); the $\alpha = 0$ case reproduces plain Hedge exactly, which is what makes this a statement about Fixed-Share and not about the particular implementation of it. Sweeping α *jointly* with the step shows why: α only ever helped when the step was too large. Both control the same quantity (how fast the algorithm forgets), and the step is the more efficient of the two instruments.

A per-expert step clock significantly worsens things. Replacing the global t with the number of rounds in which each expert was awake (which is literally the quantity the Sleeping Experts guarantee speaks about) gives a result that is *better on train but* -0.00073 *on the tail.* An expert with 6.9% coverage receives a step about 14 times larger, and the mixture chases the rare bookmakers, whose apparent quality is confounded with the difficulty of the single season in which they appear. This is the clearest demonstration of overfitting in the thesis.

Joint selection of step and calibration rate offers nothing. The two parameters do not interact: the optimal calibration rate is 0.15 in 11 of 11 rows. Joint selection on train in fact picks a worse value for OGD, since the grid is nearly flat there and the selection fits noise.

Markovian transition kernels offer nothing. All three kernels of §3.3.7(a) (uniform, towards the opening/closing twin, towards the same market phase) select $\alpha = 0$ on the chronological prefix, i.e. collapse to plain OGD. The paired tests confirm this with a mean difference of 0.000000 *exactly* and $p = 1.000$, not approximately. The finding agrees with Fixed-Share: once the step itself is tuned, no post-hoc movement of mass has anything left to correct, because both control the same quantity. At *large* α, however, the structured kernels do measurably less damage than the uniform one, i.e. the panel's structure is real and detectable, merely redundant.

The Bayesian version of Kelly does not beat the fixed fraction. The formulation of §3.5.3 is theoretically more attractive, since the shrinkage emerges from the uncertainty instead of being imposed. Paired with the fixed $\frac{1}{4}$ over the same bets, for four targets and two stake levels, it gives **6 ties and 2 significant losses** (PSC at both levels, $p = 0.003$ and $p = 0.049$), with *no* significant win. The selected concentration parameter κ is moreover unstable across targets with no evident pattern, an indication that the selection is fitting noise. The most likely explanation is that the Dirichlet model does not correctly describe the mechanism generating bookmakers' disagreements: their disagreements are not independent noise around a common truth but systematic pricing differences, so the posterior is the wrong distribution and the shrinkage it produces does not correspond to the real uncertainty.

Draws are practically unpredictable as a primary prediction. Although they constitute $\sim$26% of the matches, the best algorithm selects them as the most likely outcome in 0.4% of cases. This is a known phenomenon of football prediction, confirmed here quantitatively.

5.7 Discussion

5.7.1 What the thesis showed

Summarising the findings of this chapter, three answers emerge at different levels.

At the level of **method**, the framework works, but less than a first reading suggests. An online mixture of published odds, with no data beyond the prices themselves, consistently and significantly beats both the uniform average and the overwhelming majority of individual companies. That this is achieved with no statistical assumption about the distribution of the data, and with predictions that are by construction out of sample, is the main methodological gain of the online approach over a classical batch one. The limitation emerged from the experiment of §5.5.1: with an explicit zero-step control it turns out that 41% of the margin over the uniform average is produced by the repeated projection onto the simplex, that is by the geometry of the reduction and not by learning. The weighting does add something real, but less than one would conclude by comparing it directly against the average.

At the level of **comparison with the market**, the answer is more guarded. Over the full decade and on the full panel the sharpest bookmaker remains significantly ahead. In the configuration finally selected the difference ceases to be significant, and in the deployment test the mixture beats every opening price and ties with the bookmakers' closing prices. The real limit is not a bookmaker: Betfair Exchange, which is an exchange and therefore a market price without an operator's margin, significantly beats us. The honest statement is that we *reach* the leading edge of the bookmaker market, not that we exceed it, and that the market without an intermediary remains ahead.

At the level of **economic exploitation**, the answer is negative. Improved log-loss does not translate into profitability, because the quality difference gained is smaller than the margin the market charges. This distinction (between being a better forecaster and being able to exploit it) is, in our view, the most practically useful conclusion. The third OCO stage does not overturn it, but it clarifies what stake size can and cannot do: the learned multiplier multiplies the final bankroll where an edge exists and shrinks itself where none does, without ever creating an edge. The choice of opponent remains far more important than the choice of stake.

5.7.2 When learning pays off: heterogeneity as a precondition

The recurring pattern throughout the thesis is that learned weighting pays off when, and only when, the experts differ substantially in quality. The finding appears independently at three points:

- On the full panel, which contains 13 good closing and 13 systematically worse opening experts, all three algorithms significantly beat the plain average.
- On the homogeneous panels (opening only or closing only) Hedge and FTRL *lose* to the average, in 8 tests out of 8.
- Removing Pinnacle from the *opening* panel, even OGD falls behind the average (0.99999 against 0.99990) and significantly loses to VC_BV.

The fourth case (the closing panel without Pinnacle) does **not** support the pattern any more, and is reported for precisely that reason. There OGD ranks first (0.99919) with the average second (0.99921). The difference is 0.00002, practically zero, so the correct reading is not "weighting wins here" but "the two are indistinguishable", which still agrees with the general conclusion, since in a homogeneous panel weighting has nothing to learn.

The interpretation is that the gain comes primarily not from the algorithm but from the structure of the panel. When a real quality difference exists, the weighting finds and exploits it. When it does not, the adaptation is pure cost: the algorithm pays estimation noise with no corresponding benefit, and the average (which estimates nothing) proves more robust.

This conclusion has direct practical value for anyone who would apply the framework in another field. The criterion for a good set of experts is not their number, nor how much their numbers differ, but whether they are *produced by different processes*.

5.7.3 The character of the method

A property of the framework deserves noting that is not always made explicit. The model is not trained and then applied; there is no moment at which it "knows" and subsequently

acts. At every match it states a claim, the outcome refutes it to some degree, and the claim is revised. The refutation *is* the material of learning, literally: the gradient that updates the weights equals the loss of the prediction against what happened. Without error nothing is learned, and the weights at any moment are not the answer but the current state of an unresolved disagreement.

This property was also confirmed empirically, where we did not expect it. We examined whether the mixture eventually "settles" somewhere and quietens down, given that the step decays as $1/\sqrt{t}$. It does not. The step does fall by a factor of 277 from the first round to the last, but the mean movement of the weight vector per round is 0.0353, 0.0226, 0.0227, 0.0284 over the four quarters: the last moves *more* than the second and the third, at a point where the step is two hundred and seventy times smaller. The reason is that the movement is dominated by the projection onto the simplex every time the awake set changes, not by the step: the same phenomenon that §5.5.1 isolates quantitatively, showing that the projection on its own, with no learning at all, produces 41% of the mixture's margin over the uniform average.

Relatedly, the theoretically superior alternative (the Aggregating Algorithm, which exploits the exp-concavity of the log-loss and achieves regret $\leq \ln N$ with a constant $\eta = 1$) performs in practice *worse than the uniform average*. The explanation is precisely that its bound concerns a *static* comparator, while the panel is not static: Pinnacle's coverage collapses towards the end of the period, and an algorithm that concentrates quickly on a few experts is punished exactly by an expert that disappears.

Chapter 6

Conclusions and future work

This thesis treated the betting companies that quote a football match as experts, and learned online how to combine their forecasts with three Online Convex Optimization algorithms adapted to experts that do not quote every match. Two further online stages calibrate the combined forecast and size the stake of a bet. Over 76,584 matches from 22 leagues and ten seasons, every comparison was tested with a moving block bootstrap, and a deployment test with a real date cut checked what a model trained up to the end of 2024 would have done in 2025–26. This chapter summarises the findings, interprets them, states the limits of the method and proposes future work.

6.1 Conclusions

The main conclusions of the present work are the following.

1. **The OCO framework applies successfully to betting markets**. An online mixture of published odds significantly beats the uniform average and the overwhelming majority of individual companies, with no assumption about the distribution of the data.

2. **The mixture reaches but does not exceed the market's leading edge**. In the configuration selected, the difference from the sharpest bookmaker ceases to be statistically significant. In the deployment test it significantly beats every opening price and ties with the bookmakers' closing prices. The limit is set by Betfair Exchange, which is not a bookmaker but an exchange, that is a market price without an operator's margin: that one significantly beats us.

3. **A substantial part of the apparent superiority is not learning**. Under the Sleeping Experts reduction the weight vector moves even with a zero step, because the projection is redone onto a simplex of different dimension every time the awake set changes. With an explicit zero-step control, 41% of the margin over the uniform average is attributed to this geometry and not to the algorithm. Every future comparison against the uniform average ought to carry the same control.

4. **Panel heterogeneity is a precondition, not a detail**. Learned weighting pays off only when the experts differ substantially; on a homogeneous set the plain average is at least as robust. The finding is independently confirmed at three points, while

at a fourth (the closing panel without Pinnacle) the two end up indistinguishable (§5.7.2).

5. **Worst-case theoretical steps are not optimal in practice**, and they in fact miss in opposite directions per algorithm. They protect against an adversary that betting odds are not. The only exception is the third OCO stage, where the algorithm that exploits the structure of the loss accepts the theoretical constant uncorrected.

6. **Better prediction does not mean profit**. The quality difference gained falls short of the market's margin, and whatever edge exists concerns specific companies, not a general strategy.

7. **In the third OCO stage, *what* is learned decides the outcome**. Learning the stake directly fails systematically, because the comparator "best constant stake" does not contain the closed form and because it throws away the probability the two preceding stages have just produced. Learning the *multiplier* on top of the closed form ties with the fixed coefficient on the closing targets and significantly beats it on the opening ones, by 8 wins to 2 losses, learning to stake more against some opponents and less against others without ever being told who the opponent is. It is also the only place in the thesis where theory's a-priori constant came out right without correction, precisely because the algorithm that uses it exploits the structure of the loss.

8. **Methodological discipline in comparisons is decisive**. Several intermediate "findings" turned out to be artefacts of comparing over different sets of matches, or of selection on a flat grid. The requirement of common rounds and the reporting of a held-out tail is what caught them. The sharpest instance concerns the step multipliers themselves: read off the held-out tail, two of the three appear mistuned, and the thesis carried that caveat for a time; read off the chronological prefix, which is what the protocol specifies, all three are already at their optimum. Selecting on the tail is selecting on the test set, and here the two criteria disagree.

6.2 The odds as a social relation

It is easy to read the weights as abstract parameters. Each expert here is a business, with its own costs, its own risk tolerance and its own business model, and the differences the algorithm learns are differences between those. Pinnacle is ahead not because it necessarily has a better prediction model but because it operates on a low margin and does not restrict winning players, so that its price is continuously corrected by informed bettors instead of their being shut out; in the 2025–26 window the best forecaster is not even a bookmaker but Betfair Exchange, where the price carries no operator margin; and over ten seasons the market did *not* become more accurate (no series shows a significant trend in log-loss or ECE, p from 0.21 to 1.00) but it did become more expensive, the overround rising from 4.42% to 6.00% ($p = 0.0007$).

The odds appear as a property of the match. De-vigging institutionalises the impression: you strip the margin in order to "recover" the true probability, as though beneath the commercial form there were a determinate number that the house distorts and you restore. There is none. Remove Pinnacle from the panel and the "probability" changes,

because what was being measured was not knowledge about football but position in the market. The technique itself confesses as much: Shin's parameter z (§3.2.2) **is** the estimated proportion of informed bettors, that is a social relation, and it enters the method as a *nuisance parameter* to be removed. This is what Marx [32] called the fetishism of the commodity: a definite social relation between people assumes the form of a property of a thing. Here the relation between the house, the informed and everyone else assumes the form of a number attached to a football match, and the fetishism operates *inside* the formalism rather than outside it.

It does not follow that there is nothing to be known. The uncertainty term of Murphy's decomposition is improved by no method and no amount of capital, and the closing price beats the opening price systematically. Something resists, and that something is the match, which occurs whatever anyone prices it at. The precise statement is therefore not that the odds lack objectivity but that **they are not objective about what they appear to be about**: there is no probability of the match waiting to be read off; what there is, is an objective expression of a relation, and it is measured, compared and ranked reproducibly. The fetish is not that the number is false but that we misattribute what it is a number of. If appearance and essence coincided directly, Marx [33] observed, all science would be superfluous; here the distance between them is itself a measurement.

Who pays, and why the market got dearer without getting better. The microstructure model of §3.2.2 already supplies the mechanism: the market maker charges a margin sufficient for the gains from the uninformed to cover the losses to the informed. But "informed" is not a mental state; it is a position with respect to the **material means**: data feeds, capital enough to move a line, and accounts that are not closed when you win. That same model makes the margin an increasing function of the proportion of informed traders, which appears at first to conflict with the measurement, since a market increasingly exposed to informed money should also have become more accurate. The two reconcile once it is asked *how* an operator may respond to informed money. It can absorb it, letting the price be corrected and growing more accurate, which is Pinnacle's model and the reason Pinnacle is hardest to beat. Or it can exclude it and raise the price instead, which leaves accuracy exactly where it was and the overround higher. A decade of rising margins with flat accuracy is the signature of the second response, and it is why better prediction does not convert into profit: the quality gained is measured against a toll that moves against it.

Profit and loss is the house's category, not ours. For the house no question of truth arises, only of the balance sheet, and the mechanism that secures it is not better forecasting but control of access. The thesis measures both sides of this. The significant edge appears against two targets only, WHC ($38.52\times$, $p < 0.001$) and IW ($7.89\times$, $p = 0.008$), both recreational-facing, while against the one house that does not exclude winners the simulation *loses* significantly ($0.0002\times$, $p < 0.001$). The edge exists exactly where it cannot be exercised and is absent exactly where it could be, and the reason is the same on both sides: the sharp house has the right price *because* it lets informed money correct it, the soft houses have the wrong one *because* they drive that money away. What separates the houses is not how well they predict but whom they let bet and at what price, which is Levitt's (2004) [30] observation that betting markets are organised around the bettor rather than around the price. The odds, read this way, are not a statement about the match but **an offer, revocable against whoever accepts it too well**. Account restrictions are not

measured here, since the data are prices and not customer outcomes; what is measured is the concentration of the edge at the soft houses and its absence at the sharp one. The revocation itself has been documented elsewhere: Kaunitz et al. (2017) [26] bet a profitable odds-based strategy with real money for five months and report that the response was not a correction of the prices but stake limits and account closures directed at them as clients. That is the complement of our measurement rather than a repetition of it, and the two together close the argument: the edge concentrates where access is controlled, and it is withdrawn when exercised.

No teleology, and the path is not a straight line. Every regret bound in the thesis assumes the opposite: comparison against the best **fixed** weight vector, which is the same presupposition as de-vigging, a determinate property where there is a relation in motion. The refutation is measured. The Aggregating Algorithm, holding the strongest bound in the thesis and one independent of the horizon, performs worse than the uniform average, because Pinnacle's coverage collapses and whoever concentrates quickly is punished by an expert that disappears; the schedule designed for convergence loses significantly to the non-decaying one for FTRL ($p = 0.001$); and the weights move more in the final quarter than in the middle ones, at a point where the step is orders of magnitude smaller. Nor does the thesis's own knowledge advance linearly: nine ideas were tried and rejected, findings reverse when the window changes, and a caveat about the step multipliers proved unfounded as soon as the selection criterion was corrected (§5.1). The path moves in cycles and settles into no system; the tree of knowledge, in Bitsakis's (2006) [2] image, stays green precisely because it is never finished. Characteristically, Fixed-Share, the explicit attempt to replace the fixed comparator with a sequence, does not help, because the movement **was already there** inside the step. The formalism was tracking motion from the outset while declaring a static aim: the fetish was not in the equations but in what we took them to be measuring.

That is what is at stake practically. A strictly positivist protocol would have delivered three wrong conclusions: that the entire margin over the uniform average is a product of learning, when 41% of it is not (§5.5.1); that variety among the experts suffices, when on a homogeneous panel the learned weighting loses in 8 tests out of 8 (§5.7.2); and that the market's hierarchy is a matter of technical capability, when it is a matter of margin, of access, and of who is permitted to win. All three were avoided by the same move: refusing to identify what is measured with what is real, and looking for the explanation in the relations that produce the measurement.

6.3 Limitations

The following limitations are stated explicitly.

Window choice is a form of selection. Three time windows were examined and the result improved in our favour with every narrowing. This is exactly the pattern that produces spurious findings, and for that reason all three are reported, the less favourable alongside the favourable.

Statistical power in the narrow windows. The deployment test rests on 8,151 matches against 71,818 for the full comparison. The confidence intervals are correspondingly wider, so a "tie" partly means "less data" and not only "equal quality".

The merging of identities is an inference. The VC/BetVictor merge rests on non-overlapping temporal presence, not on an official source. There may well be comparable rebrands that were not identified.

The simulation ignores liquidity. Account limits, stake restrictions and available liquidity are not modelled: precisely the mechanisms by which companies restrict winning players in practice. The simulation shows the *statistical existence* of an edge, not viable profitability.

The causality audit is not exhaustive. One leakage channel was found and fixed in the leave-one-out design, but there may be others at less obvious points.

The causal interpretations are plausible, not proved. The explanation, for example, of the margin increase as a commercial decision is consistent with the data but is not directly tested by it.

6.4 Future work

A live odds feed. Everything in this thesis is computed on a historical file, which means every result is a statement about what the method *would* have done. The framework itself does not require that: the three stages are online and causal, each round costs $O(N \log N)$, and the state is a single weight vector plus two scalars, so the whole chain runs in real time against a streaming odds API. What such an implementation would add is not speed but the two things a file cannot supply. First, the prices would be the ones actually available for staking rather than the ones recorded afterwards, which is precisely the gap §6.3 identifies between a statistical and an exploitable edge: account limits, stake caps and liquidity become observable instead of assumed away. Second, the opening/closing distinction, which this thesis can only treat as two static snapshots, becomes the continuous object it really is: the measurement of §5.2.3, that the closing advantage is concentrated in the top quintile of line movement, suggests the interesting quantity is not the final price but the trajectory, and only a live feed records it.

A second sport. Extending to basketball or tennis would show whether the framework generalises beyond football. The cost is mainly a new data pipeline, since the algorithms do not change. A two-outcome sport, where the structure of the problem simplifies, is of particular interest.

Bayesian Model Averaging as a baseline. A full BMA baseline would give an empirical point of comparison against the online algorithms, and would further illuminate the failure of the Aggregating Algorithm.

Appendix A

Full result tables

This appendix gives in full the tables that the chapters of the main text present in abridged form. It is generated automatically from the result files, so that it cannot drift away from them.

A.1 Ranking and statistical significance

Table A.1: Full Tier-A ranking, ten seasons.

Rounds	log-loss (cal.)	Brier (raw)	Brier (cal.)	raw_rps	calibrated_rps	Acc. (raw)	Acc.
71,818	0.99685	0.59592	0.59577	0.20263	0.20256	0.50611	0.50611
76,584	0.99693	0.59612	0.59586	0.20270	0.20257	0.50662	0.50662
76,584	0.99773	0.59674	0.59637	0.20297	0.20278	0.50649	0.50649
76,584	0.99774	0.59674	0.59638	0.20297	0.20279	0.50653	0.50653
76,584	0.99813	0.59703	0.59665	0.20311	0.20291	0.50593	0.50593
53,267	0.99910	0.59732	0.59711	0.20314	0.20301	0.50705	0.50705
49,755	0.99919	0.59769	0.59728	0.20316	0.20293	0.50624	0.50624
68,397	1.0003	0.59833	0.59806	0.20372	0.20359	0.50355	0.50355
65,470	1.0005	0.59860	0.59818	0.20387	0.20368	0.50309	0.50309
71,284	1.0005	0.59838	0.59824	0.20376	0.20369	0.50309	0.50309
57,185	1.0006	0.59899	0.59826	0.20412	0.20379	0.50214	0.50214
72,752	1.0007	0.59873	0.59833	0.20381	0.20360	0.50355	0.50355
76,438	1.0007	0.59852	0.59832	0.20373	0.20362	0.50377	0.50377

Table A.2: Every series in the panel, with no coverage threshold.

n_seasons_active	single_season	hedge_weight_last_awake	Rounds	log-loss	Brier	RPS	Acc.
10	no	0.00000	71,818	0.99714	0.59592	0.20263	0.50611
1	yes	0.16631	7,680	0.99730	0.59594	0.20251	0.51029
10	no	—	76,584	0.99745	0.59612	0.20270	0.50662
2	no	0.04504	14,885	0.99783	0.59656	0.20253	0.50890
1	yes	0.16620	7,681	0.99786	0.59605	0.20253	0.50814
3	no	0.00005	20,972	0.99797	0.59682	0.20303	0.50644
1	yes	0.14584	5,587	0.99819	0.59667	0.20317	0.50743
10	no	—	76,584	0.99842	0.59674	0.20297	0.50649
10	no	—	76,584	0.99843	0.59674	0.20297	0.50653
10	no	—	76,584	0.99886	0.59703	0.20311	0.50593
1	yes	0.15156	5,257	0.99906	0.59732	0.20323	0.50409
7	no	0.00009	53,267	0.99953	0.59732	0.20314	0.50705
1	yes	0.11580	7,508	0.99955	0.59776	0.20303	0.50759
6	no	0.00024	45,424	0.99973	0.59757	0.20326	0.50590
7	no	0.00014	49,755	0.99998	0.59769	0.20316	0.50624
6	no	0.00043	42,366	1.0002	0.59787	0.20349	0.50555

(continued)

n_seasons_active	single_season	hedge_weight_last_awake	Rounds	log-loss	Brier	RPS	Acc.
1	yes	0.16597	7,673	1.0004	0.59809	0.20354	0.50541
9	no	0.00000	68,397	1.0007	0.59833	0.20372	0.50355
10	no	0.00000	71,284	1.0008	0.59838	0.20376	0.50309
1	yes	0.11542	7,524	1.0009	0.59874	0.20326	0.50704
5	no	0.18974	34,038	1.0009	0.59820	0.20376	0.50385
10	no	0.00000	76,438	1.0011	0.59852	0.20373	0.50377
9	no	0.00000	65,470	1.0012	0.59860	0.20387	0.50309
10	no	0.00000	72,752	1.0014	0.59873	0.20381	0.50355
8	no	0.00001	57,185	1.0019	0.59899	0.20412	0.50214
1	yes	0.11836	7,309	1.0021	0.59938	0.20381	0.50581
1	yes	0.16922	7,505	1.0027	0.59958	0.20402	0.50393
1	yes	0.11393	7,619	1.0027	0.59991	0.20390	0.50532
1	yes	0.14862	5,393	1.0044	0.60099	0.20444	0.50028
2	no	0.04495	14,763	1.0072	0.60309	0.20526	0.49604

Table A.3: The full battery of significance tests.

Comparison	n	Mean diff.	CI low	CI high	p
OGD vs PSC (raw)	71,818	0.00065	0.00044	0.00086	0.00000
OGD vs PSC (calibrated)	71,818	0.00043	0.00020	0.00064	0.00000
OGD vs B365C (raw)	53,267	-0.00063	-0.00093	-0.00034	0.00000
OGD vs Hedge (raw)	76,584	-0.00097	-0.00118	-0.00075	0.00000
OGD vs FTRL (raw)	76,584	-0.00098	-0.00120	-0.00077	0.00000
OGD vs Uniform average (raw)	76,584	-0.00141	-0.00167	-0.00115	0.00000
Hedge vs Uniform average (raw)	76,584	-0.00044	-0.00062	-0.00026	0.00000
FTRL vs Uniform average (raw)	76,584	-0.00043	-0.00061	-0.00025	0.00000
Hedge: raw vs calibrated	76,584	0.00069	0.00040	0.00096	0.00000
OGD: raw vs calibrated	76,584	0.00053	0.00027	0.00078	0.00000
FTRL: raw vs calibrated	76,584	0.00069	0.00040	0.00096	0.00000
Uniform average: raw vs calibrated	76,584	0.00073	0.00044	0.00101	0.00000

A.2 Panel and de-vig method

Table A.4: Panel × algorithm — proportional normalisation.

Panel	Algorithm	n_bookmakers	Rounds	log-loss	Brier	RPS	Acc.	De-vig
closing	FTRL	13	76,521	0.99761	0.59621	0.20272	0.50658	Basic
full	FTRL	26	76,584	0.99843	0.59674	0.20297	0.50653	Basic
opening	FTRL	13	76,557	1.0009	0.59842	0.20375	0.50394	Basic
closing	Hedge	13	76,521	0.99762	0.59622	0.20272	0.50659	Basic
full	Hedge	26	76,584	0.99842	0.59674	0.20297	0.50649	Basic
opening	Hedge	13	76,557	1.0009	0.59842	0.20375	0.50402	Basic
closing	OGD	13	76,521	0.99722	0.59598	0.20263	0.50648	Basic
full	OGD	26	76,584	0.99745	0.59612	0.20270	0.50662	Basic
opening	OGD	13	76,557	1.0008	0.59832	0.20370	0.50391	Basic
closing	Uniform average	13	76,521	0.99748	0.59612	0.20269	0.50672	Basic
full	Uniform average	26	76,584	0.99886	0.59703	0.20311	0.50593	Basic
opening	Uniform average	13	76,557	1.0008	0.59836	0.20373	0.50398	Basic

Table A.5: Panel × algorithm — Shin normalisation.

Panel	Algorithm	n_bookmakers	Rounds	log-loss	Brier	RPS	Acc.	De-vig
closing	FTRL	13	76,521	0.99714	0.59596	0.20260	0.50659	Shin
full	FTRL	26	76,584	0.99782	0.59642	0.20282	0.50646	Shin
opening	FTRL	13	76,557	1.0003	0.59812	0.20361	0.50385	Shin
closing	Hedge	13	76,521	0.99714	0.59596	0.20260	0.50659	Shin
full	Hedge	26	76,584	0.99781	0.59641	0.20282	0.50645	Shin
opening	Hedge	13	76,557	1.0003	0.59812	0.20361	0.50387	Shin
closing	OGD	13	76,521	0.99689	0.59581	0.20255	0.50658	Shin
full	OGD	26	76,584	0.99709	0.59595	0.20262	0.50682	Shin
opening	OGD	13	76,557	1.0002	0.59804	0.20357	0.50389	Shin
closing	Uniform average	13	76,521	0.99703	0.59589	0.20258	0.50674	Shin
full	Uniform average	26	76,584	0.99824	0.59670	0.20295	0.50598	Shin
opening	Uniform average	13	76,557	1.0002	0.59805	0.20358	0.50393	Shin

Table A.6: Each algorithm against the plain average, proportional.

Panel	Algorithm	Metric	lower_is_better	n	De-vig	Mean diff.	CI low	CI high	p
opening	Hedge	log_loss	yes	76,557	Basic	0.00008	0.00001	0.00014	0.01900
opening	Hedge	brier	yes	76,557	Basic	0.00006	0.00002	0.00009	0.00000
opening	Hedge	accuracy	no	76,557	Basic	0.00004	-0.00021	0.00027	0.81700
opening	OGD	log_loss	yes	76,557	Basic	-0.00004	-0.00013	0.00006	0.43300
opening	OGD	brier	yes	76,557	Basic	-0.00004	-0.00010	0.00001	0.14000
opening	OGD	accuracy	no	76,557	Basic	-0.00007	-0.00043	0.00033	0.75900
opening	FTRL	log_loss	yes	76,557	Basic	0.00008	0.00002	0.00014	0.01500
opening	FTRL	brier	yes	76,557	Basic	0.00006	0.00002	0.00010	0.00200
opening	FTRL	accuracy	no	76,557	Basic	-0.00004	-0.00027	0.00020	0.78200
closing	Hedge	log_loss	yes	76,521	Basic	0.00014	0.00010	0.00018	0.00000
closing	Hedge	brier	yes	76,521	Basic	0.00009	0.00007	0.00012	0.00000
closing	Hedge	accuracy	no	76,521	Basic	-0.00013	-0.00046	0.00018	0.42300
closing	OGD	log_loss	yes	76,521	Basic	-0.00026	-0.00036	-0.00016	0.00000
closing	OGD	brier	yes	76,521	Basic	-0.00014	-0.00021	-0.00008	0.00000
closing	OGD	accuracy	no	76,521	Basic	-0.00025	-0.00071	0.00022	0.29700
closing	FTRL	log_loss	yes	76,521	Basic	0.00013	0.00009	0.00018	0.00000
closing	FTRL	brier	yes	76,521	Basic	0.00009	0.00006	0.00012	0.00000
closing	FTRL	accuracy	no	76,521	Basic	-0.00014	-0.00046	0.00017	0.36700
full	Hedge	log_loss	yes	76,584	Basic	-0.00044	-0.00062	-0.00026	0.00000
full	Hedge	brier	yes	76,584	Basic	-0.00030	-0.00042	-0.00017	0.00000
full	Hedge	accuracy	no	76,584	Basic	0.00056	-0.00014	0.00125	0.12300
full	OGD	log_loss	yes	76,584	Basic	-0.00141	-0.00167	-0.00115	0.00000
full	OGD	brier	yes	76,584	Basic	-0.00091	-0.00108	-0.00074	0.00000
full	OGD	accuracy	no	76,584	Basic	0.00069	-0.00025	0.00162	0.15600
full	FTRL	log_loss	yes	76,584	Basic	-0.00043	-0.00061	-0.00025	0.00000
full	FTRL	brier	yes	76,584	Basic	-0.00029	-0.00041	-0.00017	0.00000
full	FTRL	accuracy	no	76,584	Basic	0.00060	-0.00010	0.00131	0.10800

Table A.7: Each algorithm against the plain average, Shin.

Panel	Algorithm	Metric	lower_is_better	n	De-vig	Mean diff.	CI low	CI high	p
opening	Hedge	log_loss	yes	76,557	Shin	0.00011	0.00004	0.00018	0.00000
opening	Hedge	brier	yes	76,557	Shin	0.00008	0.00003	0.00012	0.00000
opening	Hedge	accuracy	no	76,557	Shin	-0.00005	-0.00029	0.00018	0.69700
opening	OGD	log_loss	yes	76,557	Shin	0.00002	-0.00008	0.00013	0.73600
opening	OGD	brier	yes	76,557	Shin	-0.00001	-0.00006	0.00005	0.82600
opening	OGD	accuracy	no	76,557	Shin	-0.00004	-0.00042	0.00035	0.89600
opening	FTRL	log_loss	yes	76,557	Shin	0.00011	0.00004	0.00019	0.00000
opening	FTRL	brier	yes	76,557	Shin	0.00008	0.00003	0.00012	0.00000
opening	FTRL	accuracy	no	76,557	Shin	-0.00008	-0.00033	0.00017	0.55900
closing	Hedge	log_loss	yes	76,521	Shin	0.00011	0.00007	0.00016	0.00000
closing	Hedge	brier	yes	76,521	Shin	0.00008	0.00005	0.00010	0.00000
closing	Hedge	accuracy	no	76,521	Shin	-0.00014	-0.00046	0.00017	0.37900
closing	OGD	log_loss	yes	76,521	Shin	-0.00014	-0.00023	-0.00004	0.00500

(continued)

Panel	Algorithm	Metric	lower_is_better	n	De-vig	Mean diff.	CI low	CI high	p
closing	OGD	brier	yes	76,521	Shin	-0.00007	-0.00013	-0.00001	0.01600
closing	OGD	accuracy	no	76,521	Shin	-0.00016	-0.00060	0.00029	0.51300
closing	FTRL	log_loss	yes	76,521	Shin	0.00011	0.00006	0.00015	0.00000
closing	FTRL	brier	yes	76,521	Shin	0.00007	0.00004	0.00010	0.00000
closing	FTRL	accuracy	no	76,521	Shin	-0.00014	-0.00046	0.00017	0.37900
full	Hedge	log_loss	yes	76,584	Shin	-0.00043	-0.00062	-0.00023	0.00000
full	Hedge	brier	yes	76,584	Shin	-0.00029	-0.00042	-0.00016	0.00000
full	Hedge	accuracy	no	76,584	Shin	0.00047	-0.00025	0.00116	0.21900
full	OGD	log_loss	yes	76,584	Shin	-0.00115	-0.00140	-0.00090	0.00000
full	OGD	brier	yes	76,584	Shin	-0.00075	-0.00092	-0.00059	0.00000
full	OGD	accuracy	no	76,584	Shin	0.00084	-0.00010	0.00175	0.07900
full	FTRL	log_loss	yes	76,584	Shin	-0.00042	-0.00061	-0.00023	0.00000
full	FTRL	brier	yes	76,584	Shin	-0.00028	-0.00041	-0.00015	0.00000
full	FTRL	accuracy	no	76,584	Shin	0.00048	-0.00024	0.00118	0.20300

A.3 Configurations

Table A.8: Stacking of configurations.

Config.	Stage	full_log_loss	train	tail
baseline (full 26, basic)	raw	0.99779	0.99860	0.99536
baseline (full 26, basic)	calibrated	0.99728	0.99801	0.99508
closing only, basic	raw	0.99733	0.99801	0.99528
closing only, basic	calibrated	0.99693	0.99756	0.99506
full 26, Shin	raw	0.99745	0.99822	0.99512
full 26, Shin	calibrated	0.99731	0.99804	0.99512
A: closing + Shin	raw	0.99702	0.99767	0.99506
A: closing + Shin	calibrated	0.99692	0.99754	0.99507
B: A minus single-season	raw	0.99702	0.99767	0.99509
B: A minus single-season	calibrated	0.99692	0.99754	0.99509
PSC alone	raw	0.99714	0.99776	0.99526
PSC alone	calibrated	0.99685	0.99746	0.99502

Table A.9: Paired tests of configurations.

Stage	Against	n	Mean diff.	CI low	CI high	p
raw	PSC alone	71,818	0.00065	0.00044	0.00086	0.00000
calibrated	PSC alone	71,818	0.00043	0.00020	0.00064	0.00000
raw	PSC alone	71,818	0.00019	0.00009	0.00029	0.00000
raw	baseline (full 26, basic)	71,818	-0.00046	-0.00064	-0.00029	0.00000
calibrated	PSC alone	71,818	0.00009	-0.00003	0.00020	0.14300
calibrated	baseline (full 26, basic)	71,818	-0.00034	-0.00053	-0.00016	0.00000
raw	PSC alone	71,818	0.00031	0.00008	0.00054	0.01100
raw	baseline (full 26, basic)	71,818	-0.00034	-0.00046	-0.00022	0.00000
calibrated	PSC alone	71,818	0.00046	0.00021	0.00069	0.00000
calibrated	baseline (full 26, basic)	71,818	0.00004	-0.00002	0.00009	0.18100
raw	PSC alone	71,818	-0.00012	-0.00025	0.00000	0.05700
raw	baseline (full 26, basic)	71,818	-0.00077	-0.00100	-0.00056	0.00000
calibrated	PSC alone	71,818	0.00007	-0.00004	0.00018	0.23400

(continued)

Stage	Against	n	Mean diff.	CI low	CI high	p
calibrated	baseline (full 26, basic)	71,818	-0.00036	-0.00056	-0.00016	0.00000
raw	PSC alone	71,818	-0.00011	-0.00024	0.00001	0.07200
raw	baseline (full 26, basic)	71,818	-0.00077	-0.00099	-0.00055	0.00000
calibrated	PSC alone	71,818	0.00008	-0.00003	0.00018	0.18100
calibrated	baseline (full 26, basic)	71,818	-0.00035	-0.00055	-0.00015	0.00000

A.4 Without Pinnacle

Table A.10: Ranking without Pinnacle, three panels.

rank	Series	Type	Coverage %	log-loss (raw)	log-loss (cal.)	train	tail
1	FTRL	algorithm	100.0000	0.99854	0.99771	1.0011	0.98747
2	Hedge	algorithm	100.0000	0.99854	0.99771	1.0011	0.98747
3	OGD	algorithm	100.0000	0.99858	0.99782	1.0013	0.98735
4	Uniform average	algorithm	100.0000	0.99927	0.99837	1.0016	0.98862
5	VC_BV	bookmaker	89.3098	1.0004	0.99987	1.0028	0.99113
6	WH	bookmaker	85.4878	1.0012	1.0003	1.0031	0.99191
7	BW	bookmaker	94.9963	1.0011	1.0004	1.0032	0.99187
8	B365	bookmaker	99.8094	1.0008	1.0004	1.0033	0.99173
9	IW	bookmaker	74.6696	1.0017	1.0005	1.0035	0.99146
1	OGD	algorithm	69.5916	0.99997	0.99919	1.0041	0.98459
2	Uniform average	algorithm	69.5916	1.0000	0.99921	1.0041	0.98462
3	FTRL	algorithm	69.5916	1.0000	0.99921	1.0041	0.98463
4	Hedge	algorithm	69.5916	1.0000	0.99921	1.0041	0.98463
5	VC_BVC	bookmaker	59.3127	0.99981	0.99929	1.0043	0.98426
6	B365C	bookmaker	69.5537	1.0001	0.99955	1.0044	0.98493
7	IWC	bookmaker	44.4453	1.0007	0.99957	1.0042	0.98558
8	BWC	bookmaker	64.9679	1.0005	0.99968	1.0047	0.98463
9	WHC	bookmaker	55.3196	1.0008	1.0002	1.0050	0.98579
1	VC_BV	bookmaker	89.3098	1.0004	0.99987	1.0028	0.99113
2	Uniform average	algorithm	99.9647	1.0007	0.99990	1.0028	0.99130
3	FTRL	algorithm	99.9647	1.0007	0.99991	1.0028	0.99131
4	Hedge	algorithm	99.9647	1.0007	0.99991	1.0028	0.99131
5	OGD	algorithm	99.9647	1.0007	0.99999	1.0029	0.99136
6	WH	bookmaker	85.4878	1.0012	1.0003	1.0031	0.99191
7	BW	bookmaker	94.9963	1.0011	1.0004	1.0032	0.99187
8	B365	bookmaker	99.8094	1.0008	1.0004	1.0033	0.99173
9	IW	bookmaker	74.6696	1.0017	1.0005	1.0035	0.99146

Table A.11: Paired tests without Pinnacle.

Bookmaker	n	Verdict	Mean diff.	CI low	CI high	p
IW	54,877	OGD better	-0.00315	-0.00366	-0.00262	0.00000
WH	54,877	OGD better	-0.00259	-0.00311	-0.00208	0.00000
BW	54,877	OGD better	-0.00255	-0.00309	-0.00205	0.00000
B365	54,877	OGD better	-0.00223	-0.00273	-0.00176	0.00000
VC_BV	54,877	OGD better	-0.00181	-0.00230	-0.00134	0.00000
WHC	32,210	OGD better	-0.00088	-0.00135	-0.00039	0.00000
IWC	32,210	OGD better	-0.00077	-0.00108	-0.00047	0.00000
BWC	32,210	OGD better	-0.00051	-0.00085	-0.00017	0.00700
B365C	32,210	tie (not significant)	-0.00011	-0.00040	0.00021	0.50500
VC_BVC	32,210	tie (not significant)	0.00016	-0.00015	0.00043	0.28400
IW	54,877	OGD better	-0.00105	-0.00130	-0.00076	0.00000
WH	54,877	OGD better	-0.00048	-0.00074	-0.00019	0.00200

(continued)

Bookmaker	n	Verdict	Mean diff.	CI low	CI high	p
BW	54,877	OGD better	-0.00045	-0.00070	-0.00020	0.00000
B365	54,877	tie (not significant)	-0.00012	-0.00033	0.00007	0.20300
VC_BV	54,877	VC_BV better	0.00030	0.00009	0.00050	0.00400

A.5 Recent window and deployment test

Table A.12: Ranking, five seasons.

Phase	Coverage %	Rounds	log-loss (raw)	log-loss (cal.)	Brier (raw)	Brier (cal.)	Acc.
closing	87.9119	34,065	0.99471	0.99439	0.59424	0.59410	0.50976
—	100.0000	38,749	0.99556	0.99495	0.59482	0.59452	0.51090
closing	79.6975	30,882	0.99554	0.99498	0.59483	0.59456	0.50955
closing	72.1361	27,952	0.99575	0.99502	0.59490	0.59450	0.51009
closing	91.9353	35,624	0.99609	0.99504	0.59513	0.59460	0.51039
closing	99.9974	38,748	0.99602	0.99543	0.59500	0.59471	0.51107
—	100.0000	38,749	0.99677	0.99591	0.59559	0.59516	0.50920
—	100.0000	38,749	0.99689	0.99604	0.59567	0.59524	0.50956
—	100.0000	38,749	0.99692	0.99607	0.59569	0.59526	0.50954
opening	71.9090	27,864	0.99848	0.99771	0.59677	0.59634	0.50660
opening	87.6952	33,981	0.99831	0.99794	0.59665	0.59649	0.50552
opening	79.0833	30,644	0.99870	0.99801	0.59696	0.59660	0.50702
opening	91.5327	35,468	0.99981	0.99885	0.59765	0.59716	0.50665
opening	99.8400	38,687	0.99951	0.99896	0.59736	0.59709	0.50660

Table A.13: Significance tests, five seasons.

Comparison	n	Mean diff.	CI low	CI high	p
OGD vs PSC (raw)	34,065	0.00065	0.00037	0.00097	0.00000
OGD vs PSC (calibrated)	34,065	0.00036	0.00008	0.00067	0.01400
OGD vs B365C (raw)	38,748	-0.00047	-0.00086	-0.00012	0.01400
OGD vs Hedge (raw)	38,749	-0.00133	-0.00166	-0.00102	0.00000
OGD vs FTRL (raw)	38,749	-0.00136	-0.00169	-0.00104	0.00000
OGD vs Uniform average (raw)	38,749	-0.00121	-0.00151	-0.00092	0.00000
Hedge vs Uniform average (raw)	38,749	0.00012	0.00004	0.00019	0.00600
FTRL vs Uniform average (raw)	38,749	0.00015	0.00006	0.00023	0.00200
Hedge: raw vs calibrated	38,749	0.00085	0.00036	0.00133	0.00100
OGD: raw vs calibrated	38,749	0.00061	0.00021	0.00100	0.00100
FTRL: raw vs calibrated	38,749	0.00086	0.00037	0.00134	0.00100
Uniform average: raw vs calibrated	38,749	0.00086	0.00036	0.00135	0.00100

Table A.14: The market in the tail, five seasons.

rank	Phase	Coverage %	log-loss	Bookmaker	n	Mean diff.	CI low	CI high	p
1.0000	—	100.0000	1.0001	—	—	—	—	—	—
2.0000	closing	85.0674	1.0002	—	—	—	—	—	—
3.0000	closing	77.7447	1.0003	—	—	—	—	—	—
4.0000	closing	75.5458	1.0003	—	—	—	—	—	—
5.0000	closing	100.0000	1.0007	—	—	—	—	—	—
6.0000	—	100.0000	1.0008	—	—	—	—	—	—
7.0000	—	100.0000	1.0009	—	—	—	—	—	—

(continued)

rank	Phase	Coverage %	log-loss	Bookmaker	n	Mean diff.	CI low	CI high	p
8.0000	—	100.0000	1.0009	—	—	—	—	—	—
9.0000	opening	75.2847	1.0026	—	—	—	—	—	—
10.0000	opening	99.8485	1.0030	—	—	—	—	—	—
11.0000	opening	84.7697	1.0032	—	—	—	—	—	—
12.0000	opening	77.1075	1.0081	—	—	—	—	—	—
—	opening	—	—	BFE	7,699	-0.00796	-0.01093	-0.00527	0.00000
—	opening	—	—	BW	7,699	-0.00306	-0.00479	-0.00141	0.00100
—	opening	—	—	B365	7,699	-0.00283	-0.00454	-0.00118	0.00100
—	opening	—	—	PS	7,699	-0.00244	-0.00406	-0.00094	0.00200
—	closing	—	—	B365C	7,699	-0.00057	-0.00146	0.00031	0.20500
—	closing	—	—	PSC	7,699	-0.00015	-0.00056	0.00029	0.50300
—	closing	—	—	BFEC	7,699	-0.00013	-0.00092	0.00074	0.79700
—	closing	—	—	BWC	7,699	-0.00009	-0.00082	0.00064	0.81500

Table A.15: Deployment test 2025–26.

rank	Series	Type	Coverage %	log-loss
1	BFEC	closing	96.1811	1.0004
2	OGD closing-only, online	αλγόριθμος	100.0000	1.0011
3	OGD πλήρες panel, online	αλγόριθμος	100.0000	1.0013
4	BWC	closing	75.5400	1.0015
5	OGD closing-only, παγωμένο	αλγόριθμος	100.0000	1.0015
6	B365C	closing	100.0000	1.0017
7	OGD πλήρες panel, παγωμένο	αλγόριθμος	100.0000	1.0018
8	B365	opening	99.7840	1.0037
9	BW	opening	75.1253	1.0042
10	BFE	opening	95.3171	1.0076

Table A.16: Paired tests, deployment 2025–26.

Bookmaker	Phase	n	Verdict	Mean diff.	CI low	CI high	p
B365	opening	8,151	αλγόριθμος	-0.00241	-0.00382	-0.00102	0.00100
B365C	closing	8,151	ισοπαλία	-0.00036	-0.00121	0.00054	0.39000
BFE	opening	8,151	αλγόριθμος	-0.00629	-0.00878	-0.00386	0.00000
BFEC	closing	8,151	BFEC	0.00089	0.00016	0.00166	0.01300
BW	opening	8,151	αλγόριθμος	-0.00287	-0.00431	-0.00141	0.00000
BWC	closing	8,151	ισοπαλία	-0.00017	-0.00087	0.00058	0.68500
PS	opening	6,857	αλγόριθμος	-0.00228	-0.00408	-0.00058	0.00700
PSC	closing	6,892	ισοπαλία	-0.00013	-0.00058	0.00030	0.56800
B365	opening	8,151	αλγόριθμος	-0.00188	-0.00313	-0.00066	0.00500
B365C	closing	8,151	ισοπαλία	0.00018	-0.00080	0.00122	0.75100
BFE	opening	8,151	αλγόριθμος	-0.00575	-0.00821	-0.00337	0.00000
BFEC	closing	8,151	BFEC	0.00143	0.00046	0.00238	0.00200
BW	opening	8,151	αλγόριθμος	-0.00233	-0.00358	-0.00112	0.00000
BWC	closing	8,151	ισοπαλία	0.00037	-0.00043	0.00123	0.34200
PS	opening	6,857	αλγόριθμος	-0.00219	-0.00407	-0.00042	0.01500
PSC	closing	6,892	ισοπαλία	-0.00004	-0.00038	0.00030	0.79600
B365	opening	8,151	αλγόριθμος	-0.00260	-0.00417	-0.00106	0.00200
B365C	closing	8,151	ισοπαλία	-0.00055	-0.00137	0.00030	0.19300
BFE	opening	8,151	αλγόριθμος	-0.00647	-0.00904	-0.00389	0.00000
BFEC	closing	8,151	ισοπαλία	0.00070	-0.00000	0.00138	0.05200

(continued)

Bookmaker	Phase	n	Verdict	Mean diff.	CI low	CI high	p
BW	opening	8,151	αλγόριθμος	-0.00306	-0.00466	-0.00140	0.00100
BWC	closing	8,151	ισοπαλία	-0.00036	-0.00107	0.00042	0.36100
PS	opening	6,857	αλγόριθμος	-0.00223	-0.00418	-0.00036	0.01900
PSC	closing	6,892	ισοπαλία	-0.00009	-0.00038	0.00021	0.53200
B365	opening	8,151	αλγόριθμος	-0.00220	-0.00374	-0.00068	0.00700
B365C	closing	8,151	ισοπαλία	-0.00014	-0.00097	0.00076	0.71400
BFE	opening	8,151	αλγόριθμος	-0.00607	-0.00860	-0.00353	0.00000
BFEC	closing	8,151	BFEC	0.00111	0.00028	0.00192	0.00300
BW	opening	8,151	αλγόριθμος	-0.00265	-0.00419	-0.00110	0.00000
BWC	closing	8,151	ισοπαλία	0.00005	-0.00061	0.00076	0.85000
PS	opening	6,857	αλγόριθμος	-0.00223	-0.00422	-0.00035	0.01900
PSC	closing	6,892	ισοπαλία	-0.00009	-0.00031	0.00013	0.44400

A.6 Value betting

Table A.17: Full sweep of training panel × target × normalisation.

De-vig	Train panel	Phase	Bookmaker	Stage	Bets	log-growth	CI low	CI high	p
Basic	closing-trained	closing	PSC	raw	20,580	-0.00012	-0.00019	-0.00005	0.00100
Basic	closing-trained	closing	PSC	calibrated	19,497	-0.00003	-0.00010	0.00004	0.36800
Basic	closing-trained	closing	B365C	raw	7,941	0.00000	-0.00012	0.00012	0.99100
Basic	closing-trained	closing	B365C	calibrated	7,643	0.00001	-0.00013	0.00014	0.96800
Basic	closing-trained	closing	BWC	raw	3,134	0.00022	-0.00004	0.00049	0.09100
Basic	closing-trained	closing	BWC	calibrated	6,920	0.00016	0.00001	0.00033	0.04200
Basic	closing-trained	closing	VC_BVC	raw	5,142	0.00000	-0.00015	0.00018	0.93600
Basic	closing-trained	closing	VC_BVC	calibrated	4,561	0.00010	-0.00010	0.00030	0.33900
Basic	closing-trained	closing	WHC	raw	6,633	0.00055	0.00026	0.00082	0.00000
Basic	closing-trained	closing	WHC	calibrated	7,253	0.00061	0.00028	0.00088	0.00000
Basic	mixed-trained	opening	B365	raw	34,022	0.00060	0.00046	0.00072	0.00000
Basic	mixed-trained	opening	B365	calibrated	35,894	0.00072	0.00057	0.00086	0.00000
Basic	mixed-trained	opening	BW	raw	24,313	0.00074	0.00060	0.00090	0.00000
Basic	mixed-trained	opening	BW	calibrated	29,845	0.00082	0.00066	0.00098	0.00000
Basic	mixed-trained	closing	PSC	raw	42,881	-0.00028	-0.00038	-0.00019	0.00000
Basic	mixed-trained	closing	PSC	calibrated	39,997	-0.00021	-0.00033	-0.00009	0.00000
Basic	mixed-trained	opening	PS	raw	45,926	0.00053	0.00042	0.00064	0.00000
Basic	mixed-trained	opening	PS	calibrated	48,919	0.00068	0.00056	0.00080	0.00000
Basic	mixed-trained	opening	VC_BV	raw	25,631	0.00061	0.00046	0.00075	0.00000
Basic	mixed-trained	opening	VC_BV	calibrated	29,112	0.00072	0.00057	0.00087	0.00000
Basic	mixed-trained	opening	WH	raw	22,036	0.00071	0.00055	0.00088	0.00000
Basic	mixed-trained	opening	WH	calibrated	27,493	0.00077	0.00061	0.00094	0.00000
Basic	mixed-trained	opening	IW	raw	16,562	0.00088	0.00070	0.00108	0.00000
Basic	mixed-trained	opening	IW	calibrated	21,808	0.00096	0.00076	0.00117	0.00000
Basic	mixed-trained	closing	B365C	raw	9,570	-0.00011	-0.00023	0.00002	0.09300
Basic	mixed-trained	closing	B365C	calibrated	7,970	-0.00007	-0.00020	0.00008	0.32800
Basic	mixed-trained	closing	BWC	raw	3,694	0.00008	-0.00014	0.00032	0.52000
Basic	mixed-trained	closing	BWC	calibrated	6,607	0.00009	-0.00006	0.00024	0.24700
Basic	mixed-trained	closing	VC_BVC	raw	6,521	-0.00012	-0.00025	0.00003	0.10600
Basic	mixed-trained	closing	VC_BVC	calibrated	4,653	-0.00006	-0.00025	0.00012	0.45500
Basic	mixed-trained	closing	WHC	raw	6,852	0.00044	0.00016	0.00068	0.00000
Basic	mixed-trained	closing	WHC	calibrated	6,565	0.00056	0.00027	0.00083	0.00000
Basic	opening-trained	opening	B365	raw	15,836	-0.00009	-0.00017	-0.00001	0.04100
Basic	opening-trained	opening	B365	calibrated	10,829	-0.00008	-0.00017	0.00001	0.10100
Basic	opening-trained	opening	BW	raw	4,611	-0.00008	-0.00021	0.00006	0.24800
Basic	opening-trained	opening	BW	calibrated	7,841	0.00001	-0.00010	0.00011	0.82000
Basic	opening-trained	closing	PSC	raw	57,078	-0.00049	-0.00061	-0.00036	0.00000
Basic	opening-trained	closing	PSC	calibrated	56,645	-0.00044	-0.00057	-0.00031	0.00000
Basic	opening-trained	opening	PS	raw	29,835	-0.00012	-0.00018	-0.00007	0.00000
Basic	opening-trained	opening	PS	calibrated	26,675	-0.00005	-0.00010	0.00000	0.05500
Basic	opening-trained	opening	VC_BV	raw	9,508	-0.00015	-0.00026	-0.00005	0.00700

(continued)

De-vig	Train panel	Phase	Bookmaker	Stage	Bets	log-growth	CI low	CI high	p
Basic	opening-trained	opening	VC_BV	calibrated	8,083	-0.00006	-0.00018	0.00005	0.24500
Basic	opening-trained	opening	WH	raw	5,793	-0.00007	-0.00024	0.00010	0.42500
Basic	opening-trained	opening	WH	calibrated	5,931	0.00005	-0.00012	0.00024	0.54500
Basic	opening-trained	opening	IW	raw	3,101	0.00042	0.00015	0.00067	0.00400
Basic	opening-trained	opening	IW	calibrated	7,443	0.00028	0.00006	0.00048	0.00800
Basic	opening-trained	closing	B365C	raw	27,138	-0.00066	-0.00084	-0.00049	0.00000
Basic	opening-trained	closing	B365C	calibrated	27,203	-0.00061	-0.00079	-0.00042	0.00000
Basic	opening-trained	closing	BWC	raw	19,475	-0.00075	-0.00094	-0.00058	0.00000
Basic	opening-trained	closing	BWC	calibrated	20,931	-0.00071	-0.00092	-0.00051	0.00000
Basic	opening-trained	closing	VC_BVC	raw	22,741	-0.00066	-0.00085	-0.00046	0.00000
Basic	opening-trained	closing	VC_BVC	calibrated	22,708	-0.00061	-0.00082	-0.00041	0.00000
Basic	opening-trained	closing	WHC	raw	16,306	-0.00067	-0.00087	-0.00045	0.00000
Basic	opening-trained	closing	WHC	calibrated	16,588	-0.00065	-0.00086	-0.00042	0.00000
Shin	closing-trained	closing	PSC	raw	13,136	0.00000	-0.00008	0.00008	0.94700
Shin	closing-trained	closing	PSC	calibrated	18,398	-0.00000	-0.00008	0.00007	0.97400
Shin	closing-trained	closing	B365C	raw	5,198	0.00007	-0.00008	0.00022	0.42000
Shin	closing-trained	closing	B365C	calibrated	7,374	0.00003	-0.00010	0.00016	0.68700
Shin	closing-trained	closing	BWC	raw	2,535	0.00039	0.00008	0.00074	0.01400
Shin	closing-trained	closing	BWC	calibrated	6,511	0.00018	0.00002	0.00036	0.02900
Shin	closing-trained	closing	VC_BVC	raw	2,900	0.00016	-0.00009	0.00042	0.20400
Shin	closing-trained	closing	VC_BVC	calibrated	4,097	0.00013	-0.00008	0.00036	0.24500
Shin	closing-trained	closing	WHC	raw	4,980	0.00083	0.00042	0.00117	0.00100
Shin	closing-trained	closing	WHC	calibrated	6,675	0.00065	0.00032	0.00095	0.00000
Shin	mixed-trained	opening	B365	raw	31,621	0.00068	0.00055	0.00081	0.00000
Shin	mixed-trained	opening	B365	calibrated	34,959	0.00070	0.00056	0.00084	0.00000
Shin	mixed-trained	opening	BW	raw	23,893	0.00083	0.00068	0.00099	0.00000
Shin	mixed-trained	opening	BW	calibrated	28,919	0.00083	0.00067	0.00099	0.00000
Shin	mixed-trained	closing	PSC	raw	35,750	-0.00024	-0.00035	-0.00013	0.00000
Shin	mixed-trained	closing	PSC	calibrated	39,367	-0.00021	-0.00032	-0.00009	0.00000
Shin	mixed-trained	opening	PS	raw	44,003	0.00062	0.00051	0.00073	0.00000
Shin	mixed-trained	opening	PS	calibrated	48,190	0.00066	0.00054	0.00078	0.00000
Shin	mixed-trained	opening	VC_BV	raw	23,908	0.00073	0.00057	0.00088	0.00000
Shin	mixed-trained	opening	VC_BV	calibrated	28,218	0.00072	0.00056	0.00086	0.00000
Shin	mixed-trained	opening	WH	raw	21,430	0.00080	0.00063	0.00098	0.00000
Shin	mixed-trained	opening	WH	calibrated	26,564	0.00075	0.00059	0.00093	0.00000
Shin	mixed-trained	opening	IW	raw	17,465	0.00094	0.00074	0.00112	0.00000
Shin	mixed-trained	opening	IW	calibrated	21,369	0.00096	0.00076	0.00114	0.00000
Shin	mixed-trained	closing	B365C	raw	6,413	-0.00008	-0.00023	0.00006	0.25300
Shin	mixed-trained	closing	B365C	calibrated	7,786	-0.00005	-0.00019	0.00009	0.46700
Shin	mixed-trained	closing	BWC	raw	2,740	0.00016	-0.00012	0.00046	0.28100
Shin	mixed-trained	closing	BWC	calibrated	6,132	0.00013	-0.00003	0.00031	0.12000
Shin	mixed-trained	closing	VC_BVC	raw	3,871	-0.00010	-0.00029	0.00010	0.30000
Shin	mixed-trained	closing	VC_BVC	calibrated	4,216	-0.00003	-0.00023	0.00016	0.66800
Shin	mixed-trained	closing	WHC	raw	4,774	0.00073	0.00037	0.00105	0.00100
Shin	mixed-trained	closing	WHC	calibrated	5,914	0.00059	0.00028	0.00089	0.00000
Shin	opening-trained	opening	B365	raw	8,796	-0.00008	-0.00017	0.00003	0.18800
Shin	opening-trained	opening	B365	calibrated	10,578	-0.00010	-0.00019	-0.00000	0.04900
Shin	opening-trained	opening	BW	raw	2,315	0.00006	-0.00011	0.00029	0.40000
Shin	opening-trained	opening	BW	calibrated	7,441	0.00004	-0.00007	0.00015	0.44200
Shin	opening-trained	closing	PSC	raw	55,765	-0.00046	-0.00058	-0.00033	0.00000
Shin	opening-trained	closing	PSC	calibrated	56,441	-0.00044	-0.00057	-0.00030	0.00000
Shin	opening-trained	opening	PS	raw	18,357	-0.00006	-0.00011	-0.00000	0.03800
Shin	opening-trained	opening	PS	calibrated	24,953	-0.00006	-0.00011	0.00000	0.05400
Shin	opening-trained	opening	VC_BV	raw	4,268	-0.00002	-0.00015	0.00012	0.72200
Shin	opening-trained	opening	VC_BV	calibrated	7,333	-0.00006	-0.00018	0.00005	0.26500
Shin	opening-trained	opening	WH	raw	2,996	0.00020	-0.00007	0.00051	0.15500
Shin	opening-trained	opening	WH	calibrated	6,418	0.00002	-0.00014	0.00021	0.73700
Shin	opening-trained	opening	IW	raw	3,579	0.00048	0.00020	0.00074	0.00000
Shin	opening-trained	opening	IW	calibrated	7,179	0.00032	0.00011	0.00052	0.00100
Shin	opening-trained	closing	B365C	raw	26,159	-0.00063	-0.00081	-0.00045	0.00000
Shin	opening-trained	closing	B365C	calibrated	27,097	-0.00061	-0.00081	-0.00042	0.00000
Shin	opening-trained	closing	BWC	raw	19,224	-0.00074	-0.00093	-0.00055	0.00000
Shin	opening-trained	closing	BWC	calibrated	20,970	-0.00071	-0.00091	-0.00051	0.00000
Shin	opening-trained	closing	VC_BVC	raw	21,857	-0.00064	-0.00084	-0.00043	0.00000
Shin	opening-trained	closing	VC_BVC	calibrated	22,573	-0.00061	-0.00081	-0.00041	0.00000
Shin	opening-trained	closing	WHC	raw	15,400	-0.00066	-0.00088	-0.00044	0.00000
Shin	opening-trained	closing	WHC	calibrated	16,333	-0.00064	-0.00086	-0.00041	0.00000

Table A.18: Sweep over all bookmakers.

Bookmaker	Stage	Bets	Bankroll	log-growth	CI low	CI high	p
B365	raw	15,836	0.24472	-0.00009	-0.00017	-0.00001	0.04100
B365	calibrated	10,829	0.41079	-0.00008	-0.00017	0.00001	0.10100
BW	raw	4,611	0.67602	-0.00008	-0.00021	0.00006	0.24800
BW	calibrated	7,841	1.0913	0.00001	-0.00010	0.00011	0.82000
PSC	raw	42,881	0.00000	-0.00028	-0.00038	-0.00019	0.00000
PSC	calibrated	39,997	0.00021	-0.00021	-0.00033	-0.00009	0.00000
PS	raw	29,835	0.02460	-0.00012	-0.00018	-0.00007	0.00000
PS	calibrated	26,675	0.26030	-0.00005	-0.00010	0.00000	0.05500
VC_BV	raw	9,508	0.23716	-0.00015	-0.00026	-0.00005	0.00700
VC_BV	calibrated	8,083	0.59737	-0.00006	-0.00018	0.00005	0.24500
WH	raw	5,793	0.68312	-0.00007	-0.00024	0.00010	0.42500
WH	calibrated	5,931	1.3757	0.00005	-0.00012	0.00024	0.54500
IW	raw	3,101	3.7046	0.00042	0.00015	0.00067	0.00400
IW	calibrated	7,443	7.8901	0.00028	0.00006	0.00048	0.00800
B365C	raw	9,570	0.36496	-0.00011	-0.00023	0.00002	0.09300
B365C	calibrated	7,970	0.58286	-0.00007	-0.00020	0.00008	0.32800
BWC	raw	3,694	1.3448	0.00008	-0.00014	0.00032	0.52000
BWC	calibrated	6,607	1.8050	0.00009	-0.00006	0.00024	0.24700
VC_BVC	raw	6,521	0.46013	-0.00012	-0.00025	0.00003	0.10600
VC_BVC	calibrated	4,653	0.75253	-0.00006	-0.00025	0.00012	0.45500
WHC	raw	6,852	19.7908	0.00044	0.00016	0.00068	0.00000
WHC	calibrated	6,565	38.5182	0.00056	0.00027	0.00083	0.00000

Table A.19: Value betting, five seasons.

Bookmaker	Phase	Stage	Bets	Bankroll	log-growth	CI low	CI high	p
B365	opening	raw	5,183	0.68039	-0.00007	-0.00023	0.00005	0.21600
B365	opening	calibrated	4,493	0.61559	-0.00011	-0.00028	0.00005	0.17800
B365C	closing	raw	4,621	0.87490	-0.00003	-0.00018	0.00015	0.73300
B365C	closing	calibrated	5,537	0.77807	-0.00005	-0.00021	0.00014	0.65000
B365C	closing	raw	6,305	0.22316	-0.00024	-0.00040	-0.00006	0.01500
B365C	closing	calibrated	7,362	0.32589	-0.00015	-0.00032	0.00002	0.08800
BW	opening	raw	1,108	0.85016	-0.00015	-0.00041	0.00020	0.44000
BW	opening	calibrated	4,546	1.1965	0.00004	-0.00011	0.00019	0.64500
BWC	closing	raw	1,377	2.0538	0.00052	0.00006	0.00099	0.01900
BWC	closing	calibrated	5,751	2.6509	0.00017	-0.00000	0.00036	0.05100
BWC	closing	raw	2,307	1.0231	0.00001	-0.00030	0.00037	0.93100
BWC	closing	calibrated	6,674	1.2275	0.00003	-0.00013	0.00020	0.69800
IW	opening	raw	562	1.0875	0.00015	-0.00028	0.00071	0.52200
IW	opening	calibrated	4,645	1.4991	0.00009	-0.00010	0.00027	0.33500
IWC	closing	raw	1,332	2.6726	0.00074	0.00025	0.00122	0.00200
IWC	closing	calibrated	5,092	8.5053	0.00042	0.00016	0.00068	0.00000
IWC	closing	raw	1,981	1.3847	0.00016	-0.00020	0.00056	0.42100
IWC	closing	calibrated	5,947	5.7760	0.00029	0.00006	0.00054	0.01000
PS	opening	raw	13,103	0.09545	-0.00018	-0.00026	-0.00009	0.00000
PS	opening	calibrated	12,643	0.20826	-0.00012	-0.00022	-0.00003	0.01100
PSC	closing	raw	14,512	0.15081	-0.00013	-0.00022	-0.00004	0.00400
PSC	closing	calibrated	15,306	0.61266	-0.00003	-0.00012	0.00006	0.49900
PSC	closing	raw	17,953	0.04992	-0.00017	-0.00026	-0.00007	0.00000
PSC	closing	calibrated	18,526	0.21303	-0.00008	-0.00019	0.00002	0.13400

(continued)

Bookmaker	Phase	Stage	Bets	Bankroll	log-growth	CI low	CI high	p
VC_BV	opening	raw	2,408	0.46501	-0.00032	-0.00049	-0.00014	0.00000
VC_BV	opening	calibrated	1,503	0.83650	-0.00012	-0.00041	0.00017	0.41800
VC_BVC	closing	raw	3,094	1.0480	0.00002	-0.00019	0.00023	0.90100
VC_BVC	closing	calibrated	3,283	1.0843	0.00002	-0.00022	0.00026	0.90500
VC_BVC	closing	raw	4,523	0.22559	-0.00033	-0.00052	-0.00012	0.00100
VC_BVC	closing	calibrated	4,722	0.34505	-0.00023	-0.00042	-0.00003	0.02000
WH	opening	raw	2,023	1.4058	0.00017	-0.00020	0.00056	0.35900
WH	opening	calibrated	2,817	1.6474	0.00018	-0.00012	0.00052	0.25600
WHC	closing	raw	4,327	17.0701	0.00066	0.00026	0.00101	0.00100
WHC	closing	calibrated	5,517	33.6750	0.00064	0.00029	0.00096	0.00000
WHC	closing	raw	4,352	4.7664	0.00036	0.00004	0.00064	0.02900
WHC	closing	calibrated	5,471	8.5626	0.00039	0.00008	0.00070	0.01800

A.7 Extensions and negative results

Table A.20: Fixed-Share by α.

α	Rounds	log-loss	Brier	RPS	Acc.
0.00000	76,584	0.99842	0.59674	0.20297	0.50649
0.00500	76,584	0.99885	0.59702	0.20310	0.50597
0.01000	76,584	0.99885	0.59702	0.20310	0.50595
0.02000	76,584	0.99885	0.59703	0.20311	0.50595
0.05000	76,584	0.99886	0.59703	0.20311	0.50594

Table A.21: Weights by expert and α (at each expert's last awake round).

Bookmaker	0.0	0.005	0.01	0.02	0.05
1XB	0.16922	0.09999	0.09999	0.09999	0.09999
1XBC	0.16620	0.10000	0.10000	0.10000	0.10000
BF	0.16597	0.09998	0.09999	0.09999	0.09999
BFC	0.16631	0.10000	0.10000	0.10000	0.10000
BFD	0.11393	0.06232	0.06244	0.06247	0.06248
BFDC	0.11542	0.06236	0.06246	0.06248	0.06249
BMGM	0.11836	0.06235	0.06245	0.06247	0.06248
BMGMC	0.11580	0.06234	0.06246	0.06249	0.06249
CL	0.14862	0.06293	0.06264	0.06258	0.06254
CLC	0.15156	0.06345	0.06278	0.06263	0.06258
LBC	0.14584	0.06243	0.06246	0.06248	0.06249

Table A.22: Per-league specialisation.

League	League	Matches	Unified	Specialist	Spec. wins	top3_trusted_bookmakers
SC0	Scotland Premiership	2,231	0.93590	0.93635	no	CLC, BFEC, BFD

(continued)

League	League	Matches	Unified	Specialist	Spec. wins	top3_trusted_bookmakers
I1	Italy Serie A	3,800	0.94636	0.94692	no	CL, BF, LBC
E0	England Premier League	3,800	0.94928	0.94978	no	BFEC, WHC, CLC
SP1	Spain La Liga	3,800	0.96476	0.96534	no	LBC, BFC, PSC
D1	Germany Bundesliga	3,060	0.97734	0.97823	no	BMGMC, LBC, 1XBC
F1	France Ligue 1	3,477	0.98084	0.98132	no	LBC, 1XBC, BFC

Table A.23: Performance by outcome class.

Series	Type	Outcome	n	Acc.	log-loss	Brier
Hedge	algorithm	1 (Home win)	33,280	0.82058	0.76764	0.43084
Hedge	algorithm	X (Draw)	20,080	0.00249	1.3149	0.83571
Hedge	algorithm	2 (Away win)	23,224	0.49216	1.0555	0.62784
OGD	algorithm	1 (Home win)	33,280	0.81830	0.76463	0.42884
OGD	algorithm	X (Draw)	20,080	0.00418	1.3134	0.83548
OGD	algorithm	2 (Away win)	23,224	0.49440	1.0579	0.62889
FTRL	algorithm	1 (Home win)	33,280	0.82061	0.76769	0.43088
FTRL	algorithm	X (Draw)	20,080	0.00249	1.3150	0.83570
FTRL	algorithm	2 (Away win)	23,224	0.49225	1.0554	0.62781
Uniform average	algorithm	1 (Home win)	33,280	0.82263	0.76722	0.43061
Uniform average	algorithm	X (Draw)	20,080	0.00159	1.3159	0.83603
Uniform average	algorithm	2 (Away win)	23,224	0.48816	1.0567	0.62887
B365	bookmaker	1 (Home win)	33,216	0.82557	0.76485	0.42878
B365	bookmaker	X (Draw)	20,050	0.00165	1.3235	0.84164
B365	bookmaker	2 (Away win)	23,172	0.47695	1.0609	0.63147
B365C	bookmaker	1 (Home win)	22,894	0.81729	0.76566	0.42918
B365C	bookmaker	X (Draw)	13,984	0.00279	1.3275	0.84407
B365C	bookmaker	2 (Away win)	16,389	0.50394	1.0464	0.62166
BW	bookmaker	1 (Home win)	31,644	0.82755	0.76947	0.43235
BW	bookmaker	X (Draw)	19,102	0.00162	1.3208	0.83778
BW	bookmaker	2 (Away win)	22,006	0.47333	1.0576	0.63048
BWC	bookmaker	1 (Home win)	21,384	0.81673	0.77242	0.43422
BWC	bookmaker	X (Draw)	13,106	0.00351	1.3192	0.83683
BWC	bookmaker	2 (Away win)	15,265	0.50292	1.0447	0.62136
IW	bookmaker	1 (Home win)	24,912	0.82671	0.78010	0.43964
IW	bookmaker	X (Draw)	14,973	0.00147	1.3050	0.82684
IW	bookmaker	2 (Away win)	17,300	0.46809	1.0589	0.63125
PS	bookmaker	1 (Home win)	30,943	0.81944	0.76503	0.42925
PS	bookmaker	X (Draw)	18,694	0.00332	1.3182	0.83832
PS	bookmaker	2 (Away win)	21,647	0.48247	1.0638	0.63295
PSC	bookmaker	1 (Home win)	31,186	0.81668	0.76231	0.42734
PSC	bookmaker	X (Draw)	18,837	0.00695	1.3127	0.83592
PSC	bookmaker	2 (Away win)	21,795	0.49314	1.0604	0.62971
VC_BV	bookmaker	1 (Home win)	29,698	0.82480	0.76751	0.43096
VC_BV	bookmaker	X (Draw)	17,933	0.00112	1.3178	0.83746
VC_BV	bookmaker	2 (Away win)	20,766	0.47799	1.0605	0.63120
WH	bookmaker	1 (Home win)	28,469	0.82528	0.77363	0.43519
WH	bookmaker	X (Draw)	17,150	0.00146	1.3108	0.83259
WH	bookmaker	2 (Away win)	19,851	0.47438	1.0600	0.63078

Table A.24: Calibration per season.

Series	Comparison	n	Mean diff.	CI low	CI high	p
Hedge	per-season vs raw	76,584	-0.00056	-0.00081	-0.00030	0.00000
Hedge	per-season vs global	76,584	0.00013	-0.00003	0.00031	0.10800
OGD	per-season vs raw	76,584	-0.00041	-0.00064	-0.00017	0.00100
OGD	per-season vs global	76,584	0.00012	-0.00005	0.00030	0.15800
FTRL	per-season vs raw	76,584	-0.00056	-0.00082	-0.00030	0.00000
FTRL	per-season vs global	76,584	0.00013	-0.00003	0.00031	0.10900
Uniform average	per-season vs raw	76,584	-0.00059	-0.00087	-0.00032	0.00000
Uniform average	per-season vs global	76,584	0.00014	-0.00004	0.00032	0.13700
B365	per-season vs raw	76,438	-0.00031	-0.00054	-0.00007	0.00700
B365	per-season vs global	76,438	0.00009	-0.00008	0.00026	0.28300
B365C	per-season vs raw	53,267	-0.00033	-0.00059	-0.00007	0.01200
B365C	per-season vs global	53,267	0.00010	-0.00004	0.00025	0.15400
BW	per-season vs raw	72,752	-0.00061	-0.00091	-0.00031	0.00000
BW	per-season vs global	72,752	0.00014	-0.00005	0.00032	0.14900
BWC	per-season vs raw	49,755	-0.00065	-0.00106	-0.00022	0.00100
BWC	per-season vs global	49,755	0.00014	-0.00008	0.00036	0.20500
IW	per-season vs raw	57,185	-0.00110	-0.00150	-0.00069	0.00000
IW	per-season vs global	57,185	0.00016	-0.00006	0.00039	0.14600
PS	per-season vs raw	71,284	-0.00017	-0.00039	0.00005	0.14000
PS	per-season vs global	71,284	0.00011	-0.00009	0.00030	0.24600
PSC	per-season vs raw	71,818	-0.00016	-0.00033	0.00001	0.08200
PSC	per-season vs global	71,818	0.00013	-0.00003	0.00028	0.10800
VC_BV	per-season vs raw	68,397	-0.00039	-0.00066	-0.00013	0.00000
VC_BV	per-season vs global	68,397	0.00009	-0.00009	0.00028	0.30500
WH	per-season vs raw	65,470	-0.00061	-0.00095	-0.00026	0.00000
WH	per-season vs global	65,470	0.00010	-0.00011	0.00033	0.37300

Table A.25: Calibration with a warm start.

Series	n	Mean diff.	CI low	CI high	p
Hedge	76,584	-0.00066	-0.00093	-0.00037	0.00000
Hedge	76,584	0.00003	-0.00005	0.00012	0.47600
Hedge	76,584	-0.00010	-0.00021	0.00001	0.08500
OGD	76,584	-0.00048	-0.00070	-0.00026	0.00000
OGD	76,584	0.00005	-0.00001	0.00011	0.13600
OGD	76,584	-0.00008	-0.00023	0.00007	0.31300
FTRL	76,584	-0.00066	-0.00094	-0.00038	0.00000
FTRL	76,584	0.00003	-0.00005	0.00012	0.47900
FTRL	76,584	-0.00010	-0.00021	0.00001	0.08600
Uniform average	76,584	-0.00069	-0.00097	-0.00040	0.00000
Uniform average	76,584	0.00004	-0.00005	0.00013	0.41800
Uniform average	76,584	-0.00010	-0.00024	0.00004	0.15700
B365	76,438	-0.00036	-0.00061	-0.00011	0.00500
B365	76,438	0.00004	-0.00009	0.00016	0.54800
B365	76,438	-0.00005	-0.00015	0.00004	0.27000
B365C	53,267	-0.00040	-0.00068	-0.00011	0.00500
B365C	53,267	0.00004	-0.00004	0.00012	0.37300
B365C	53,267	-0.00007	-0.00018	0.00005	0.27600

(continued)

Series	n	Mean diff.	CI low	CI high	p
BW	72,752	-0.00071	-0.00106	-0.00039	0.00000
BW	72,752	0.00004	-0.00011	0.00018	0.60100
BW	72,752	-0.00010	-0.00020	0.00001	0.06400
BWC	49,755	-0.00074	-0.00116	-0.00030	0.00100
BWC	49,755	0.00005	-0.00014	0.00025	0.59500
BWC	49,755	-0.00009	-0.00019	0.00002	0.10000
IW	57,185	-0.00122	-0.00161	-0.00082	0.00000
IW	57,185	0.00004	-0.00004	0.00012	0.38800
IW	57,185	-0.00012	-0.00031	0.00005	0.17900
PS	71,284	-0.00022	-0.00044	0.00000	0.05600
PS	71,284	0.00006	-0.00009	0.00021	0.38900
PS	71,284	-0.00005	-0.00015	0.00004	0.32200
PSC	71,818	-0.00022	-0.00038	-0.00005	0.00900
PSC	71,818	0.00007	-0.00001	0.00014	0.07400
PSC	71,818	-0.00006	-0.00016	0.00004	0.25400
VC_BV	68,397	-0.00043	-0.00066	-0.00020	0.00000
VC_BV	68,397	0.00006	-0.00001	0.00013	0.10700
VC_BV	68,397	-0.00004	-0.00021	0.00013	0.63800
WH	65,470	-0.00067	-0.00100	-0.00035	0.00000
WH	65,470	0.00003	-0.00009	0.00016	0.57100
WH	65,470	-0.00006	-0.00023	0.00009	0.44400

Table A.26: Mann–Kendall trend tests.

Series	Metric	S	Z	p	trend
Algorithm average	ece	-9	-0.71554	0.47427	decreasing
B365	ece	-3	-0.17889	0.85803	decreasing
BW	ece	5	0.35777	0.72051	increasing
Bookmaker average	ece	7	0.53666	0.59151	increasing
FTRL	ece	-11	-0.89443	0.37109	decreasing
Hedge	ece	-11	-0.89443	0.37109	decreasing
OGD	ece	-15	-1.2522	0.21050	decreasing
PS	ece	3	0.17889	0.85803	increasing
PSC	ece	-3	-0.17889	0.85803	decreasing
Uniform average	ece	-11	-0.89443	0.37109	decreasing
Algorithm average	log_loss	1	0.00000	1.0000	increasing
B365	log_loss	9	0.71554	0.47427	increasing
BW	log_loss	7	0.53666	0.59151	increasing
Bookmaker average	log_loss	7	0.53666	0.59151	increasing
FTRL	log_loss	1	0.00000	1.0000	increasing
Hedge	log_loss	1	0.00000	1.0000	increasing
OGD	log_loss	3	0.17889	0.85803	increasing
PS	log_loss	3	0.17889	0.85803	increasing
PSC	log_loss	1	0.00000	1.0000	increasing
Uniform average	log_loss	1	0.00000	1.0000	increasing
B365	overround_pct	31	2.6833	0.00729	increasing

(continued)

Series	Metric	S	Z	p	trend
BW	overround_pct	23	1.9677	0.04910	increasing
Bookmaker average	overround_pct	39	3.3988	0.00068	increasing
PS	overround_pct	31	2.6833	0.00729	increasing
PSC	overround_pct	33	2.8622	0.00421	increasing

Table A.27: Step-multiplier sweep.

family	setting	full_log_loss	tail	Series
Hedge x mult	0.00100	0.99885	0.99881	—
Hedge x mult	0.00500	0.99879	0.99877	—
Hedge x mult	0.01000	0.99873	0.99873	—
Hedge x mult	0.02000	0.99862	0.99867	—
Hedge x mult	0.05000	0.99845	0.99866	—
Hedge x mult	0.10000	0.99840	0.99873	—
Hedge x mult	0.25000	0.99842	0.99883	—
Hedge x mult	0.50000	0.99844	0.99891	—
Hedge x mult	0.75000	0.99845	0.99894	—
Hedge x mult	1.0000	0.99845	0.99896	—
Hedge x mult	1.5000	0.99846	0.99899	—
Hedge x mult	2.0000	0.99847	0.99901	—
Hedge x mult	5.0000	0.99854	0.99916	—
Hedge x mult	10.0000	0.99864	0.99934	—
Hedge x mult	25.0000	0.99881	0.99969	—
Hedge x mult	50.0000	0.99896	1.0002	—
Hedge x mult	100.0000	0.99907	1.0005	—
Hedge x mult	200.0000	0.99914	1.0007	—
Hedge x mult	500.0000	0.99919	1.0008	—
Hedge x mult	1,000	0.99922	1.0008	—
OGD x mult	1.0000	0.99826	0.99803	—
OGD x mult	2.0000	0.99822	0.99800	—
OGD x mult	5.0000	0.99814	0.99794	—
OGD x mult	10.0000	0.99804	0.99789	—
OGD x mult	25.0000	0.99784	0.99780	—
OGD x mult	50.0000	0.99761	0.99770	—
OGD x mult	75.0000	0.99749	0.99766	—
OGD x mult	90.0000	0.99747	0.99765	—
OGD x mult	100.0000	0.99745	0.99765	—
OGD x mult	110.0000	0.99745	0.99765	—
OGD x mult	125.0000	0.99745	0.99766	—
OGD x mult	150.0000	0.99748	0.99769	—
OGD x mult	200.0000	0.99758	0.99778	—
OGD x mult	500.0000	0.99822	0.99858	—
OGD x mult	1,000	0.99852	0.99895	—
Hedge const eta	0.00001	0.99878	0.99878	—
Hedge const eta	0.00003	0.99866	0.99875	—
Hedge const eta	0.00010	0.99847	0.99882	—
Hedge const eta	0.00030	0.99845	0.99892	—
Hedge const eta	0.00100	0.99848	0.99901	—
Hedge const eta	0.00300	0.99856	0.99920	—
Hedge const eta	0.01000	0.99874	0.99957	—
Hedge const eta	0.03000	0.99898	1.0003	—

(continued)

family	setting	full_log_loss	tail	Series
Hedge const eta	0.10000	0.99913	1.0007	—
Hedge const eta	0.30000	0.99922	1.0008	—
Hedge const eta	0.50000	0.99927	1.0008	—
Hedge const eta	1.0000	0.99932	1.0010	—
Hedge const eta	2.0000	0.99934	1.0011	—
OGD const eta	0.00010	0.99829	0.99806	—
OGD const eta	0.00030	0.99828	0.99804	—
OGD const eta	0.00100	0.99824	0.99799	—
OGD const eta	0.00300	0.99816	0.99792	—
OGD const eta	0.01000	0.99800	0.99781	—
OGD const eta	0.03000	0.99771	0.99767	—
OGD const eta	0.10000	0.99760	0.99795	—
OGD const eta	0.30000	0.99827	0.99882	—
OGD const eta	1.0000	0.99861	0.99916	—
OGD const eta	3.0000	0.99871	0.99911	—
OGD const eta	10.0000	0.99881	0.99920	—
OGD const eta	30.0000	0.99891	0.99951	—
OGD const eta	100.0000	0.99904	0.99959	—
FTRL const eta	0.00000	0.99886	0.99882	—
FTRL const eta	0.00000	0.99886	0.99882	—
FTRL const eta	0.00000	0.99886	0.99882	—
FTRL const eta	0.00000	0.99886	0.99882	—
FTRL const eta	0.00000	0.99885	0.99881	—
FTRL const eta	0.00000	0.99883	0.99880	—
FTRL const eta	0.00000	0.99877	0.99875	—
FTRL const eta	0.00000	0.99863	0.99876	—
FTRL const eta	0.00001	0.99841	0.99879	—
FTRL const eta	0.00003	0.99845	0.99894	—
FTRL const eta	0.00010	0.99846	0.99896	—
FTRL const eta	0.00030	0.99849	0.99903	—
FTRL const eta	0.00100	0.99858	0.99926	—
FTRL const eta	0.00300	0.99869	0.99963	—
FTRL const eta	0.01000	0.99906	1.0005	—
FTRL const eta	0.10000	0.99926	1.0008	—
FTRL const eta	1.0000	0.99934	1.0011	—
FTRL x mult	0.00100	0.99881	0.99879	—
FTRL x mult	0.00500	0.99865	0.99872	—
FTRL x mult	0.01000	0.99852	0.99879	—
FTRL x mult	0.02000	0.99841	0.99878	—
FTRL x mult	0.05000	0.99843	0.99889	—
FTRL x mult	0.10000	0.99845	0.99895	—
FTRL x mult	0.25000	0.99845	0.99896	—
FTRL x mult	0.50000	0.99846	0.99899	—
FTRL x mult	0.75000	0.99848	0.99903	—
FTRL x mult	1.0000	0.99850	0.99907	—
FTRL x mult	1.5000	0.99853	0.99915	—
FTRL x mult	2.0000	0.99854	0.99920	—
FTRL x mult	5.0000	0.99867	0.99946	—
FTRL x mult	10.0000	0.99889	0.99976	—
FTRL x mult	25.0000	0.99913	1.0005	—
FTRL x mult	50.0000	0.99920	1.0006	—
FTRL x mult	100.0000	0.99923	1.0007	—
FTRL x mult	200.0000	0.99926	1.0008	—
FTRL x mult	500.0000	0.99930	1.0009	—

(continued)

family	setting	full_log_loss	tail	Series
FTRL x mult	1,000	0.99932	1.0010	—
OGD honest M	4.6052	0.99819	0.99797	M=log(100)=4.61
OGD honest M	3.7630	0.99817	0.99796	M=3.76 (observed max)
OGD adaptive	—	0.99781	0.99842	AdaGrad per-expert
OGD adaptive	—	0.99815	0.99795	AdaGrad scalar
shipped	—	0.99842	0.99883	Hedge (shipped)
shipped	—	0.99745	0.99765	OGD (shipped)
shipped	—	0.99843	0.99889	FTRL (shipped)
shipped	—	0.99886	0.99882	Uniform average

Table A.28: Comparison of time windows.

Series	Type	Coverage %	Rounds	log-loss (raw)	log-loss (cal.)	window
PSC	bookmaker	87.9119	34,065	0.99471	0.99439	5yr
OGD	algorithm	100.0000	38,749	0.99556	0.99495	5yr
VC_BVC	bookmaker	79.6975	30,882	0.99554	0.99498	5yr
WHC	bookmaker	72.1361	27,952	0.99575	0.99502	5yr
BWC	bookmaker	91.9353	35,624	0.99609	0.99504	5yr
B365C	bookmaker	99.9974	38,748	0.99602	0.99543	5yr
Uniform average	algorithm	100.0000	38,749	0.99677	0.99591	5yr
Hedge	algorithm	100.0000	38,749	0.99689	0.99604	5yr
FTRL	algorithm	100.0000	38,749	0.99692	0.99607	5yr
WH	bookmaker	71.9090	27,864	0.99848	0.99771	5yr
PS	bookmaker	87.6952	33,981	0.99831	0.99794	5yr
VC_BV	bookmaker	79.0833	30,644	0.99870	0.99801	5yr
BW	bookmaker	91.5327	35,468	0.99981	0.99885	5yr
B365	bookmaker	99.8400	38,687	0.99951	0.99896	5yr
PSC	bookmaker	93.7768	71,818	0.99714	0.99685	10yr
OGD	algorithm	100.0000	76,584	0.99745	0.99693	10yr
Hedge	algorithm	100.0000	76,584	0.99842	0.99773	10yr
FTRL	algorithm	100.0000	76,584	0.99843	0.99774	10yr
Uniform average	algorithm	100.0000	76,584	0.99886	0.99813	10yr
B365C	bookmaker	69.5537	53,267	0.99953	0.99910	10yr
BWC	bookmaker	64.9679	49,755	0.99998	0.99919	10yr
VC_BV	bookmaker	89.3098	68,397	1.0007	1.0003	10yr
WH	bookmaker	85.4878	65,470	1.0012	1.0005	10yr
PS	bookmaker	93.0795	71,284	1.0008	1.0005	10yr
IW	bookmaker	74.6696	57,185	1.0019	1.0006	10yr
BW	bookmaker	94.9963	72,752	1.0014	1.0007	10yr
B365	bookmaker	99.8094	76,438	1.0011	1.0007	10yr

Table A.29: Information arrival by quintile.

stratifier	stratum	n	Comparison	Mean diff.	CI low	CI high	p
line movement	0	10,650	closing vs opening	-0.00044	-0.00084	0.00000	0.05200
line movement	1	10,649	closing vs opening	-0.00072	-0.00140	-0.00000	0.05000
line movement	2	10,649	closing vs opening	-0.00200	-0.00296	-0.00095	0.00000
line movement	3	10,649	closing vs opening	-0.00452	-0.00596	-0.00301	0.00000
line movement	4	10,649	closing vs opening	-0.01158	-0.01403	-0.00909	0.00000
Shin z anomaly	0	15,315	closing vs opening	-0.00295	-0.00397	-0.00187	0.00000
Shin z anomaly	1	15,311	closing vs opening	-0.00371	-0.00486	-0.00250	0.00000

(continued)

stratifier	stratum	n	Comparison	Mean diff.	CI low	CI high	p
Shin z anomaly	2	15,313	closing vs opening	-0.00281	-0.00401	-0.00169	0.00000
Shin z anomaly	3	15,303	closing vs opening	-0.00440	-0.00562	-0.00314	0.00000
Shin z anomaly	4	15,252	closing vs opening	-0.00498	-0.00641	-0.00353	0.00000

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